Bounds for the argument of the Riemann zeta function and a prime-corrected Riemann–von Mangoldt approximation of its zeros

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Abstract

The imaginary parts γ_k of all nontrivial zeros of the Riemann zeta function, counted with multiplicity, are approximated by a strictly increasing sequence τ_k^{C} defined by the Riemann–von Mangoldt main term and the phases of approximately $\sqrt{T}$ Euler factors at height T. The points are defined by sublevel measures rather than by choosing roots of a possibly nonmonotone equation. Without assuming the Riemann hypothesis (RH),

$$\frac{1}{\#\mathcal{K}_T}\sum_{k\in\mathcal{K}_T}|\tau_k^{\mathrm{C}}-\gamma_k|^2=O(\log^{-2}T),\qquad \mathcal{K}_T=\{k:T<\gamma_k\le 2T\}.$$

Thus the root-mean-square error is bounded by a constant multiple of the mean spacing $2\pi/\log T$ between zeros at height T. Combined with Selberg's second-moment theorem, an exact integral identity valid without RH and retaining repeated imaginary parts gives

$$\frac{1}{\#\mathcal{K}_T}\sum_{k\in\mathcal{K}_T}|\gamma_k-\tau_k|^2=\frac{2\log\log T}{\log^2 T}(1+o(1)),\qquad N_0(\tau_k)=k-\tfrac{1}{2}.$$

Consequently, the arithmetic correction $\tau_k^{\mathrm{C}}-\tau_k$ has the same leading second-moment asymptotic, and the corrected ordinal equation has an $O(1)$ mean-square defect at the actual imaginary parts γ_k.

Independently, an admissible-exponent-profile version of Backlund's method gives

$$\limsup_{T\to\infty}\frac{|S(T)|}{\log T}\le 0.0968160\ldots,\qquad \limsup_{k\to\infty}|\tau_k-\gamma_k|\le 0.60832,$$

and the explicit unconditional estimate

$$|S(T)|\le 0.09783\log T+0.7214\log\log T+10.6\qquad (T\ge e).$$

Its leading coefficient is smaller than that of Bellotti–Wong, while the complete bound is smaller only for $\log T>2425.8385\ldots$.

MSC 2020: 11M06, 11M26. **Keywords:** Riemann zeta function; Riemann hypothesis (RH); argument of the zeta function; Backlund's method; exponent pairs; zeros of the zeta function; Euler phases.

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1 Introduction

The imaginary parts γ_k of the nontrivial zeros of the Riemann zeta function are located only on average by the Riemann–von Mangoldt formula: the points τ_k defined by $N_0(\tau_k) = k - \frac{1}{2}$ (1.2), where N_0 is the smooth main term of the zero-counting function, discard the oscillating term $S(t) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + it)$. The primes enter through S. The Euler product writes $\zeta(s)$ as a product of one factor $(1-p^{-s})^{-1}$ for each prime p. The *Euler phase of the prime* p is $-\frac{1}{\pi} \arg(1-p^{-1/2-it})$, the normalized phase of this factor at $s = \frac{1}{2} + it$, and $S(t)$ is, formally, the sum of the Euler phases of all primes. Truncated at the primes $p \le x$, with x a small power of t, this sum approximates S in mean square, unconditionally (Selberg, Tsang, Fujii; see below). Lemma 7.6 extends this to about $\sqrt{t}$ primes. The *arithmetic correction* studied here adds the Euler phases of finitely many primes to the main term (Φ_m in (1.10), interpolated as Φ in (1.11)). The zeros are then approximated by points τ_k^{C} attached to the levels $N_0 + \Phi = k - \frac{1}{2}$, instead of the roots τ_k of $N_0(t) = k - \frac{1}{2}$.

The main result constructs the strictly increasing sequence τ_k^{C} through the sublevel measures (1.12) of $N_0+\Phi$, where Φ is defined by the Euler phases of finitely many primes. Unconditionally, the root-mean-square error of τ_k^{C} with respect to γ_k on dyadic blocks is $O(1/\log T)$, a bounded multiple of the mean zero spacing (Theorem 1.5). The key technical feature is the sublevel-set definition: it remains valid even when the prime-corrected counting function $N_0 + \Phi$ is not monotone, where the equation $N_0 + \Phi = k - \frac{1}{2}$ may have no unique root.

An exact defect identity supplies an independent comparison. Without RH, the uncorrected points τ_k differ from the γ_k by root-mean-square $\sqrt{2 \log\log T}/\log T\,(1+o(1))$, the factor $\log\log T$ coming from Selberg's mean square of S. Thus the arithmetic correction removes the factor

$\log\log T$ from the proved mean-square upper order. Further consequences proved below describe the arithmetic correction itself and the defect of the ordinal equation when evaluated at the actual imaginary parts γ_k.

The other results concern the argument $S(T)$ of the zeta function. A profile-based application of Backlund's method yields an asymptotic coefficient $0.0968160\ldots$, a resulting upper limit 0.60832 for the distance between the points τ_k of the Riemann–von Mangoldt equation and the actual imaginary parts γ_k, and a fully explicit estimate $|S(T)| \le 0.09783\log T + 0.7214\log\log T + 10.6$. The leading coefficient is smaller than Bellotti–Wong's; the complete estimate improves upon theirs only for $\log T > 2425.8385\ldots$.

Several related approaches supplement the main term $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}$ of the Riemann–von Mangoldt formula using primes or values of ζ. They differ from the present construction in the correcting term and in whether RH is assumed.

For approaches that use ζ, explicit bounds in the critical strip (Yang [44]; Bellotti [2]) give unconditional bounds for $S(T)$ and hence for $N - N_0$ (Trudgian [40], [41]; Bellotti and Wong [3, Thm. 1.1, Cor. 1.5]). These control the size of $N - N_0$, not the position of the imaginary parts.

França and LeClair [8, eq. (20)] write the n-th imaginary part as a solution of $N_0(t) + S(t) = n - \frac{1}{2}$ with the exact S; they do not assume RH, but they assume that $\lim_{\delta\to 0^+}\arg\zeta(\frac{1}{2}+\delta+it)$ exists for every t, and they do not prove that the equation has exactly one solution for each n [8, §9]. In approaches using primes, S is approximated by a sum over primes $p \le x$, in mean square and unconditionally (Selberg [37], Tsang [42], Fujii [11]).

Kargin [22, Thm. 1.1] takes t_k with $N_0(t_k) = k - \frac{1}{2}$ and proves, without RH, that $\gamma_k - t_k$, suitably normalized, is asymptotically Gaussian (for his enumeration of the zeros, which agrees with the one used here when RH holds; see Section 1.2), a limit law over windows of k and not a bound for each imaginary part.

Korolev [23, Thm. 4.2] proves, without RH, Selberg's formulas for the moments of the Gram displacements Δ_k^{G}, the integers that locate each γ_k among the Gram points, where the smooth term N_0 takes integer values up to the Stirling remainder.

Gonek [13] builds from a finite Euler product a function ζ_X whose zeros up to height t number $\frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} - \frac{1}{\pi}\sum_{n\le X^2}\Lambda_X(n)\frac{\sin(t\log n)}{n^{1/2}\log n} + O_X(1)$ for $X < \exp(C_3\log t/\Phi(t))$, in his notation, under RH [13, Thm. 7.5, p. 2175], and these zeros converge to those of ζ as $X\to\infty$, again under RH and without a rate [13, Thm. 9.1, p. 2181].

LeClair [24] replaces S by a finite number of Euler factors, a free parameter, on the basis of a conjectured random-walk behavior of sums over primes; that work is primarily numerical and does not state an error bound.

The present article is unconditional, uses about $\sqrt{t}$ primes at height t, a number tied to the height rather than a free parameter, and bounds the mean-square error of approximations τ_k^{C} that are increasing and indexed by construction.

1.1 Inverting the Riemann–von Mangoldt formula

Let $\rho_k = \beta_k + i\gamma_k$ be the nontrivial zeros of the Riemann zeta function with positive imaginary parts, listed with multiplicity and $0 < \gamma_1 \le \gamma_2 \le \cdots$, and let $N(t) = \#\{k : \gamma_k \le t\}$. For $t > 0$, $t \notin \{\gamma_k\}$, the Riemann–von Mangoldt formula reads [39, §9.3]

$$N(t) = N_0(t) + S(t) + \frac{\varpi(t)}{\pi}, \qquad N_0(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}, \qquad S(t) = \frac{1}{\pi}\arg\zeta\big(\tfrac{1}{2}+it\big), \quad (1.1)$$

where $\varpi(t) = O(1/t)$ is the Stirling remainder of the Riemann–Siegel theta function (Lemma 4.1). At $t \in \{\gamma_k\}$ this paper uses the right-hand convention $S(t) = S(t^+)$, compatible with the right-continuous counting function N. Different conventions are used in the literature (see [39, §9.3]). All bounds below have continuous majorants and extend to both one-sided limits by approximation through points $t \notin \{\gamma_k\}$. The function N_0 is strictly increasing on $(2\pi,\infty)$, with

$N_0(2\pi) = -\frac{1}{8}$ and $N_0(t) \to \infty$ as $t \to \infty$. Thus for every integer $k \geq 1$ there is exactly one $\tau_k > 2\pi$ with

$$N_0(\tau_k) = k - \tfrac{1}{2}. \tag{1.2}$$

The sequence (τ_k) solves the counting equation with the oscillating term S removed. In terms of the Gram points g_m, defined by $\vartheta(g_m) = m\pi$ for $m \geq -1$ ($g_{-1} = 9.66\ldots$), one has $N_0(g_m) = m + 1 - \varpi(g_m)/\pi$, because $\vartheta/\pi + 1 = N_0 + \varpi/\pi$ (Lemma 4.1). Since $|\varpi|/\pi < \frac{1}{2}$, we have $N_0(g_{k-2}) < k - \frac{1}{2} < N_0(g_{k-1})$ for every $k \geq 1$. Strict monotonicity thus gives $g_{k-2} < \tau_k < g_{k-1}$. The imaginary part γ_k lies in the same Gram interval, $\gamma_k \in (g_{k-2}, g_{k-1}]$, exactly when the displacement Δ_k^{G} of Remark 7.7 vanishes. This is Gram's law in the index convention adopted here for the k-th imaginary part (the Gram–Selberg phenomenon of [23, Def. 1.4]), and it fails for a positive proportion of k [23, p. 58]. The resulting sequence gives a continuous counterpart of the Gram scale, read at the half-integer levels.

The object of this paper is to determine how far these points lie from the imaginary parts γ_k, and how much arithmetic information can bring them, in root-mean-square, within a bounded number of mean spacings $2\pi/\log\gamma_k$, with the intermediate objective of obtaining a better approximation of τ_k (or modifications thereof) to γ_k. The first question concerns the size of S; the second concerns its approximation by the Euler phases of the primes.

1.2 Results

Throughout, write $\mathfrak{a} = \limsup_{T\to\infty} |S(T)|/\log T$. An *admissible profile* is a continuous function $\mathfrak{e} : [\frac{1}{2}, 1] \to [0, \infty)$ with $\mathfrak{e}(1) = 0$ such that, for every $\varepsilon > 0$,

$$\zeta(\sigma + it) \ll_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon} \qquad \text{uniformly for } \tfrac{1}{2} \leq \sigma \leq 1,\ |t| \geq 2. \tag{1.3}$$

Its *reflected extension* is

$$\mathfrak{e}^*(\sigma) = \begin{cases} \frac{1}{2} - \sigma, & \sigma < 0, \\ \frac{1}{2} - \sigma + \mathfrak{e}(1-\sigma), & 0 \leq \sigma < \frac{1}{2}, \\ \mathfrak{e}(\sigma), & \frac{1}{2} \leq \sigma \leq 1, \\ 0, & \sigma > 1, \end{cases} \tag{1.4}$$

a continuous function on $\mathbb{R}$ (at $\sigma = 0$ both formulas give $\frac{1}{2}$ because $\mathfrak{e}(1) = 0$).

Theorem 1.1 (Asymptotic bound)**.** *For every admissible profile $\mathfrak{e}$,*

$$\mathfrak{a} \leq \Lambda(\mathfrak{e}) := \frac{1}{2\pi\log 2} \int_0^1 \frac{\mathfrak{e}^*(\sigma)}{\sqrt{\sigma(2-\sigma)}}\, d\sigma. \tag{1.5}$$

Let $\mathfrak{e}_{\mathrm{B}}$ be the piecewise linear function through the nine points

$$\begin{gathered} \left(\tfrac{1}{2}, \tfrac{13}{84}\right),\ \left(\tfrac{139}{194}, \tfrac{13}{194}\right),\ \left(\tfrac{173}{207}, \tfrac{13}{414}\right),\ \left(\tfrac{773}{854}, \tfrac{13}{854}\right),\ \left(\tfrac{820}{867}, \tfrac{13}{1734}\right), \\ \left(\tfrac{3387}{3494}, \tfrac{13}{3494}\right),\ \left(\tfrac{1149}{1169}, \tfrac{13}{7014}\right),\ \left(\tfrac{13921}{14054}, \tfrac{13}{14054}\right),\ (1, 0). \end{gathered} \tag{1.6}$$

The nodes are the points $(\lambda_j - \kappa_j, \kappa_j)$ for $(\kappa_j, \lambda_j) = A^j(\frac{13}{84}, \frac{55}{84})$, $0 \leq j \leq 7$, followed by $(1, 0)$. Bourgain's cited exponent pairs include an arbitrary positive ε in both coordinates; the limiting profile is obtained after taking $\varepsilon \to 0^+$, as justified in the proof of Corollary 1.2.

Corollary 1.2 (Bourgain profile)**.** $\mathfrak{a} \leq \Lambda(\mathfrak{e}_{\mathrm{B}}) = 0.0968160063\ldots$

Theorem 1.3 (Explicit bound)**.** *For every $T \geq e$,*

$$|S(T)| \leq 0.09783 \log T + 0.7214 \log\log T + 10.6, \tag{1.7}$$

$$\left|N(T) - N_0(T)\right| \leq 0.09783 \log T + 0.7214 \log\log T + 10.7. \tag{1.8}$$

The coefficient 0.09783 comes from the value 0.0978292318… of the Backlund quotient (5.10) for the explicit profile $\mathfrak{e}_{\mathrm{E}}$ of Section 4.1, at fixed parameters $c = 1.000002$, $R = 1.000005$. Among the explicit unconditional bounds compared in Table 1, the coefficient 0.09783 is the smallest leading coefficient in a bound of the form $|S(T)| \le A_1 \log T + A_2 \log\log T + A_3$ for all $T \ge e$.

The previous values are 0.112 (Trudgian, 2014), 0.110 (Platt and Trudgian, 2015), 0.1038 (Hasanalizade, Shen and Wong, 2022) and 0.10076 (Bellotti and Wong, 2025). Table 1 lists the whole history, from von Mangoldt (1905).

For S, Bellotti and Wong [3, Cor. 1.5] prove $|S(T)| \le 0.10076 \log T + \min\{0.24460 \log\log T + 7.20844,\ 1.68845 \log\log T + 1.50956\}$ for $T \ge e$ (their Theorem 1.1, for N, is discussed in Section 1.3). The bound (1.7) is the smaller of the two exactly when $\log T > 2425.8385\ldots$. The minimum of the two bounds holds for all $T \ge e$.

Corollary 1.4 (Distance to the imaginary parts). *(a)* $\limsup_{k\to\infty} |\tau_k - \gamma_k| \le 2\pi\Lambda(\mathfrak{e}_{\mathrm{B}}) = 0.6083129\ldots < 0.60832$.

(b) For every $k \ge 1$, with $\mathfrak{m}_k := \min(\tau_k, \gamma_k) > 2\pi$,

$$|\tau_k - \gamma_k| \le \frac{2\pi\big(0.09783 \log\gamma_k + 0.7214 \log\log\gamma_k + 10.2\big)}{\log(\mathfrak{m}_k/2\pi)}. \tag{1.9}$$

Now, for the localization by primes let $p_1 < p_2 < \cdots$ be the primes, let

$$\Phi_m(t) = -\frac{1}{\pi}\sum_{\nu=1}^{m} \arg\big(1 - p_\nu^{-1/2-it}\big) = -\frac{1}{\pi}\sum_{\nu=1}^{m}\sum_{r\ge1} \frac{\sin(rt\log p_\nu)}{r\, p_\nu^{r/2}} \tag{1.10}$$

be the sum of the Euler phases of the first m primes, the Euler phase of the prime p being $-\frac{1}{\pi}\arg(1 - p^{-1/2-it})$ (principal argument; $m \ge 0$, and $\Phi_0 = 0$ as an empty sum). For every fixed m, the series over r converges absolutely: $|p^{-1/2-it}| < 1$ and $\Re(1 - p^{-1/2-it}) > 0$, so the principal logarithm satisfies $\mathrm{Log}(1-z) = -\sum_{r\ge1} z^r/r$. Let $m_j = \lfloor 2^{j/2} \rfloor \ge 0$ for $j \in \mathbb{Z}$, and on the dyadic block $2^j \le t \le 2^{j+1}$ let

$$\Phi(t) = \Big(2 - \frac{t}{2^j}\Big)\Phi_{m_j}(t) + \Big(\frac{t}{2^j} - 1\Big)\Phi_{m_{j+1}}(t), \qquad \mathcal{F}(t) = N_0(t) + \Phi(t). \tag{1.11}$$

We define the corrected points by

$$\tau_k^{\mathrm{C}} = \int_0^\infty \mathbf{1}_{\{\mathcal{F}(t) < k - \frac12\}}\, dt \qquad (k \ge 1). \tag{1.12}$$

Theorem 1.5 (Localization by primes). *The numbers τ_k^{C} are finite and $0 < \tau_1^{\mathrm{C}} < \tau_2^{\mathrm{C}} < \cdots$. For $k \ge 2$ (equivalently $\tau_k \ge 2\pi e$, since $N_0(2\pi e) = \frac78$), τ_k^{C} is determined by k and by the primes p_ν with $\nu \le \lfloor 2\sqrt{\tau_k} \rfloor$. There are absolute constants C and T_0 such that, with $\mathcal{K}_T = \{k : T < \gamma_k \le 2T\}$,*

$$\sum_{k\in\mathcal{K}_T} (\tau_k^{\mathrm{C}} - \gamma_k)^2 \le C\,\frac{T}{\log T}, \qquad \text{hence} \qquad \frac{1}{\#\mathcal{K}_T}\sum_{k\in\mathcal{K}_T} (\tau_k^{\mathrm{C}} - \gamma_k)^2 \le \frac{C'}{\log^2 T} \qquad (T \ge T_0), \tag{1.13}$$

with C' absolute.

Corollary 1.6. *For every $\mathsf{A} > 0$ and $T \ge T_0$, $\#\{k \in \mathcal{K}_T : |\tau_k^{\mathrm{C}} - \gamma_k| > \mathsf{A}\cdot\frac{2\pi}{\log T}\} \le \frac{C''}{\mathsf{A}^2}\,\#\mathcal{K}_T$, with C'' absolute.*

The sequence τ_k does not attain this mean-square order, and its leading mean-square asymptotic is known.

Theorem 1.7 (Mean square without primes)**.** *Without any hypothesis on the zeros, as* $T \to \infty$*,*

$$\frac{1}{\#\mathcal{K}_T} \sum_{k \in \mathcal{K}_T} (\tau_k - \gamma_k)^2 = \frac{2 \log\log T}{\log^2 T} \big(1 + o(1)\big), \qquad \textit{hence} \qquad \sum_{k \in \mathcal{K}_T} (\tau_k - \gamma_k)^2 \asymp \frac{T \log\log T}{\log T}. \tag{1.14}$$

Theorem 1.7 is an asymptotic equality: the root-mean-square of $\tau_k - \gamma_k$ is $\frac{\sqrt{2 \log\log T}}{2\pi}$ mean spacings, no more and no less. The Euler phases of the primes in Theorem 1.5 remove the factor $\log\log T$ from the proved mean-square upper order.

The proof (Section 8) rests on an exact identity, valid for every absolutely continuous function F, between the sum of $(k - \frac{1}{2} - F(\gamma_k))^2$ over the imaginary parts and the integral of $(N - F)^2 F'$, together with Selberg's mean square of S. It does not use the transfer inequality of Section 7.

Kargin [22, Thm. 1.1] considers the points t_k defined by $N_0(t_k) = k - \frac{1}{2}$, which are the τ_k of (1.2), and proves, without assuming RH, that

$$\frac{(\gamma_k - \tau_k) \log \tau_k}{\sqrt{2 \log\log \tau_k}} \tag{1.15}$$

is asymptotically standard normal, with convergence of moments, when k is uniform in a window $N \le k \le N + \lfloor N^\theta \rfloor - 1$, $\frac{1}{2} < \theta \le 1$. His γ_k enumerate the zeros with real part at least $\frac{1}{2}$, which is the enumeration used here when RH holds.

Korolev [23, Thm. 4.2] proves Selberg's formulas for the moments of the Gram displacements Δ_k^{G} over short ranges of k. Since

$$|s_k + \Delta_k^{\mathrm{G}}| < 1 \tag{1.16}$$

(Remark 7.7), his second moment leads to the same value of $\langle s_k^2 \rangle_{\mathcal{K}_T}$. Neither work states (1.14) for the full enumeration of the zeros. The present route of Section 8 uses neither Gram intervals nor probabilistic limit theorems, and the identity behind it holds for every C^1 function in place of N_0, monotone or not. These antecedents address the same scale of ordinal fluctuations but do not state the present full-multiplicity mean-square identity.

Theorem 1.5 is unconditional and makes no assumption about the real parts of the zeros (RH) or about their multiplicities. The order of the points follows from the inclusion of sublevel sets as the level increases. In this way, no root selection or Newton step is needed.

1.3 Relation to the literature

Bounds of the form $A_1 \log T + A_2 \log\log T + A_3$ for $|N(T) - \frac{T}{2\pi} \log \frac{T}{2\pi e}|$ go back to von Mangoldt [43], and for $|S(T)|$ to Backlund [1]. Table 1 lists them.

The leading coefficient was lowered by Trudgian to 0.112 (the value of the published version, as listed in [17, Table 1] and [3, Table 1]; the preprint [41, Thm. 1] states 0.111), by Platt and Trudgian to 0.110 [32, Cor. 1], by Hasanalizade, Shen and Wong to 0.1038 [17, Cor. 1.2], and by Bellotti and Wong to 0.10076 [3, Cor. 1.5].

The proof of [32, Cor. 1] rests on [41] (see [3, §5, footnote 8]) and on the bound for $\zeta(\frac{1}{2} + it)$ of [32, Thm. 1], which is affected by the error in the lemma of Cheng and Graham pointed out in [34, p. 3]. (Fiori [9] corrected the Phragmén–Lindelöf lemma [41, Lemma 3]; the version used here is his Corollary 3, stated as Theorem 3.4.) These proofs use a disc argument together with Jensen's formula. The present proof changes the treatment of the reflected exponents on the left of the critical line.

Bellotti and Wong give two separate bounds. Their Theorem 1.1 is for the counting function without the constant $\frac{7}{8}$: $|N(T) - \frac{T}{2\pi} \log \frac{T}{2\pi e}| \le 0.10076 \log T + 0.24460 \log\log T + 8.08344$ for $T \ge e$. Their Corollary 1.5 is the bound for $|S(T)|$ quoted after Theorem 1.3, with the constant 7.20844 in the first term of the minimum, and $8.08344 - 7.20844 = \frac{7}{8}$.

By (1.1), $N(T) - \frac{T}{2\pi} \log \frac{T}{2\pi e} = S(T) + \frac{7}{8} + \varpi(T)/\pi$ with $|\varpi(T)|/\pi < 0.07$ (Lemma 4.1), so (1.7) is to be compared with their Corollary 1.5, and (1.8), in which N_0 contains $\frac{7}{8}$, with the bound

year	authors	A_1	A_2	A_3	range
1905	von Mangoldt [43]	0.4320	1.9167	13.0788	$T \geq 28.558$
1913	Grossmann [14]	0.2907	1.7862	7.0120	$T \geq 50$
1916	Backlund [1]	0.1370	0.4430	5.2250	$T \geq 2$
1941	Rosser [36]	0.1370	0.4430	2.4630	$T \geq 2$
2012	Trudgian [40]	0.1700	0	2.8730	$T \geq e$
2014	Trudgian [41]	0.1120	0.2780	3.3850	$T \geq e$
2015	Platt, Trudgian [32]	0.1100	0.2900	3.165	$T \geq e$
2022	Hasanalizade, Shen, Wong [17]	0.1038	0.2573	9.4925	$T \geq e$
2025	Bellotti, Wong [3]	0.10076	0.24460	8.08344	$T \geq e$
2026	Casaña-Giner [This work]	0.09783	0.7214	11.55	$T \geq e$

Table 1: Explicit bounds $\left|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\right| \leq A_1 \log T + A_2 \log\log T + A_3$. The rows up to 2025 are those of [3, Table 1], with the year of publication, checked against [43, p. 1] (0.43200, 1.91662, 13.07873), [1, (39) and pp. 355, 375] (0.137, 0.443, $4.350 + \frac{7}{8}$; Backlund's $Q(T)$ is $N(T) - N_0(T)$, p. 354), Grossmann's constants as quoted in [1, p. 355] (0.29062, 1.7862, $6.13695 + \frac{7}{8}$), and [40, Thm. 1] (0.17, $1.998 + \frac{7}{8}$). Backlund states his inequality (39) for every $T \geq 2$ [1, p. 375]; [3, Table 1] lists $T \geq 200$, the range of his earlier bound [1, p. 355]; for [17] the constant term is the corrected value of [3, §1, footnote 1] (9.3675 in [17, Cor. 1.2]); the preprint [41] has 0.111 and 0.275 in place of 0.1120 and 0.2780 [41, Thm. 1, Cor. 1]. The last row follows from (1.7), (1.1) and $|\varpi(T)|/\pi < 0.07$ for $T \geq e$ (Lemma 4.1): $10.6 + \frac{7}{8} + 0.07 = 11.545 < 11.55$.

$|N(T) - N_0(T)| \leq 0.10076 \log T + 0.24460 \log\log T + 7.28$ that follows from their Corollary 1.5 and Lemma 4.1.

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Bellotti and Wong already take the center and the radius of the disc close to 1 ($c = 1.000225$, $r = 1.000605$ in [3, Table 2], chosen numerically at fixed T_0; the limit $c, R \to 1$ of Theorem 1.1 is the asymptotic form of the same choice, and is not taken there) and use on $[\frac{1}{2}, 1]$ the same explicit profile $\mathfrak{e}_{\mathrm{E}}$ as Theorem 1.3.

To the left of $\frac{1}{2}$, they bound ζ by linear interpolation of exponents between $\sigma = 0$ and $\sigma = \frac{1}{2}$ [3, §3.1]. The improvement comes from the exact reflection (1.4) given by the functional equation, pointwise (Lemma 4.3). It lies below the chord because $\mathfrak{e}_{\mathrm{E}}$ is convex.

With $c, R \to 1$ the Backlund quotient of the profile of [3] tends to $0.10073687\ldots$, that of the reflected profile to $0.09782899\ldots$ (Appendix A). Of the reduction of the leading coefficient from 0.10076 to 0.09783, the limit $c, R \to 1$ accounts for $0.10076 - 0.10073687\ldots = 0.0000231\ldots$ and the reflected profile for $0.10073687\ldots - 0.09782899\ldots = 0.0029078\ldots$ (the remaining $0.09783 - 0.09782899\ldots < 10^{-6}$ is the cost of fixed c, R and of rounding). The improvement is thus essentially due to the reflected profile.

Bellotti and Wong remark that 0.10076 is nearly optimal among the values that their general theorem gives with the explicit bounds for ζ now available [3, §1].

Theorem 1.3 agrees with that remark: it uses the same bounds for ζ on $[\frac{1}{2}, 1]$ and a disc of the same kind, and lowers the coefficient only through the treatment of ζ to the left of $\frac{1}{2}$, which lies outside the scope of the remark. Below a value that is nearly optimal for the established method, a further gain requires a change in the method.

The reduction obtained, 0.00293, is of the same size as the previous step $0.1038 \to 0.10076$. At fixed c, R very close to 1, this gives a larger constant term. For the Lindelöf profile $\mathfrak{e} = 0$ the functional (1.5) is $0.0786252\ldots$, and $\Lambda(\mathfrak{e}) \geq \Lambda(0)$ for every admissible $\mathfrak{e}$ (Remark 5.2).

The present analysis does not determine whether other choices of disc parameters yield a smaller asymptotic coefficient.

The approximation of S by Euler phases is Selberg's [37].

Its mean-value form, the continuous counterpart of Theorem 1.5, is classical: for x a small fixed power of T, the moments of $S(t)$ minus the phase of the primes $p \le x$ are bounded on average over $[T, 2T]$, a result attributed to Selberg and Tsang in [7, §3, before (3.6)], with reference to [42, Thm. 5.1].

It is used here in Fujii's form, Theorem 3.8(a), assembled from [11, §3], together with Fujii's moments for $S(t+h) - S(t)$ [11, Main Theorem]. Its extension to about $\sqrt{t}$ primes, counted by number, follows from the mean value theorem of Montgomery and Vaughan (Theorem 3.9) and is carried out in Lemma 7.6.

The new ingredient of Theorem 1.5 is the passage from the counting function to the imaginary parts themselves: the sum is over the zeros, and neither the monotonicity of $\mathcal{F}$ nor any hypothesis on the zeros is used. This passage is Theorem 7.3. França and LeClair [8] study equations $N_0(t) + S(t) = k - \frac{1}{2}$ for the individual imaginary parts, with the exact S. That each of them has exactly one solution is not proved there.

Two prime-based constructions provide a comparison with Theorem 1.5. LeClair [24, §VI, eq. (49)] replaces S by the phase of a finite number of Euler factors, that number being a free numerical parameter, and computes very high zeros by root-finding near the previous approximation. The aim there is numerical, and no error bound is stated. Gonek [13] builds functions ζ_X from a smoothed Euler product over $n \le X^2$ (in this paragraph X is his parameter); for large t the zeros of $\zeta_X(\frac{1}{2} + it)$ are the points where a phase function built from primes crosses the levels $(2k+1)\pi$ [13, §10]. His results on these zeros assume RH: their number and simplicity when X is small with respect to t [13, Thm. 8.1], and their convergence to the imaginary parts of the zeros of ζ as $X \to \infty$, on a fixed compact interval and without a rate [13, Thm. 9.1]. Theorem 1.5 has the following properties:

1. it is unconditional;
2. it is quantitative: the root-mean-square error is $O(1/\log T)$, and Corollary 1.6 bounds the proportion of large deviations;
3. the number of primes is tied to the height, about $\sqrt{t}$ primes at height t, instead of being a free parameter;
4. τ_k^{C} is the measure of a sublevel set, so the points are increasing and carry the index k by construction; in [24] and [13] the solutions are indexed by selection or by counting.

By comparison, Theorem 1.5 controls only the mean square, whereas Gonek's Theorem 9.1, under RH, concerns every imaginary part.

Monotonicity. The bound (1.13) does not follow directly from the results of Gonek [13] and LeClair [24]: deducing it from them would require, among other things, a proof that $\mathcal{F}$ is monotone. The available arguments for monotonicity assume RH [13, Thm. 8.1 and §10]; the present criterion avoids RH and does not require monotonicity. In this paragraph X denotes a cutoff for the primes, as in [13]. If $\mathcal{F}$, which is continuous, were strictly increasing on $[2\pi, \infty)$, then for $k \ge 2$ the equation $\mathcal{F}(t) = k - \frac{1}{2}$ would have exactly one solution, equal to τ_k^{C} (on $(0, 2\pi]$, $\mathcal{F} < \frac{7}{8} + \frac{1}{2} < \frac{3}{2}$ by (7.1)); monotonicity alone does not bound $\tau_k^{\mathrm{C}} - \gamma_k$, since $\mathcal{F}'$ may be small in a long gap, and the passage from the counting error $N - \mathcal{F}$ to the positions is Theorem 7.3 in either case. For points defined as solutions, however, monotonicity is needed, and the two requirements, bounded error and monotonicity, pull in opposite directions.

1. *Few primes give monotonicity but not bounded error.* Gonek's zeros of $\zeta_X(\frac{1}{2}+it)$ are simple and are counted because his phase is increasing, which he proves for X small with respect to t and under RH [13, Thm. 8.1 and §10]. In that range the prime approximation does not have bounded error: by Selberg's approximation [11, §3] and Theorem 3.9, the mean square over $[T, 2T]$ of S plus $\frac{1}{\pi}\sum_{p \le X} p^{-1/2}\sin(t\log p)$ is $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O\big(1 + (\log\frac{\log T}{\log X})^{1/2}\big)$

for $2 \le X \le T^{1/40}$ (Theorem 3.8(a) with $\mathsf{k} = 1$, $\alpha_{\mathrm{F}} = \frac{1}{2}$, $H = T$, $Y = T^{1/40}$, and Theorem 3.9 for the primes in (X, Y); with $P(t) = \sum_{X<p<Y} p^{-1/2-it}$, the sine sum is $-\operatorname{Im} P$, $(\operatorname{Im} P)^2 = \frac{1}{2}|P|^2 - \frac{1}{2}\operatorname{Re} P^2$, and $|\int_T^{2T} P(t)^2\,dt| \ll Y$ as in item 2). This tends to infinity when $\log X = o(\log T)$.

2. *Many primes give bounded error without establishing monotonicity.* Bounded error needs $\log X \gg \log T$. Then the derivative of the sum of the Euler phases of the primes is not negligible relative to N_0' in mean square: by Theorem 3.9, the mean square over $[T, 2T]$ of $\frac{1}{\pi}\sum_{p\le X} p^{-1/2}\log p\cos(t\log p)$ is $\frac{1}{2\pi^2}\sum_{p\le X}\frac{\log^2 p}{p}(1+O(\frac{p}{T})) \asymp \log^2 X \asymp \log^2 T$ for $2 \le X \le T$ with $\log X \asymp \log T$, the order of $N_0'(T)^2$. (With $P(t) = \sum_{p\le X} p^{-1/2}\log p\, p^{-it}$, the cosine sum is $\operatorname{Re} P$ and $(\operatorname{Re} P)^2 = \frac{1}{2}|P|^2 + \frac{1}{2}\operatorname{Re} P^2$. Theorem 3.9 gives the mean of $|P|^2$, and $|\int_T^{2T} P(t)^2\,dt| \le \frac{2}{\log 4}\big(\sum_{p\le X} p^{-1/2}\log p\big)^2 \ll X$, since $|\int_T^{2T}(pq)^{-it}\,dt| \le 2/\log(pq)$. Divided by T, this is absorbed in the remainder $\sum_{p\le X}\frac{\log^2 p}{p}O(\frac{p}{T}) \gg \frac{X}{T}$.) No argument presently available establishes monotonicity of $\mathcal{F}$ in this range and the derivative estimate gives little information about whether $\mathcal{F}$ is monotone.

At a level where $\mathcal{F}$ is not monotone, the equation may have several solutions. If solutions are indexed by counting, every spurious pair shifts all the later indices, so the errors accumulate along the sequence. Theorem 1.5 neither proves nor uses monotonicity of $\mathcal{F}$. The number τ_k^{C} is the distribution function $\lambda \mapsto \operatorname{meas}\{t > 0 : \mathcal{F}(t) < \lambda\}$ evaluated at $\lambda = k - \frac{1}{2}$. Equivalently, it is the position at which the increasing rearrangement of $\mathcal{F}$ reaches this level, with the generalized-inverse convention ([15, §10.12] define the rearrangement in decreasing order as the inverse of the distribution function; the increasing one is defined in the same way). If $\mathcal{F}$ is replaced on a bounded interval I by another function with the same minimum and maximum on I, then τ_k^{C} changes only for the levels $k - \frac{1}{2}$ with $\min_I \mathcal{F} < k - \frac{1}{2} \le \max_I \mathcal{F}$: for the other levels $\operatorname{meas}(I \cap \{\mathcal{F} < k - \frac{1}{2}\})$ is 0 or $|I|$ in both cases. Theorem 7.3, which holds for every measurable f, bounds the total effect by the fourth moment of $\lfloor \mathcal{F} + \frac{1}{2} \rfloor - N$ and the third moment of the gaps. The sublevel construction thus takes the place of a monotonicity argument.

Precedents of the construction. The definition (1.12) evaluates the distribution function $D_{\mathcal{F}}(\lambda) = \operatorname{meas}\{t > 0 : \mathcal{F}(t) < \lambda\}$ at $\lambda = k - \frac{1}{2}$. With the standard generalized-inverse convention, the increasing rearrangement reaches this level at the position $D_{\mathcal{F}}(k - \frac{1}{2})$ (Remark 7.4).

The imaginary parts themselves are the same construction applied to N, since $\gamma_k = \operatorname{meas}\{N < k - \frac{1}{2}\}$, and Theorem 7.3 compares the two. Rearrangements and quantile functions are classical [15, Ch. X]. An application of this construction to the imaginary parts of the zeros, or an inequality of the form (7.5), is not known to the author to have been published as of 30 October 2026.

As of 30 October 2026, the author is not aware of published results that combine the unconditional construction by the Euler phases of finitely many primes, increasing sublevel-defined approximations of the imaginary parts, and the mean-square localization in Theorem 1.5. The bounds in Theorems 1.1 and 1.3 are compared quantitatively with their cited antecedents above.

1.4 Organization

Section 2 fixes the notation. Section 3 collects the results from the literature, with the precise form in which they are used. Section 4 proves the explicit lemmas on ϖ, χ and ζ.

Section 5 proves Theorems 1.1 and 1.3 and Corollary 1.2; Section 6 proves Corollary 1.4; Section 7 proves Theorem 1.5 and Corollary 1.6. Section 8 proves the exact defect identity and Theorem 1.7.

Section 9 proves consequences for the prime correction, the ordinal defect and a deterministic certificate for simple critical-line zeros. Section 10 records the scope and limitations of these results.

Appendix A contains the program that computes the constants, with its output; Appendix B lists, for each result taken from the literature, where it is stated and how it is used.

Dependence of the results. Theorem 1.1 follows from Proposition 5.1 (Backlund's method, Theorem 3.5) and Lemma 4.3; Corollary 1.2 from Theorem 1.1 and Theorems 3.4, 3.7; Theorem 1.3 from Proposition 5.1, Lemma 4.4 (which uses Theorems 3.3, 3.4 and Lemma 4.3), Theorem 3.5(c), Theorem 3.6 and Lemma 4.1; Corollary 1.4 and Proposition 6.2 from Theorem 1.3, Corollary 1.2, Lemma 6.1, Theorem 3.1 and Lemma 4.1.

Theorem 1.5 follows from Theorem 3.1, Lemma 4.1 (in the weak form of Remark 4.2), Proposition 7.1, Theorem 7.3 (a deterministic estimate based on Lemma 7.2), Lemma 7.5 (Theorems 3.1, 3.8(b)) and Lemma 7.6 (Theorems 3.8(a), 3.9, 3.10); Corollary 1.6 from Theorem 1.5.

Theorem 1.7 follows from Theorem 8.1 (a deterministic identity), Theorem 3.11, Theorem 3.1, Lemma 4.1, Lemma 6.1 and the proof of Corollary 1.4(a).

2 Notation

Every symbol listed below has one meaning throughout, except the symbols explicitly declared local where they are introduced (the paragraph on Gonek in Section 1, Section 3, two paragraphs of Section 1.3, Remark 4.2, Lemma 7.2 and its proof, Theorem 7.3 and Remark 7.4, Section 8, Section 9, and Theorem 9.5).

Dummy variables of integration and summation are local. The letter π denotes only $3.14159\ldots$. Log and ph are the principal logarithm and argument ($-\pi < \mathrm{ph} \le \pi$). The implied constants in $\ll$ and O are absolute unless a subscript indicates a dependence. In the definition of an admissible exponent profile, the constant may also depend on the chosen profile.

Conventions at the imaginary parts. The zeros are listed with multiplicity, so N is right continuous, $N(\gamma^+) = N(\gamma)$, and $N(\gamma^+) - N(\gamma^-)$ is the multiplicity of the imaginary part γ (the number of zeros with that imaginary part, counted with multiplicity).

For $t > 0$, $t \notin \{\gamma_k\}$, $S(t) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + it)$ with the argument obtained by continuous variation along the segments from 2 to $2 + it$ and from $2 + it$ to $\frac{1}{2} + it$, so S is continuous on each open interval between consecutive distinct imaginary parts.

At an imaginary part γ the one-sided limits $S(\gamma^\pm)$ exist, $S(\gamma^+) - S(\gamma^-)$ is the multiplicity of γ (because N_0 and ϖ are continuous in (1.1)), and this article adopts the right-continuous convention $S(\gamma) = S(\gamma^+)$. This convention is chosen to agree with $N(\gamma) = \#\{k : \gamma_k \le \gamma\}$; alternative conventions, including midpoint values, occur in the literature (see [39, §9.3]).

Other conventions at the imaginary parts (such as the mean of the two limits) change S on a null set and do not affect the integrals of Theorems 3.8 and 3.9.

Lemma 2.1. *Let B be a continuous function on an open interval $J \subset (0, \infty)$. If $|S(t)| \le B(t)$ for every $t \in J \setminus \{\gamma_k\}$, then $|S(\gamma^-)| \le B(\gamma)$ and $|S(\gamma^+)| \le B(\gamma)$ for every $\gamma \in J \cap \{\gamma_k\}$.*

Proof. The set $\{\gamma_k\}$ has no accumulation point in $(0, \infty)$, and the one-sided limits $S(\gamma^\pm)$ exist. Take $t_n \to \gamma$ from the left (from the right) with $t_n \in J \setminus \{\gamma_k\}$. Then $|S(t_n)| \le B(t_n)$, and $n \to \infty$ gives the claim, since B is continuous at γ. □

All bounds for $|S(T)|$ in this article are proved first for $T \notin \{\gamma_k\}$ and have right sides continuous in T, so Lemma 2.1 gives the same bounds for $S(\gamma^-)$ and $S(\gamma^+)$. Hence, by (1.1), they apply to the corresponding one-sided values $G(\gamma^\pm) = S(\gamma^\pm) + \varpi(\gamma)/\pi$, where $G = N - N_0$.

Multiplicities enter Lemma 6.1 and Proposition 6.2 only through the inequalities $N(\gamma_k^-) + 1 \le k \le N(\gamma_k^+)$.

symbol	meaning (where defined)
$s = \sigma + it$	complex variable, $\sigma, t \in \mathbb{R}$
$\rho_k = \beta_k + i\gamma_k$	zeros with $\gamma_k > 0$, nondecreasing, with multiplicity
$N(t)$, $N(t^{\pm})$	$\#\{k : \gamma_k \le t\}$ (right continuous); one-sided limits
ϑ, ϖ	Riemann–Siegel theta function; $\varpi(t) = \vartheta(t) - \frac{t}{2}\log\frac{t}{2\pi e} + \frac{\pi}{8}$ (Lemma 4.1)
N_0, $\mathcal{L}$	$N_0(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$
$S(t)$, G, $G_k^{\pm}$	$\frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$, by continuous variation from 2 to $2 + it$ to $\frac{1}{2} + it$; $G = N - N_0 = S + \varpi/\pi$; $G(\gamma_k^{\pm})$
τ_k, s_k, ξ_k, $\mathfrak{m}_k$	point defined by $N_0(\tau_k) = k - \frac{1}{2}$ (1.2); $N_0(\tau_k) - N_0(\gamma_k)$; mean-value point; $\min(\tau_k, \gamma_k)$ (Section 6)
g_m, Δ_k^{G}	Gram points; displacement of γ_k with respect to them (Remark 7.7)
$\chi(s)$, $\mathcal{R}(z)$	factor of the functional equation, $\zeta(s) = \chi(s)\zeta(1-s)$; Stirling remainder (Theorem 3.2)
z, z_i, y, ω_i, ϕ	local variables of Lemmas 4.1–4.3
$\mathfrak{a}$	$\limsup_{T\to\infty} \lvert S(T)\rvert/\log T$
$\mathfrak{e}$, $\mathfrak{e}^*$, $\Lambda(\mathfrak{e})$, 0^*	admissible profile (1.3); reflected extension (1.4); bounding functional (1.5); $\max(\frac{1}{2} - \sigma, 0)$
$\mathfrak{e}_{\mathrm{B}}$, $\mathfrak{e}_{\mathrm{E}}$, $\mathfrak{e}_{\varepsilon'}$	Bourgain profile (1.6); explicit profile (4.5); approximating profile (Section 5.2)
(κ, λ), (κ_j, λ_j), $A(\cdot)$	exponent pairs; Bourgain's pair and its images; van der Corput's A-process
σ_m, ς_i, e_i, ς_j'	$1 - m/(2^m - 2)$, $m \ge 4$ (Yang's lines); nodes and exponents in Lemma 4.4; nodes in Section 5.2
Q, Ξ	shift in Theorem 3.4 ($Q = 10$ in all uses); $\Xi(s) = (s-1)\zeta(s)$ in Lemma 4.4
c, R, $\mathbb{D}(c, R)$, η	center, radius and disc of Backlund's method; $c = 1 + \eta$, $R = 1 + 2\eta$ in Section 5.1
$\sigma^\sharp$, $\delta^\sharp$, $\mathfrak{d}$	$c + (c - \frac{1}{2})^2/R$; $2c - \sigma^\sharp - \frac{1}{2}$; $\log(R/(c - \frac{1}{2}))$
$\mathfrak{f}_n$, w, $\mathcal{Z}_n(c, R)$	Backlund function (3.3); its zeros; weighted zero count (3.4)
$\mathcal{E}(T, d)$	error term of Bennett et al. in Backlund's trick (Theorem 3.5)
Ψ, θ	even majorant in Theorem 3.5(a); angle, $s = c + Re^{i\theta}$
$\mathfrak{h}$, K, Ω, $\mathcal{I}_{\mathfrak{h}}$	exponent function, constant and factor in (5.1); $\int_0^\pi \mathfrak{h}(c + R\cos\theta)\, d\theta$
C_1, C_2, $C_3(T)$	$\mathcal{I}_{\mathfrak{e}_{\mathrm{E}}^*}/(2\pi\mathfrak{d})$, $1/(2\mathfrak{d})$, constant term (Section 5.3)
A_1, A_2, A_3	generic coefficients of explicit bounds for S and for N (Section 1.3)
p, p_ν, π	primes; the ν-th prime; 3.14159…
Φ_m, m_j, Φ, $\Phi^{(3)}$	Euler phase (1.10), prime budget $\lfloor 2^{j/2} \rfloor$, interpolated phase (1.11), prime powers $r \ge 3$ in (7.11)
$\mathcal{F}$, τ_k^{C}, $\mathcal{T}$	sublevel function $N_0 + \Phi$; sublevel construction (1.12); $\tau_k^{\mathrm{C}} = \mathcal{T}(\tau_k)$
$\mathcal{K}_T$, $\langle\cdot\rangle_{\mathcal{K}_T}$	$\{k : T < \gamma_k \le 2T\}$; mean over $\mathcal{K}_T$
a, b, ℓ_0, ℓ_1	window endpoints; $N(a)$, $N(b)$ (Theorem 7.3)
D_ℓ, $\mathcal{M}D$, Θ	gaps $\gamma_{\ell+1} - \gamma_\ell$; discrete maximal function over intervals inside a window $[A, B]$, level (Lemma 7.2)
f, x_k^f, q_f, E_k, u_k, v_k	measurable model of N, its sublevel points; $\lvert\lfloor f + \frac{1}{2}\rfloor - N\rvert$, disagreement set; its measures to the right and left of γ_k (Theorem 7.3)
F, u_k, Ψ_j	absolutely continuous model of N, its defects $k - \frac{1}{2} - F(\gamma_k)$, and $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j}dx$ (Theorem 8.1)
L_T, $\mathcal{A}_h$, v	$\log(T/2\pi)$, $\{t \in [T, 4T] : N(t+h) = N(t)\}$, hL_T (Lemma 7.5)
U, Y, X, V	dyadic height 2^j; $U^{1/200}$, p_m; $\sum_{Y\le p\le X} 1/p$ (Lemma 7.6)
$\mathscr{D}_1$, $\mathscr{D}_2$, d_1, d_2, $\mathfrak{c}_n$, $\mathfrak{c}_v'$	prime polynomials (7.11); numbers of ordered prime factorizations, coefficients (Lemma 7.6)
k, α_{F}, H, h	order, exponent, interval length and shift in Fujii's theorem (Theorem 3.8)
C, C', C'', C_0, T_0, A	absolute constants and thresholds, not the same at each occurrence; A is the multiplier of the mean spacing in Corollary 1.6

3 Results from the literature

The following statements are used as they stand; each was checked in the source indicated. Within this section the letters $\sigma_\pm$, A, B, C, C', α_i, β_i, Ξ, $\varkappa$ and b_n are local. The letters k, α_{F}, H and h have the meaning of Section 2.

Theorem 3.1 ([39], §§9.3, 9.4, 9.12)**.** *For $t > 0$, $t \notin \{\gamma_k\}$, $N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t)$, where $\vartheta(t) = \operatorname{Im}\operatorname{Log}\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ (the branch of $\operatorname{Log}\Gamma$ that is real on $(0,\infty)$). Moreover $S(t) = O(\log t)$ and $\gamma_{k+1} - \gamma_k \to 0$ as $k \to \infty$.*

Theorem 3.2 (Stirling's formula with remainder, [6, §5.11(ii)])**.** *For $|\operatorname{ph} z| < \pi$, $\operatorname{Log}\Gamma(z) = (z - \frac{1}{2})\operatorname{Log} z - z + \frac{1}{2}\log(2\pi) + \mathcal{R}(z)$ with $|\mathcal{R}(z)| \le \sec^2(\frac{1}{2}\operatorname{ph} z)/(12|z|)$.*

Theorem 3.3 (Explicit bounds for ζ)**.** *For real t:*

(a) $|\zeta(\frac{1}{2} + it)| \le 0.618|t|^{1/6}\log|t|$ for $3 < |t| \le e^{105}$ [19], and $|\zeta(\frac{1}{2} + it)| \le 66.7|t|^{27/164}$ for $|t| > e^{105}$ [30];

(b) for every integer m with $4 \le m \le 9$, with $\sigma_m = 1 - m/(2^m - 2)$ and T_m the threshold of [44, (3.4)]: $|\zeta(\sigma_m + it)| \le 1.546|t|^{1/(2^m-2)}\log|t|$ for $|t| \ge T_m$ [44, §3.2], and $|\zeta(\sigma_m + it)| \le 1.878|t|^{1/(2^m-2)}\log|t|$ for $3 \le |t| \le T_m$ ([44, §3.1] with the correction of Remark 4.5(ii)); in particular $|\zeta(\sigma_m + it)| \le 2|t|^{1/(2^m-2)}\log|t|$ for $|t| \ge 3$. [44, Thm. 1.1] states the first bound for $|t| \ge 3$ and every $m \ge 4$; only the form given here is used;

(c) $|\zeta(1 + it)| \le \log|t|$ for $|t| \ge 3$ [29, Thm. 1.1].

Inputs (a) and (b) are the display preceding (2.3) and the bound (2.4) in [3, §2]. For the 1-line, [3, (2.1)] combines [29] with [18, Thm. 1] for $\exp(3070) < t \le \exp(3.69 \cdot 10^8)$ and with $|\zeta(1 + it)| \le 58.096\log^{2/3} t$ for $t > \exp(3.69 \cdot 10^8)$, which they call a direct consequence of [2, Theorem 1]. That theorem, [2, Thm. 1.1], states $|\zeta(\sigma + it)| \le 70.6995\,|t|^{4.43795(1-\sigma)^{3/2}}\log^{2/3}|t|$ for $\frac{1}{2} \le \sigma \le 1$, $|t| \ge 3$, which at $\sigma = 1$ gives the constant 70.6995; the constant 58.096 is [25, Lemma 5] ($|t| \ge 3$), obtained there by running the argument of [2] with other parameters. On the 1-line only (c) is needed, which holds for every $|t| \ge 3$, so none of these bounds enters Theorem 1.3.

Theorem 3.4 (Phragmén–Lindelöf principle, [9, Cor. 3])**.** *Let $\sigma_- < \sigma_+$ and $Q > 1 - \sigma_-$ be real. Let Ξ be holomorphic on $\sigma_- \le \operatorname{Re} s \le \sigma_+$ with $|\Xi(s)| < C\exp(C'e^{\varkappa|t|})$ there, for some $C, C' > 0$ and $0 < \varkappa < \pi/(\sigma_+ - \sigma_-)$. Suppose that, with $A, B, \alpha_1, \alpha_2, \beta_1 \ge 0$ and $\alpha_1 \ge \beta_1$,*

$$|\Xi(s)| \le \begin{cases} A|Q+s|^{\alpha_1}|\operatorname{Log}(Q+s)|^{\alpha_2} & (\operatorname{Re} s = \sigma_-), \\ B|Q+s|^{\beta_1}|\operatorname{Log}(Q+s)|^{\alpha_2} & (\operatorname{Re} s = \sigma_+). \end{cases} \tag{3.1}$$

Then for $\sigma_- \le \operatorname{Re} s \le \sigma_+$,

$$|\Xi(s)| \le \bigl(A|Q+s|^{\alpha_1}\bigr)^{\frac{\sigma_+ - \operatorname{Re} s}{\sigma_+ - \sigma_-}}\bigl(B|Q+s|^{\beta_1}\bigr)^{\frac{\operatorname{Re} s - \sigma_-}{\sigma_+ - \sigma_-}}|\operatorname{Log}(Q+s)|^{\alpha_2}. \tag{3.2}$$

The same power α_2 of the logarithm on both lines is essential: the version with different powers [41, Lemma 3] has a gap in its proof, corrected in [9]. With equal powers, Theorem 3.4 also follows from its case $\alpha_2 = 0$, which is Rademacher's theorem [33], applied to $\Xi(s)\,(\operatorname{Log}(Q+s))^{-\alpha_2}$: since $\operatorname{Re}(Q+s) > 1$ on the strip, $\operatorname{Log}(Q+s)$ has positive real part, so its principal power is holomorphic there, has modulus $|\operatorname{Log}(Q+s)|^{-\alpha_2}$, and is bounded by $(\log(Q+\sigma_-))^{-\alpha_2}$. Since $\log|Q+s| \le |\operatorname{Log}(Q+s)|$, a boundary bound with $\log|Q+s|$ implies the hypothesis.

For $T>0$, integers $n\ge 1$ and the disc $\mathbb{D}(c,R)=\{s:|s-c|<R\}$ put

$$\mathfrak{f}_n(s)=\tfrac12\Big[\big((s+iT-1)\zeta(s+iT)\big)^n+\big((s-iT-1)\zeta(s-iT)\big)^n\Big], \tag{3.3}$$

$$\mathcal{Z}_n(c,R)=\frac1n\sum_{w}\log\frac{R}{|w-c|}, \tag{3.4}$$

the sum over the zeros w of $\mathfrak{f}_n$ in $\mathbb{D}(c,R)$, with multiplicity.

Theorem 3.5 (Backlund's method, [3, Lemmas 2.1, 2.3, 2.4]). *Let c,R be real, $\sigma^\sharp=c+(c-\frac12)^2/R$, $\delta^\sharp=2c-\sigma^\sharp-\frac12$, and suppose*

$$c-R<\tfrac12<1<c<\sigma^\sharp<c+R,\qquad c<R,\qquad 0<\delta^\sharp<\tfrac12. \tag{3.5}$$

(a) If $\Psi:[-\pi,\pi]\to\mathbb{R}$ is even and $\Psi(\theta)\ge\frac1n\log|\mathfrak{f}_n(c+Re^{i\theta})|$ for all $n\ge1$ and θ, there are $n_1<n_2<\cdots$ with

$$\limsup_{m\to\infty}\mathcal{Z}_{n_m}(c,R)\le\log\frac{\zeta(c)}{\sqrt{(c-1)^2+T^2}\,\zeta(2c)}+\frac1\pi\int_0^\pi\Psi(\theta)\,d\theta. \tag{3.6}$$

(b) For every $n\ge1$, if $T\notin\{\gamma_k\}$,

$$\Big|\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg\big((\sigma+iT-1)\zeta(\sigma+iT)\big)\Big|\le\frac{\pi\mathcal{Z}_n(c,R)}{2\mathfrak{d}}+\frac{\mathcal{E}(T,\delta^\sharp)}{2}+\frac{3\pi}{2n}+\frac\pi4,\qquad \mathfrak{d}=\log\frac{R}{c-\frac12}, \tag{3.7}$$

where $\mathcal{E}(T,d)$ ($0\le d<\frac92$, $T\ge\frac57$) is the explicit function $E(0,d,T)$ of [4, Fig. 2].

(c) For $d\in[\frac14,\frac58]$ and $T\ge\frac57$, $\mathcal{E}(T,d)/\pi\le(640d-112)/(1536(3T-1))+2^{-10}$.

Bellotti and Wong state (a) and (b) as [3, Lemmas 2.1, 2.3] and take them from [16, Props. 3.5, 3.7], which are stated for Dedekind zeta functions ζ_K (here $K=\mathbb{Q}$). Their function $\mathcal{E}(T,d)$ is the function $E(a,d,T)$ of [4, Def. 3.3 and Fig. 2] at $a=0$ (the two expressions agree term by term), and (c) is [4, Lemma 3.4] at $a=0$.

Theorem 3.6 ([3, (1.2)], from Platt's computation [31]). *$|S(T)|\le 2.5167$ for $0\le T\le 30\,610\,046\,000$.*

Theorem 3.7 (Exponent pairs). *(a) If (κ,λ) is an exponent pair, so is $A(\kappa,\lambda)=\big(\frac{\kappa}{2\kappa+2},\frac12+\frac{\lambda}{2\kappa+2}\big)$ [20, Lemma 2.8].*

(b) If (κ,λ) is an exponent pair and $\frac12\le\sigma\le\lambda-\kappa$, then $\zeta(\sigma+it)\ll|t|^{(\kappa+\lambda-\sigma)/2}\log|t|$ for $|t|\ge2$ [20, (7.57), p. 190]; in particular $\zeta(\lambda-\kappa+it)\ll|t|^\kappa\log|t|$.

(c) For every $\varepsilon>0$, $(\frac{13}{84}+\varepsilon,\frac{55}{84}+\varepsilon)$ is an exponent pair [5, §5, Thm. 6].

Statement (c) is the form in which the result is stated in [5]. The bound $\zeta(\frac12+it)\ll|t|^{13/84+\varepsilon}$ is its consequence by (b). Part (c) is used only through (a) and (b), with ε kept in both coordinates (Section 5.2).

Theorem 3.8 (Fujii [11], for the modulus 1, with the corrections of [12, §5]). *Let $\mathsf{k}\ge1$ be an integer and $0<\alpha_{\mathrm F}\le\frac12$ be fixed, and let $T^{1/2+\alpha_{\mathrm F}}\le H\le T$. (The Main Theorem of [11, p. 225] is stated for a primitive character χ modulo q and fixed a_1,a_2 with $0<a_i\le\frac12$, under the hypotheses $q\le T^{1/4-a_1}$, $T^{1/2+a_2}\le H\le T$ and $0<h\le H-(H/\sqrt{T})^{1/8}$; both endpoints $a_2=\frac12$ and $H=T$ are admitted. Here $q=1$, χ is the principal character, $S(t,\chi)=S(t)$, and $\alpha_{\mathrm F}=a_2$; the condition on h holds for $0<h\le1$ as soon as $T\ge9$, since then $H\ge\sqrt{T}\ge3$ and $H-(H/\sqrt{T})^{1/8}\ge H-H^{1/8}\ge1$.)*

(a) (Selberg's approximation, [11, §3, pp. 232–233]: Selberg's expression with $x = T^{\alpha_F/(60k)}$, the bounds $\|B_i\| \ll k^2 H^{1/(2k)}$ for its five error terms (with $k^{1/2}$ for $i = 1, 4$ and $k^{3/2}$ for $i = 5$), and [11, Lemma 5] for the primes in $[x^3, Y]$.) For $T^{\alpha_F/(20k)} \le Y \le H^{1/k}$,

$$\int_T^{T+H} \Big| S(t) - \frac{1}{\pi} \operatorname{Im} \sum_{p<Y} p^{-1/2-it} \Big|^{2k} dt \ll_{k,\alpha_F} H. \tag{3.8}$$

(b) (Main Theorem, [11, p. 225].) For $0 < h \le 1$ and $T \ge T_0(k, \alpha_F)$, $\int_T^{T+H} (S(t+h) - S(t))^{2k} dt \ll_{k,\alpha_F} H\,(\log(2 + h\log T))^k$.

The corrections in [12, §5] require h to be bounded in the Main Theorem and change the dependence on k of some remainder terms. Neither affects (a) or (b) as stated, since $0 < h \le 1$ and $k \le 2$ is fixed in all applications below. In [12, §1, (α)] the Main Theorem is restated for $q = 1$ and $H = T$, exactly the case of (b), with $\log(3 + h\log T)$ in place of $\log(2 + h\log T)$ and with the remainder $O((Ak)^{2k} T(\log(3 + h\log T))^{k-1/2})$. Since $\log(3+x)/\log(2+x)$ decreases for $x \ge 0$, $\log(3+x) \le \frac{\log 3}{\log 2}\log(2+x)$, and that form also gives (b).

In (a), [11, Lemma 5] is stated for $\int_0^H$ and for coefficients with $|a(p)| \le A\log p/\log y$. The shift $t \mapsto t+T$ replaces $a(p)$ by $a(p)p^{-iT}$, and for $a(p) = 1$ on $x^3 \le p < Y$ one has $\log p/\log Y \ge \frac{\alpha_F/(20k)}{1/k} = \frac{\alpha_F}{20}$, so $A = 20/\alpha_F$.

Theorem 3.9 (Montgomery–Vaughan [28, Cor. 3])**.** *For complex $b_1, \dots, b_M$,*

$$\int_U^{2U} \Big| \sum_{n \le M} b_n n^{-it} \Big|^2 dt = \sum_{n \le M} |b_n|^2 \big(U + O(n)\big), \tag{3.9}$$

with an absolute implied constant. (Stated in [28, Cor. 3] for $\int_0^U$ and for every sequence with $\sum_n n|b_n|^2 < \infty$; we use only finite sums, and the integral over $[U, 2U]$ is the case $b_n \mapsto b_n n^{-iU}$.)

Theorem 3.10 (Mertens and the prime number theorem; see [27] and [38, Ch. I.1])**.** *Mertens' theorem gives $\sum_{p \le y} 1/p = \log\log y + O(1)$ for $y \ge 2$. The prime number theorem gives $p_m \asymp m\log m$ for $m \ge 2$.*

Theorem 3.11 (Selberg [37]; see [39, Notes for Chapter 9, (9.25.1), p. 245])**.** *For every integer $k \ge 1$, as $T \to \infty$,*

$$\int_0^T S(t)^{2k}\, dt \sim \frac{(2k)!}{k!\,(2\pi)^{2k}}\, T(\log\log T)^k. \tag{3.10}$$

Fujii [11, p. 225] quotes Selberg's theorem for $\int_T^{T+H}$, $T^{1/2+a} \le H \le T$, with the remainder $O_k(H(\log\log T)^{k-1/2})$. Only the asymptotic form is used: with $k = 1$, in the dyadic form (8.9) (Proposition 8.4), with $k = 2$ in Remark 7.7, and with every k in Proposition 8.6.

4 Explicit lemmas

Lemma 4.1. *For $t > 0$, $\vartheta(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8} + \varpi(t)$ with $|\varpi(t)| \le 0.53/t$. Consequently (1.1) holds for $t > 0$, $t \notin \{\gamma_k\}$.*

Proof. Put $z = \frac14 + iy$, $y = t/2 > 0$, so $\operatorname{ph} z = \frac{\pi}{2} - \arctan\frac{1}{4y} \in (0, \frac{\pi}{2})$ and $\sec^2(\frac12 \operatorname{ph} z) < 2$. By Theorem 3.2,

$$\operatorname{Im}\operatorname{Log}\Gamma(z) = y\log|z| - \tfrac14 \operatorname{ph} z - y + \operatorname{Im}\mathcal{R}(z), \qquad \log|z| = \log y + \tfrac12 \log\Big(1 + \frac{1}{16y^2}\Big). \tag{4.1}$$

Since $y\log y - y - \frac{t}{2}\log\pi = \frac{t}{2}\log\frac{t}{2\pi e}$, this gives $\varpi(t) = \frac{y}{2}\log(1 + \frac{1}{16y^2}) + \frac14 \arctan\frac{1}{4y} + \operatorname{Im}\mathcal{R}(z)$. The first term lies in $[0, \frac{1}{32y}] = [0, \frac{1}{16t}]$, the second in $[0, \frac{1}{16y}] = [0, \frac{1}{8t}]$, and $|\mathcal{R}(z)| \le \frac{2}{12|z|} \le \frac{1}{6y} = \frac{1}{3t}$. Hence $|\varpi(t)| \le (\frac{1}{16} + \frac18 + \frac13)\frac1t = \frac{25}{48t} < \frac{0.53}{t}$. The last assertion is Theorem 3.1, since $\frac{1}{\pi}\vartheta + 1 = N_0 + \frac{\varpi}{\pi}$. □

Remark 4.2 (Weak form of Lemma 4.1). The explicit constant 0.53 is needed only for (1.8) and for the inequalities $|\varpi|/\pi < \frac12$ of Section 1.1 and Remark 7.7. Theorem 1.5 and Corollary 1.6 use Lemma 4.1 only through the weak form

$$\varpi(t) = O(1/t) \qquad (t \to \infty), \tag{4.2}$$

and in fact only through the boundedness of ϖ on $[1,\infty)$ and the bound $\varpi(t+h)-\varpi(t) = O(1/t)$ for $0 < h \le 1$ in the proof of Lemma 7.5. The weak form does not need the remainder bound of Theorem 3.2. In this remark E and z are local. Put $\mathsf{E}(z) = \operatorname{Log}\Gamma(z) - (z-\frac12)\operatorname{Log} z + z$. The partial-fraction series of Γ'/Γ gives $\mathsf{E}'(z) = \frac{\Gamma'}{\Gamma}(z) - \operatorname{Log} z + \frac{1}{2z} = O(|z|^{-2})$ uniformly for $\operatorname{Re} z \ge \frac14$. Hence $\mathsf{E}(n+iy)-\mathsf{E}(n) = O(y/n^2)$, and Stirling's formula for $n!$ gives $\mathsf{E}(n) \to \log\sqrt{2\pi}$, so $\lim_{n\to\infty}\mathsf{E}(n+iy)$ is real. Integrating E' along the half-line from $\frac14+iy$ to $+\infty+iy$ then gives $\operatorname{Im}\mathsf{E}(\frac14+iy) = O(1/y)$. With $y = t/2$, this is (4.2), by the computation in the proof of Lemma 4.1 with $\mathcal{R} = \mathsf{E} - \log\sqrt{2\pi}$.

Lemma 4.3. *For $-0.01 \le \sigma \le \frac12$ and $t \ge 98$, $|\chi(\sigma+it)| \le 1.0070\,(t/2\pi)^{1/2-\sigma}$.*

Proof. Recall that $\chi(s) = \pi^{s-1/2}\Gamma(\frac{1-s}{2})/\Gamma(\frac{s}{2})$. Put $y = t/2 \ge 49$, $z_1 = \frac{1-\sigma}{2} + iy$, $z_2 = \frac{\sigma}{2} + iy$, with real parts $\omega_1 \in [\frac14, 0.505]$ and $\omega_2 \in [-0.005, \frac14]$. Since $|\Gamma(\bar z)| = |\Gamma(z)|$, $|\chi(s)| = \pi^{\sigma-1/2}|\Gamma(z_1)|/|\Gamma(z_2)|$. For $z = \omega + iy$ with $y > 0$ one has $\operatorname{ph} z = \frac\pi2 - \arctan(\omega/y)$, and Theorem 3.2 gives

$$\begin{aligned} \log|\Gamma(z)| &= (\omega-\tfrac12)\log y - \tfrac\pi2 y + \tfrac12\log 2\pi + \phi(\omega,y) + \operatorname{Re}\mathcal{R}(z), \\ \phi(\omega,y) &= \frac{\omega-\frac12}{2}\log\Big(1+\frac{\omega^2}{y^2}\Big) + y\arctan\frac{\omega}{y} - \omega. \end{aligned} \tag{4.3}$$

Subtracting, and using $\omega_1 - \omega_2 = \frac12 - \sigma$,

$$\log|\chi(s)| = (\tfrac12-\sigma)\log\frac{t}{2\pi} + \phi(\omega_1,y) - \phi(\omega_2,y) + \operatorname{Re}\big(\mathcal{R}(z_1) - \mathcal{R}(z_2)\big). \tag{4.4}$$

For $|\omega| \le 0.505$ and $|\omega - \frac12| \le 0.505$, $|\frac{\omega-1/2}{2}\log(1+\frac{\omega^2}{y^2})| \le \frac{0.505\cdot 0.256}{2y^2}$ and $|y\arctan\frac{\omega}{y} - \omega| \le \frac{|\omega|^3}{3y^2} \le \frac{0.043}{y^2}$, so $|\phi(\omega_i,y)| \le \frac{0.108}{y^2} \le \frac{0.435}{t^2}$. Next $\operatorname{ph} z_1 \in (0,\frac\pi2)$ and $\operatorname{ph} z_2 \le \frac\pi2 + \arctan\frac{0.005}{49}$, so $\sec^2(\frac12\operatorname{ph} z_i) \le 2.0005$ and $|\mathcal{R}(z_i)| \le \frac{2.0005}{12y} = \frac{2.0005}{6t}$. Hence $\log|\chi(s)| - (\frac12-\sigma)\log\frac{t}{2\pi} \le \frac{0.6669}{t} + \frac{0.87}{t^2} \le 0.00690$ for $t \ge 98$, and $e^{0.0069} < 1.0070$. □

4.1 The explicit profile

Let $\mathfrak{e}_{\mathrm{E}}$ be the piecewise linear function on $[\frac12, 1]$ through

$$\big(\tfrac12, \tfrac{27}{164}\big), \quad \big(\sigma_m, \tfrac{1}{2^m-2}\big)\ (4 \le m \le 9), \quad (1,0), \tag{4.5}$$

that is, through $(\frac57, \frac{1}{14})$, $(\frac56, \frac{1}{30})$, $(\frac{28}{31}, \frac{1}{62})$, $(\frac{17}{18}, \frac{1}{126})$, $(\frac{123}{127}, \frac{1}{254})$, $(\frac{167}{170}, \frac{1}{510})$, and let $\mathfrak{e}^*_{\mathrm{E}}$ be its reflected extension (1.4).

Lemma 4.4. *For $-0.01 \le \sigma \le 3$ and $|t| \ge 98$, $|\zeta(\sigma+it)| \le 2.74\,|t|^{\mathfrak{e}^*_{\mathrm{E}}(\sigma)}\log|t|$.*

Proof. By $\zeta(\bar s) = \overline{\zeta(s)}$ it suffices to take $t > 0$. Let $\varsigma_0 < \varsigma_1 < \cdots < \varsigma_8$ be $\frac12, \sigma_4, \dots, \sigma_9, 1, 3$, with exponents $e_0 = \frac{27}{164}$, $e_i = \frac{1}{2^{i+3}-2}$ $(1 \le i \le 6)$, $e_7 = e_8 = 0$. The e_i are nonincreasing, and $\mathfrak{e}^*_{\mathrm{E}}$ is the linear interpolation of (ς_i, e_i) on $[\frac12, 3]$. Put $Q = 10$ and $\Xi(s) = (s-1)\zeta(s)$, an entire function with $|\Xi(\sigma+it)| \ll |t|^2$ for $\frac12 \le \sigma \le 3$, $|t| \ge 1$.

Boundary values. It is enough to show that for every i and every real t,

$$|\Xi(\varsigma_i + it)| \le 2|Q+s|^{1+e_i}\log|Q+s| \qquad (s = \varsigma_i + it). \tag{4.6}$$

Suppose first that $|t|>3$. On $\sigma=\frac12$, Theorem 3.3(a) gives $|\zeta|\le 2|t|^{27/164}\log|t|$: for $|t|\le e^{105}$ because $|t|^{1/6-27/164}=|t|^{1/492}\le e^{105/492}<2/0.618$, and for $|t|>e^{105}$ because $\log|t|>105>66.7/2$. On $\sigma=\varsigma_i=\sigma_{i+3}$ $(1\le i\le 6)$, (b) gives $|\zeta|\le 2|t|^{e_i}\log|t|$. On $\sigma=1$, (c) gives $|\zeta(1+it)|\le\log|t|$. On $\sigma=3$, $|\zeta|\le\zeta(3)<2\log 3$. Since $|s-1|\le|Q+s|$ and $|t|\le|Q+s|$, (4.6) follows for $|t|>3$. Now let $|t|\le 3$. For $\sigma>0$, $(s-1)\zeta(s)=s-s(s-1)\int_1^\infty\{u\}u^{-s-1}du$, by [39, (2.1.4)] and $\int_1^\infty u^{-s-1}du=\frac1s$, so for $\frac12\le\sigma\le 3$, $|\Xi(s)|\le|s|+2|s||s-1|\le 4.243+2\cdot 4.243\cdot 3.606<34.9$, whereas the right side of (4.6) is at least $2\cdot 10.5\log 10.5>49$.

Interpolation. Theorem 3.4 on each strip $\varsigma_i\le\sigma\le\varsigma_{i+1}$ $(Q>1-\varsigma_i$, $A=B=2$, $\alpha_1=1+e_i\ge\beta_1=1+e_{i+1}$, $\alpha_2=1$; the bounds (4.6) imply its hypotheses because $\log|Q+s|\le|\operatorname{Log}(Q+s)|$ gives, for $\frac12\le\sigma\le 3$ and all t,

$$|\Xi(s)|\le 2|Q+s|^{1+\mathfrak{e}^*_{\mathrm{E}}(\sigma)}|\operatorname{Log}(Q+s)|. \tag{4.7}$$

For $t\ge 98$, $|Q+s|\le(169+t^2)^{1/2}\le t(1+84.5/t^2)$, $|\operatorname{Log}(Q+s)|\le\log|Q+s|+\frac{\pi}{2}\le\log t+84.5/t^2+\frac{\pi}{2}$ and $|s-1|\ge t$. Since $\mathfrak{e}^*_{\mathrm{E}}\le\frac{27}{164}$ on $[\frac12,3]$,

$$\begin{aligned}|\zeta(s)|&\le 2\Big(1+\frac{84.5}{98^2}\Big)^{1+\frac{27}{164}}\Big(1+\frac{84.5/98^2+\pi/2}{\log 98}\Big)t^{\mathfrak{e}^*_{\mathrm{E}}(\sigma)}\log t\\&\le 2.7167\,t^{\mathfrak{e}^*_{\mathrm{E}}(\sigma)}\log t\qquad(\tfrac12\le\sigma\le 3).\end{aligned} \tag{4.8}$$

Reflection. For $-0.01\le\sigma<\frac12$, $\zeta(s)=\chi(s)\zeta(1-s)$ and $|\zeta(1-s)|=|\zeta(1-\sigma+it)|$ with $1-\sigma\in(\frac12,1.01]$. By the previous step and Lemma 4.3, $|\zeta(s)|\le 1.0070\cdot 2.7167\,t^{\frac12-\sigma+\mathfrak{e}^*_{\mathrm{E}}(1-\sigma)}\log t\le 2.74\,t^{\mathfrak{e}^*_{\mathrm{E}}(\sigma)}\log t$, because $\frac12-\sigma+\mathfrak{e}^*_{\mathrm{E}}(1-\sigma)=\mathfrak{e}^*_{\mathrm{E}}(\sigma)$ by (1.4) (for $\sigma<0$ both sides equal $\frac12-\sigma$). □

Remark 4.5 (Robustness of the inputs). Theorem 3.3 enters Theorem 1.3 only through (4.6), that is, through bounds with constant 2 on the lines $\sigma=\varsigma_i$, and each input has room. (i) On $\sigma=\frac12$, the bound of [19] alone gives (4.6) for $3<|t|\le e^{577}$, since $|t|^{1/492}\le e^{577/492}<2/0.618$. Above e^{577} any bound $C|t|^{27/164}$ with $C\le 1154$ suffices in place of [30]. Revers [34, Thm. 1.1] gives $|\zeta(\frac12+it)|\le 0.611|t|^{1/6}\log|t|$ for $|t|\ge 3$, with a detailed treatment of the intermediate range of [19].

(ii) On $\sigma=\sigma_m$, the bound of [44] is proved for $|t|\ge T_m$, the threshold of [44, (3.4)] (with $k=m$), by exponential sums [44, §3.2], and for $3\le|t|\le T_m$ by a Phragmén–Lindelöf interpolation of $(s-1)\zeta(s)$ between $\sigma=\frac12$ and $\sigma=1$ with $Q_0=1.31$ [44, §3.1, (3.9)–(3.13)], whose conclusion is stated there with $\log|Q_0+s|$. Theorem 3.4 gives it with $|\operatorname{Log}(Q_0+s)|$.

Redoing [44, (3.9)–(3.13)] with this factor, with $A(t_0)$ as defined there and t_0 on a fine partition of $[3,T_m]$, interval arithmetic (the interval-arithmetic program of Appendix A) gives on $3\le t\le T_m$ the constant 1.878 (precisely ≤ 1.87781, attained for $m=9$) in place of 1.546 for $4\le m\le 9$, the only values of m used in Lemma 4.4.

Hence (b) holds for these m with constant 2, which is all that (4.6) uses.

(iii) On $\sigma=1$, [18, §2] give independently $|\zeta(1+it)|\le\log|t|-0.45$ for $|t|\ge 3$.

5 A disc-based estimate for $S(T)$

Proposition 5.1. *Let c,R satisfy (3.5) with $c\le 2$ and $R\le 2$, and let $T\ge 3$, $T\notin\{\gamma_k\}$. Suppose that, for a nonnegative measurable function $\mathfrak{h}$ on $[c-R,c+R]$ with $\int_0^\pi\mathfrak{h}(c+R\cos\theta)\,d\theta<\infty$, a constant $K\ge 1$ and a function $\Omega\ge 1$ on $[T-2,T+2]$ such that $t\mapsto t^{\mathfrak{h}(\sigma)}\Omega(t)$ is nondecreasing there for every σ,*

$$|\zeta(\sigma+it)|\le K\,t^{\mathfrak{h}(\sigma)}\Omega(t)\qquad(c-R\le\sigma\le c+R,\ T-2\le t\le T+2). \tag{5.1}$$

Then

$$|S(T)| \le \frac{1}{2\mathfrak{d}}\Big[\log\frac{\zeta(c)}{\zeta(2c)} + \log\frac{T+3}{T} + \frac{\mathcal{I}_{\mathfrak{h}}}{\pi}\log(T+2) + \log\Omega(T+2) + \log K\Big] + \frac{\log\zeta(\sigma^\sharp)}{\pi} + \frac{\sigma^\sharp - \frac12}{\pi T} + \frac{\mathcal{E}(T,\delta^\sharp)}{2\pi} + \frac14, \tag{5.2}$$

where $\mathcal{I}_{\mathfrak{h}} = \int_0^\pi \mathfrak{h}(c + R\cos\theta)\,d\theta$ *and* $\mathfrak{d} = \log(R/(c-\frac12))$.

Proof. Decomposition of the argument. On the segments from 2 to $2+iT$ and from $2+iT$ to $\sigma^\sharp + iT$ one has $\operatorname{Re} s > 1$, where $\log\zeta$ is holomorphic. Hence

$$\pi S(T) = \arg\zeta(\sigma^\sharp + iT) + \Delta^{1/2}_{\sigma=\sigma^\sharp}\arg\big((\sigma+iT-1)\zeta(\sigma+iT)\big) - \Delta^{1/2}_{\sigma=\sigma^\sharp}\arg(\sigma+iT-1) \tag{5.3}$$

(this is the decomposition used in [3, §3]). Here $|\arg\zeta(\sigma^\sharp+iT)| \le |\log\zeta(\sigma^\sharp+iT)| \le \log\zeta(\sigma^\sharp)$, and the change of $\arg(\sigma-1+iT)$ from $\sigma=\sigma^\sharp$ to $\frac12$ is $\arctan\frac{\sigma^\sharp-1}{T} + \arctan\frac{1}{2T} \le \frac{\sigma^\sharp-1/2}{T}$.

The majorant Ψ. For $s = c + Re^{i\theta}$ one has $\sigma = \operatorname{Re} s \in [c-R, c+R]$ and $|\operatorname{Im} s| \le R \le 2$, so $s+iT$ has imaginary part in $[T-2, T+2]$, $s-iT$ has imaginary part in $[-T-2, -T+2]$, and $|s \pm iT - 1| \le \sqrt{9+(T+2)^2} \le T+3$. Since $|\zeta(\bar s)| = |\zeta(s)|$, (5.1) and the monotonicity assumption bound both terms in (3.3) by $\big((T+3)K(T+2)^{\mathfrak{h}(\sigma)}\Omega(T+2)\big)^n$, so $\frac1n\log|\mathfrak{f}_n(s)| \le \Psi(\theta)$ with

$$\Psi(\theta) = \log(T+3) + \mathfrak{h}(c+R\cos\theta)\log(T+2) + \log\Omega(T+2) + \log K, \tag{5.4}$$

an even function of θ. Theorem 3.5(a) gives a sequence n_m with $\limsup_m \mathcal{Z}_{n_m}(c,R) \le Z^*$, where Z^* is the bracket in (5.2) (using $\sqrt{(c-1)^2+T^2} \ge T$).

Backlund's trick. Theorem 3.5(b) at $n = n_m$, letting $m \to \infty$, gives $|\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg((\sigma+iT-1)\zeta(\sigma+iT))| \le \frac{\pi Z^*}{2\mathfrak{d}} + \frac{\mathcal{E}(T,\delta^\sharp)}{2} + \frac{\pi}{4}$. Dividing the decomposition by π gives (5.2). □

5.1 Proof of Theorem 1.1

Let $\mathfrak{e}$ be admissible, $0 < \eta < 10^{-2}$, $c = 1+\eta$, $R = 1+2\eta$. Then $c-R = -\eta$, $\sigma^\sharp = \frac54 + \frac32\eta < c+R = 2+3\eta$, $\delta^\sharp = \frac14 + \frac{\eta}{2} \in [\frac14, \frac58]$ and $R/(c-\frac12) = 2$. So (3.5) holds and $\mathfrak{d} = \log 2$.

Let $\varepsilon \in (0,1)$. It is enough to establish $|\zeta(\sigma+it)| \le K_\varepsilon t^{\mathfrak{e}^*(\sigma)+\varepsilon}$ for $-0.01 \le \sigma \le 3$ and $t \ge 98$, with K_ε depending on ε only (not on η). For $\frac12 \le \sigma \le 1$ this is (1.3). For $1 \le \sigma \le 3$, $\zeta(\sigma+it) \ll \log t \ll_\varepsilon t^\varepsilon$ [39, Thm. 3.5]. For $-0.01 \le \sigma < \frac12$ it follows from Lemma 4.3, $\zeta(s) = \chi(s)\zeta(1-s)$ and the two previous cases at $1-\sigma \in (\frac12, 1.01]$, as in the last step of the proof of Lemma 4.4. Since $[c-R, c+R] = [-\eta, 2+3\eta] \subset [-0.01, 3]$, the claim covers the disc for every $\eta < 10^{-2}$. For $T \ge 100$, Proposition 5.1 applies with $\mathfrak{h} = \mathfrak{e}^* + \varepsilon$, $K = K_\varepsilon$ and $\Omega = 1$. In (5.2) the terms $\log\frac{\zeta(c)}{\zeta(2c)}$, $\log\frac{T+3}{T}$, $\log K_\varepsilon$, $\log\zeta(\sigma^\sharp)$ and $\frac{\sigma^\sharp-1/2}{\pi T}$ are $O_{\varepsilon,\eta}(1)$, and $\mathcal{E}(T,\delta^\sharp)/(2\pi) \le \frac12\big(\frac{640\delta^\sharp-112}{1536(3T-1)} + 2^{-10}\big) = O(1)$ by Theorem 3.5(c). Hence

$$|S(T)| \le \frac{\log T}{2\pi\log 2}\int_0^\pi \big(\mathfrak{e}^*(1+\eta+(1+2\eta)\cos\theta) + \varepsilon\big)d\theta + O_{\varepsilon,\eta}(1). \tag{5.5}$$

Passage to the limit. The limits are taken in the order $T \to \infty$, then $\varepsilon \to 0$, then $\eta \to 0$. The parameters η and ε do not depend on T. For fixed ε and η the term $O_{\varepsilon,\eta}(1)$ is bounded, so dividing by $\log T$ and letting $T \to \infty$ through $T \notin \{\gamma_k\}$ (at $T \in \{\gamma_k\}$, $S(T) = S(T^+)$ is a limit of values at $t \notin \{\gamma_k\}$, so the lim sup over $T \notin \{\gamma_k\}$ is $\mathfrak{a}$) gives

$$\mathfrak{a} \le \frac{1}{2\pi\log 2}\Big(\int_0^\pi \mathfrak{e}^*\big(1+\eta+(1+2\eta)\cos\theta\big)d\theta + \pi\varepsilon\Big). \tag{5.6}$$

The right side no longer involves K_ε. Let $\varepsilon \to 0$. Finally, $\mathfrak{e}^*$ is continuous on $\mathbb{R}$: at $\sigma = \frac12$ both formulas in (1.4) give $\mathfrak{e}(\frac12)$, at $\sigma = 1$ both give $\mathfrak{e}(1) = 0$, and at $\sigma = 0$ both give $\frac12$.

Hence $\mathfrak{e}^*$ is uniformly continuous on $[-1,4]$, and since the arguments $1+\eta+(1+2\eta)\cos\theta$ and $1+\cos\theta$ differ by at most 3η, the integrand converges uniformly in θ as $\eta\to 0$. Therefore $\mathfrak{a}\le \frac{1}{2\pi\log 2}\int_0^\pi \mathfrak{e}^*(1+\cos\theta)\,d\theta$. The integrand vanishes for $0\le\theta\le\frac{\pi}{2}$. On $[\frac{\pi}{2},\pi]$ substitute $\sigma=1+\cos\theta\in[0,1]$, $d\sigma=-\sin\theta\,d\theta$, $\sin\theta=\sqrt{1-(\sigma-1)^2}=\sqrt{\sigma(2-\sigma)}$. This is (1.5). □

Remark 5.2. Admissible profiles are nonnegative by definition, so $\mathfrak{e}^*\ge 0^*$ pointwise, where $0^*(\sigma)=\max(\frac12-\sigma,0)$ is the reflected extension of the zero profile, and

$$\Lambda(\mathfrak{e})\ge\Lambda(0)=\frac{1}{2\pi\log 2}\int_0^{1/2}\frac{(1/2-\sigma)\,d\sigma}{\sqrt{\sigma(2-\sigma)}}=\frac{3\sqrt3-\pi}{12\pi\log 2}=0.0786252\ldots \tag{5.7}$$

The Lindelöf hypothesis would give $\mathfrak{a}\le 0.0786253$ by Theorem 1.1, and no admissible profile makes the right side of (1.5) smaller. Whether a disc with (c,R) away from $(1,1)$ could give less is not addressed here. (With $\sigma=1+\cos\theta$ the integral is $\int_{2\pi/3}^{\pi}(-\frac12-\cos\theta)\,d\theta=\frac{\sqrt3}{2}-\frac{\pi}{6}$.)

5.2 Proof of Corollary 1.2

Let $0<\varepsilon'<\varepsilon_0$, where $\varepsilon_0>0$ is small enough that the nodes ς_j' defined below increase and $\varsigma_7'<1$ (this holds at $\varepsilon'=0$, where they are the nodes (1.6), hence for all small ε' by continuity), $(\kappa_0,\lambda_0)=(\frac{13}{84}+\varepsilon',\frac{55}{84}+\varepsilon')$ and $(\kappa_j,\lambda_j)=A^j(\kappa_0,\lambda_0)$, $1\le j\le 7$.

By Theorem 3.7(c),(a) these are exponent pairs, and $\lambda_j-\kappa_j\ge\frac12$ (it equals $\frac12$ for $j=0$, and A maps $\lambda-\kappa$ to $\frac12+\frac{\lambda-\kappa}{2\kappa+2}$). Put $\varsigma_j'=\lambda_j-\kappa_j$ $(0\le j\le 7)$, $\varsigma_8'=1$, $\kappa_8=0$.

The ς_j' increase by the choice of ε_0, and the κ_j decrease since $\kappa_{j+1}=\kappa_j/(2\kappa_j+2)<\kappa_j$. By Theorem 3.7(b) and $\zeta(1+it)\ll\log|t|$ there is a constant $A_{\varepsilon'}\ge 2$ with $|\zeta(\varsigma_j'+it)|\le A_{\varepsilon'}|t|^{\kappa_j}\log|t|$ for $|t|\ge 3$ and $0\le j\le 8$.

With $Q=10$ and $\Xi(s)=(s-1)\zeta(s)$, the argument of the proof of Lemma 4.4 (for $|t|\le 3$ the bound $|\Xi|<34.9$ from [39, (2.1.4)]; the pole of ζ at $s=1$ is removed by the factor $s-1$) gives $|\Xi(\varsigma_j'+it)|\le A_{\varepsilon'}|Q+s|^{1+\kappa_j}|\operatorname{Log}(Q+s)|$ for all real t.

Theorem 3.4 on each strip $\varsigma_j'\le\sigma\le\varsigma_{j+1}'$, with $\alpha_2=1$ on both lines, then shows that the piecewise linear function $\mathfrak{e}_{\varepsilon'}$ through the points (ς_j',κ_j), $0\le j\le 8$, is an admissible profile. By Theorem 1.1, $\mathfrak{a}\le\Lambda(\mathfrak{e}_{\varepsilon'})$.

As $\varepsilon'\to 0$ the nodes tend to (1.6), $\mathfrak{e}^*_{\varepsilon'}\to\mathfrak{e}^*_{\mathrm B}$ uniformly, and $\Lambda(\mathfrak{e}_{\varepsilon'})\to\Lambda(\mathfrak{e}_{\mathrm B})$.

The value $\Lambda(\mathfrak{e}_{\mathrm B})=0.0968160063238\ldots$ is computed in Appendix A in two ways: by 40-digit quadrature on each linear piece, and by the closed form (5.9) with $c=R=1$, which expresses $2\pi\log 2\cdot\Lambda(\mathfrak{e}_{\mathrm B})=\int_0^\pi\mathfrak{e}^*_{\mathrm B}(1+\cos\theta)\,d\theta$ as a combination with rational coefficients of π and of values of arccos and of square roots at rational points (here $\theta_h=\frac{2\pi}{3}$).

The certified ball-arithmetic computation of Appendix A gives $0.096816006\le\Lambda(\mathfrak{e}_{\mathrm B})\le 0.096816007$ and $2\pi\Lambda(\mathfrak{e}_{\mathrm B})\le 0.60832$. □

5.3 Proof of Theorem 1.3

Take $c=1.000002$ and $R=1.000005$. Then $c-R=-3\cdot10^{-6}$, $\sigma^\sharp=1.25000275\ldots$, $\delta^\sharp=0.2500012499\ldots\in[\frac14,\frac58]$, $c<R$, and (3.5) holds. Let $T\ge 100$, $T\notin\{\gamma_k\}$. For $\sigma\in[c-R,c+R]\subset[-0.01,3]$ and $t\in[T-2,T+2]\subset[98,\infty)$, Lemma 4.4 gives (5.1) with $\mathfrak{h}=\mathfrak{e}^*_{\mathrm E}$, $K=2.74$ and $\Omega(t)=\log t$.

The integral $\mathcal{I}_{\mathfrak{e}^*_{\mathrm E}}$ has a closed form. Write $\varsigma_0=\frac12<\varsigma_1<\cdots<\varsigma_7=1$ for the abscissae in (4.5), e_i for the second coordinates and $\varrho_i=(e_{i+1}-e_i)/(\varsigma_{i+1}-\varsigma_i)$ for the slopes, so that

$$\mathfrak{e}_{\mathrm E}(\sigma)=e_0+\varrho_0\big(\sigma-\tfrac12\big)+\sum_{i=1}^{6}(\varrho_i-\varrho_{i-1})\max(0,\sigma-\varsigma_i)-\varrho_6\max(0,\sigma-1)\qquad(\sigma\ge\tfrac12),\tag{5.8}$$

and by (1.4) $\mathfrak{e}^*_{\mathrm E}(\sigma)=\frac12-\sigma+\mathfrak{e}_{\mathrm E}(1-\sigma)$ for $\sigma\le\frac12$, the right side being $\frac12-\sigma$ for $\sigma<0$. As θ runs over $[0,\pi]$, $\sigma=c+R\cos\theta$ decreases from $c+R$ to $c-R<0$. Split the integral at $\theta_h=\arccos\frac{1/2-c}{R}$,

where $\sigma = \frac12$. The linear terms integrate to multiples of θ_h, $\pi - \theta_h$ and $\sin\theta_h$, and each hinge integrates exactly: for $\frac12 \le \varsigma \le 1$, with $\theta_\varsigma = \arccos\frac{\varsigma - c}{R} \le \theta_h$ and $\theta'_\varsigma = \arccos\frac{1-\varsigma-c}{R} \ge \theta_h$,

$$\begin{aligned}\int_0^{\theta_h} \max(0, c + R\cos\theta - \varsigma)\,d\theta &= (c-\varsigma)\,\theta_\varsigma + R\sin\theta_\varsigma,\\ \int_{\theta_h}^{\pi} \max(0, 1 - c - R\cos\theta - \varsigma)\,d\theta &= (1-c-\varsigma)(\pi - \theta'_\varsigma) + R\sin\theta'_\varsigma,\end{aligned} \tag{5.9}$$

with $\sin\theta_\varsigma = \sqrt{1 - ((\varsigma - c)/R)^2}$ and likewise for θ'_ς. Hence $\mathcal{I}_{\mathfrak{e}_\mathrm{E}^*}$ is a combination, with rational coefficients, of π, fifteen values of arccos at rational points and fifteen square roots of rational numbers. Evaluating it with certified ball arithmetic (Appendix A) gives

$$C_1 := \frac{\mathcal{I}_{\mathfrak{e}_\mathrm{E}^*}}{2\pi\mathfrak{d}} = 0.0978292318785\ldots, \quad C_2 := \frac{1}{2\mathfrak{d}} = 0.7213464797\ldots, \quad \log\frac{\zeta(c)}{\zeta(2c)} = 12.6246665\ldots \tag{5.10}$$

By Proposition 5.1,

$$\begin{aligned}|S(T)| \le C_1\log(T+2) + C_2\log\log(T+2) + C_2\Big[12.62467 + \log\frac{T+3}{T} + \log 2.74\Big]\\ + \frac{\log\zeta(\sigma^\sharp)}{\pi} + \frac{0.7501}{\pi T} + \frac{\mathcal{E}(T,\delta^\sharp)}{2\pi} + \frac14.\end{aligned} \tag{5.11}$$

Here $\log\zeta(\sigma^\sharp) = 1.52498\ldots$,[1] and by Theorem 3.5(c), $\mathcal{E}(T,\delta^\sharp)/(2\pi) \le \frac12\big(\frac{640\delta^\sharp - 112}{1536\cdot 299} + 2^{-10}\big) < 0.00055$. Replacing $\log(T+2)$ by $\log T$ and $\log\log(T+2)$ by $\log\log T$ costs at most $\frac{2C_1}{T} + \frac{2C_2}{T\log T}$, by the mean value theorem ($\log(T+2) - \log T \le \frac{2}{T}$, and $\log\log(T+2) - \log\log T \le \frac{2}{T\log T}$ because $(\log\log u)' = 1/(u\log u)$ decreases).

Let $C_3(T)$ be the sum of these two costs and of all terms of (5.11) other than $C_1\log(T+2) + C_2\log\log(T+2)$, with $\mathcal{E}(T,\delta^\sharp)$ replaced by the majorant of Theorem 3.5(c). Every term of $C_3(T)$ is constant or decreasing in T: $\log\frac{T+3}{T}$, $\frac{0.7501}{\pi T}$, $\frac{2C_1}{T}$ and $\frac{2C_2}{T\log T}$ decrease.

The majorant $\frac12\big(\frac{640\delta^\sharp - 112}{1536(3T-1)} + 2^{-10}\big)$ of $\mathcal{E}(T,\delta^\sharp)/(2\pi)$ decreases because $640\delta^\sharp - 112 = 48.0007\ldots > 0$. The remaining terms $C_2(12.62467 + \log 2.74)$, $\frac{\log\zeta(\sigma^\sharp)}{\pi}$ and $\frac14$ are constants. With the rounded values displayed, $C_3(100) \le 10.5987$.

The program of Appendix A gives $C_3(100) = 10.598602\ldots$, and $10.598545\ldots$ for the same sum with the exact differences $\log 102 - \log 100$ and $\log\log 102 - \log\log 100$ in place of the two costs.

Hence $|S(T)| \le C_1\log T + C_2\log\log T + C_3(100)$ for $T \ge 100$, and (1.7) follows from $0.09783 > C_1$, $0.7214 > C_2$, $10.6 > C_3(100)$. For $e \le T \le 100$, Theorem 3.6 gives $|S(T)| \le 2.5167$, smaller than the right side of (1.7). Finally (1.8) follows from (1.1) and $|\varpi(T)|/\pi \le 0.53/(\pi e) < 0.07$ (Lemma 4.1). □

6 Distance from τ_k to the imaginary parts

For $k \ge 1$ put $s_k = k - \frac12 - N_0(\gamma_k) = N_0(\tau_k) - N_0(\gamma_k)$, and write $G = N - N_0 = S + \varpi/\pi$ (Lemma 4.1), $G_k^\pm = G(\gamma_k^\pm)$.

Lemma 6.1. *For every $k \ge 1$, $G_k^- + \frac12 \le s_k \le G_k^+ - \frac12$; in particular $|s_k| \le \max(|G_k^-|, |G_k^+|) - \frac12$.*

[1] Appendix A certifies $\log\zeta(\sigma^\sharp) = 1.5249835586\ldots$ by ball arithmetic. The bound $\log\zeta(\sigma^\sharp) \le 1.52499$ also follows from $\zeta(\sigma) \le \sum_{n<N} n^{-\sigma} + (N-\frac12)^{1-\sigma}/(\sigma-1)$ with $N = 10^4$, which gives $\log\zeta(\sigma^\sharp) \le 1.5249836$. This inequality follows from $n^{-\sigma} \le \int_{n-1/2}^{n+1/2} x^{-\sigma}\,dx$, which follows from $(x-h)^{-\sigma} + (x+h)^{-\sigma} \ge 2\big((x-h)(x+h)\big)^{-\sigma/2} \ge 2x^{-\sigma}$ for $0 \le h < x$.

Proof. The indices j with $\gamma_j = \gamma_k$ are exactly $N(\gamma_k^-)+1, \dots, N(\gamma_k^+)$, so $N(\gamma_k^-)+1 \le k \le N(\gamma_k^+)$. Subtract $\frac{1}{2}+N_0(\gamma_k)$ and use the continuity of N_0. With $\mathfrak{g} = \max(|G_k^-|, |G_k^+|)$, $s_k \le G_k^+ - \frac{1}{2} \le \mathfrak{g} - \frac{1}{2}$ and $s_k \ge G_k^- + \frac{1}{2} \ge -\mathfrak{g} + \frac{1}{2}$. □

Proof of Corollary 1.4. The function N_0 is increasing on $(2\pi, \infty)$ with increasing derivative $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, and $\gamma_1 > 14 > 2\pi$, $\tau_k > 2\pi$, so $\mathfrak{m}_k > 2\pi$. By the mean value theorem, $|s_k| = \mathcal{L}(\xi_k)|\tau_k - \gamma_k| \ge \mathcal{L}(\mathfrak{m}_k)|\tau_k - \gamma_k|$ for some ξ_k between τ_k and γ_k.

(b) Inequality (1.8) holds for $G(t)$ at $t \ge e$, $t \notin \{\gamma_k\}$, and, since its right side is continuous and increasing in t, for $G_k^\pm$ (note $\gamma_k > 14 > e$). With Lemma 6.1 it gives $|s_k| \le 0.09783 \log\gamma_k + 0.7214 \log\log\gamma_k + 10.2$, and (1.9) follows from $\mathcal{L}(\mathfrak{m}_k) = \frac{1}{2\pi}\log\frac{\mathfrak{m}_k}{2\pi} > 0$.

(a) First, $\mathfrak{m}_k \ge \gamma_k/2$ for large k: if $\tau_k < \gamma_k/2$, then $|s_k| = N_0(\gamma_k) - N_0(\tau_k) \ge N_0(\gamma_k) - N_0(\gamma_k/2) \ge \frac{\gamma_k}{2}\mathcal{L}(\frac{\gamma_k}{2})$, which exceeds the bound $O(\log\gamma_k)$ of (b) for large k. Hence $\log\mathfrak{m}_k = \log\gamma_k + O(1)$, so $|\tau_k - \gamma_k| = O(1)$ by (b), and then $\log\mathfrak{m}_k = \log\gamma_k + o(1)$. For $\varepsilon > 0$, Corollary 1.2 and $\varpi(t) \to 0$ give $|G(t)| \le (\Lambda(\mathfrak{e}_B) + \varepsilon)\log t$ for large $t \notin \{\gamma_k\}$, hence $|G_k^\pm| \le (\Lambda(\mathfrak{e}_B) + \varepsilon)\log\gamma_k$, $|s_k| \le (\Lambda(\mathfrak{e}_B)+\varepsilon)\log\gamma_k$ and $|\tau_k - \gamma_k| \le |s_k|/\mathcal{L}(\mathfrak{m}_k) \le 2\pi(\Lambda(\mathfrak{e}_B)+\varepsilon)(1+o(1))$. Let $\varepsilon \to 0$. Finally, $2\pi \cdot 0.0968160063238\ldots = 0.6083129\ldots$. □

Proposition 6.2 (Limit superior of $|\tau_k - \gamma_k|$)**.** $\limsup_{k\to\infty}|\tau_k - \gamma_k| = 2\pi\mathfrak{a}$. *In particular the bound* $\limsup_k|\tau_k - \gamma_k| \le 2\pi\Lambda(\mathfrak{e}_B)$ *of Corollary 1.4(a) is equivalent, through this identity, to the bound* $\mathfrak{a} \le \Lambda(\mathfrak{e}_B)$ *of Corollary 1.2, and the Lindelöf hypothesis would give* $\limsup_k|\tau_k - \gamma_k| \le 2\pi\Lambda(0) = 0.49401\ldots$ *(Remark 5.2).*

Proof. Upper bound. By the definition of $\mathfrak{a}$, for every $\varepsilon > 0$ and all large $t \notin \{\gamma_k\}$, $|S(t)| \le (\mathfrak{a}+\varepsilon)\log t$. The proof of Corollary 1.4(a), with $\Lambda(\mathfrak{e}_B)$ replaced by $\mathfrak{a}$, gives $\limsup_k|\tau_k - \gamma_k| \le 2\pi\mathfrak{a}$. It uses Lemma 6.1, which allows multiple imaginary parts.

Lower bound. One may assume $\mathfrak{a} > 0$. Let $t > \gamma_1$, $t \notin \{\gamma_k\}$, and $k = N(t)$, so that $\gamma_k < t < \gamma_{k+1}$. If the imaginary part γ_k is multiple, k is the largest index with that imaginary part, and γ_{k+1} is the next distinct imaginary part, whatever its multiplicity. Since no imaginary part lies in (γ_k, γ_{k+1}), $N(\gamma_k^+) = k = N(\gamma_{k+1}^-)$.

By the definition of s_k, $s_k = N(\gamma_k^+) - \frac{1}{2} - N_0(\gamma_k) = G_k^+ - \frac{1}{2}$ and $s_{k+1} = N(\gamma_{k+1}^-) + \frac{1}{2} - N_0(\gamma_{k+1}) = G_{k+1}^- + \frac{1}{2}$. On (γ_k, γ_{k+1}), $G = k - N_0$ is decreasing, so $G_k^+ \ge G(t) \ge G_{k+1}^-$, hence $s_k \ge G(t) - \frac{1}{2}$ and $s_{k+1} \le G(t) + \frac{1}{2}$.

If $G(t) \ge \frac{1}{2}$, then $|s_k| \ge G(t) - \frac{1}{2}$, and if $G(t) \le -\frac{1}{2}$ then $|s_{k+1}| \ge |G(t)| - \frac{1}{2}$, so in all cases $\max(|s_k|, |s_{k+1}|) \ge |G(t)| - \frac{1}{2}$.

For $j \in \{k, k+1\}$ the mean value theorem gives $|s_j| = \mathcal{L}(\xi_j)|\tau_j - \gamma_j|$ with ξ_j between τ_j and γ_j, and $\mathcal{L}$ increases, so $|\tau_j - \gamma_j| \ge |s_j|/\mathcal{L}(\max(\tau_j, \gamma_j))$. By the proof of Corollary 1.4(a), $|\tau_j - \gamma_j| = O(1)$, and $\gamma_{k+1} - \gamma_k \to 0$ as $t \to \infty$ (Theorem 3.1; $k = N(t) \to \infty$), so $\max(\tau_j, \gamma_j) = t + O(1)$ and $2\pi\mathcal{L}(\max(\tau_j, \gamma_j)) = \log t - \log 2\pi + O(1/t)$. Finally $G(t) = S(t) + \varpi(t)/\pi$ with $\varpi(t) \to 0$ (Lemma 4.1).

Choose $t \to \infty$, $t \notin \{\gamma_k\}$, with $|S(t)| \ge (\mathfrak{a} - \varepsilon)\log t$, which exist by the definition of $\mathfrak{a}$ and the convention $S(T) = S(T^+)$. Then, for one of $j = k, k+1$,

$$|\tau_j - \gamma_j| \ge \frac{2\pi\big((\mathfrak{a} - \varepsilon)\log t - \frac{1}{2} - o(1)\big)}{\log t - \log 2\pi + O(1/t)} \longrightarrow 2\pi(\mathfrak{a} - \varepsilon), \tag{6.1}$$

and $j \to \infty$ with t. Let $\varepsilon \to 0$. Thus the limit superior of the displacement is determined by that of $|S(T)|/\log T$. □

7 Localization by primes

7.1 The construction

Each summand in (1.10) is $-\frac{1}{\pi}\arg(1 - p^{-1/2-it})$ with $|p^{-1/2-it}| \le 2^{-1/2}$, so its modulus is at most $\frac{1}{\pi}\arcsin(2^{-1/2}) = \frac{1}{4}$. For every fixed m, the inner series over r converges absolutely. The

argument bound follows geometrically by maximizing the angle of $1-z$ over $|z| \le 2^{-1/2}$, which gives $|\arg(1-z)| \le \arcsin|z|$. Therefore

$$|\Phi_m(t)| \le \frac{m}{4}. \tag{7.1}$$

On $[2^j, 2^{j+1}]$ the two definitions of Φ in (1.11) at the endpoints agree, so Φ and $\mathcal{F}$ are continuous on $(0,\infty)$, and $\mathcal{F}(t) = N_0(t) \to \frac{7}{8}$ as $t \to 0^+$ (for $t < \frac{1}{2}$ one has $m_j = m_{j+1} = 0$). Since $m_{j+1}^2 \le 2^{j+1} \le 2t$ on the block,

$$|\Phi(t)| \le \tfrac{1}{4}\sqrt{2t} \qquad (t > 0). \tag{7.2}$$

Since $N_0(\tau_k) = k - \frac{1}{2}$, (1.12) reads $\tau_k^{\mathrm{C}} = \mathcal{T}(\tau_k)$ with $\mathcal{T}(\tau) = \int_0^\infty \mathbf{1}_{\{\mathcal{F}(t) < N_0(\tau)\}} dt$: the corrected point is a function of the original point.

Proposition 7.1. *For every $k \ge 1$, τ_k^{C} is finite and $0 < \tau_k^{\mathrm{C}} < \tau_{k+1}^{\mathrm{C}}$. If $\tau_k \ge 2\pi e$, the integral (1.12) may be truncated at $2\tau_k$, and its integrand uses only the primes p_ν with $\nu \le \lfloor 2\sqrt{\tau_k} \rfloor$.*

Proof. By (7.2), $\mathcal{F}(t) \ge N_0(t) - \frac{1}{4}\sqrt{2t} \to \infty$, so each sublevel set is bounded and $\tau_k^{\mathrm{C}} < \infty$. The block containing 2π is $[4,8]$, with $m_2 = m_3 = 2$, so $\mathcal{F}(2\pi) \le N_0(2\pi) + \frac{2}{4} = -\frac{1}{8} + \frac{1}{2} = \frac{3}{8}$. Fix $k \ge 1$. Then $\mathcal{F}(2\pi) < \frac{1}{2} \le k - \frac{1}{2}$, and since $\mathcal{F}(t) \to \infty$ there is $t_1 > 2\pi$ with $\mathcal{F}(t_1) > k + \frac{1}{2}$. By the intermediate value theorem $\mathcal{F}$ takes the value k in $(2\pi, t_1)$, so the set $\{k - \frac{1}{2} < \mathcal{F} < k + \frac{1}{2}\} = \mathcal{F}^{-1}\big((k - \frac{1}{2}, k + \frac{1}{2})\big)$ is nonempty, and it is open because $\mathcal{F}$ is continuous. Every nonempty open subset of $\mathbb{R}$ contains a nondegenerate interval and therefore has positive Lebesgue measure. Hence $\tau_{k+1}^{\mathrm{C}} - \tau_k^{\mathrm{C}} = \mathrm{meas}\{k - \frac{1}{2} \le \mathcal{F} < k + \frac{1}{2}\} \ge \mathrm{meas}\{k - \frac{1}{2} < \mathcal{F} < k + \frac{1}{2}\} > 0$. Likewise $\tau_1^{\mathrm{C}} \ge \mathrm{meas}\{\mathcal{F} < \frac{1}{2}\} > 0$, because $\{\mathcal{F} < \frac{1}{2}\}$ is open and contains 2π.

If $\tau_k \ge 2\pi e$ and $t \ge 2\tau_k$, then $N_0(t) - N_0(\tau_k) \ge (t - \tau_k)\mathcal{L}(\tau_k) \ge \frac{t}{4\pi} > \frac{1}{4}\sqrt{2t}$ (as $t \ge 4\pi e > 2\pi^2$), so $\mathcal{F}(t) > k - \frac{1}{2}$. For $t \le 2\tau_k$ the budgets used satisfy $m^2 \le 4\tau_k$, hence $m \le 2\sqrt{\tau_k}$. □

7.2 A deterministic transfer inequality

Lemma 7.2. *Let $A \le B$ be integers and $D = (D_\ell)_{A \le \ell \le B}$ nonnegative. For $A \le \ell \le B$ let $(\mathcal{M}D)_\ell = \max\{\frac{1}{b-a+1}\sum_{r=a}^b D_r : A \le a \le \ell \le b \le B\}$. Then $\sum_{\ell=A}^B (\mathcal{M}D)_\ell^3 \le 24 \sum_{\ell=A}^B D_\ell^3$.*

Proof. Rising sun. In this proof g, $\mathcal{G}$ and $\mathcal{P}_\pm$ are local. Let g be real on $[A,B] \cap \mathbb{Z}$, and let $\mathcal{P}_+(A', B)$ be the set of $j \in [A', B]$ for which $\sum_{r=j}^b \mathsf{g}_r \ge 0$ for some $b \in [j, B]$. One can show that $\sum_{j \in \mathcal{P}_+(A',B)} \mathsf{g}_j \ge 0$ by induction on $B - A'$ (this is the discrete form of F. Riesz's rising sun lemma [35]). If $A' \notin \mathcal{P}_+(A', B)$, then $\mathcal{P}_+(A', B) = \mathcal{P}_+(A'+1, B)$. Otherwise choose $b^* \in [A', B]$ maximizing $\mathcal{G}(b) = \sum_{r=A'}^b \mathsf{g}_r$. Then $\mathcal{G}(b^*) \ge 0$, and for $A' < j \le b^*$ also $\sum_{r=j}^{b^*} \mathsf{g}_r = \mathcal{G}(b^*) - \mathcal{G}(j-1) \ge 0$. Hence $[A', b^*] \subset \mathcal{P}_+(A', B)$, $\mathcal{P}_+(A', B) = [A', b^*] \cup \mathcal{P}_+(b^*+1, B)$, and $\sum_{\mathcal{P}_+(A',B)} \mathsf{g} = \mathcal{G}(b^*) + \sum_{\mathcal{P}_+(b^*+1,B)} \mathsf{g} \ge 0$. The same argument from the right gives $\sum_{j \in \mathcal{P}_-} \mathsf{g}_j \ge 0$ for the set $\mathcal{P}_-$ of $\ell \in [A, B]$ with $\sum_{r=a}^\ell \mathsf{g}_r \ge 0$ for some $a \in [A, \ell]$.

Weak type. Let $\Theta > 0$ and $\mathsf{g} = D - \Theta$. Then $\Theta\, \#\mathcal{P}_+ \le \sum_{\mathcal{P}_+} D \le \sum D$ and likewise for $\mathcal{P}_-$. If $(\mathcal{M}D)_\ell \ge \Theta$, some $[a,b] \ni \ell$ has $\sum_{r=a}^b \mathsf{g}_r \ge 0$. Splitting at ℓ, either $\sum_{r=a}^\ell \mathsf{g}_r \ge 0$, so $\ell \in \mathcal{P}_-$, or $\ell < b$ and $\sum_{r=\ell+1}^b \mathsf{g}_r > 0$, so $\ell + 1 \in \mathcal{P}_+$. Hence $\Theta\, \#\{\ell : (\mathcal{M}D)_\ell \ge \Theta\} \le 2 \sum_\ell D_\ell$.

Strong type. Let $D''_\ell = D_\ell \mathbf{1}_{\{2D_\ell \ge \Theta\}}$. Every term of the remaining part $D - D''$ is $< \Theta/2$, so every average of $D - D''$ over an interval of indices is $< \Theta/2$, that is, $\mathcal{M}(D - D'') < \Theta/2$. Since $\mathcal{M}$ is subadditive, $(\mathcal{M}D)_\ell \ge \Theta$ implies $(\mathcal{M}D'')_\ell \ge \Theta/2$, and $\Theta\, \#\{\mathcal{M}D \ge \Theta\} \le 4 \sum_\ell D_\ell \mathbf{1}_{\{2D_\ell \ge \Theta\}}$. Hence

$$\sum_\ell (\mathcal{M}D)_\ell^3 = \int_0^\infty 3\Theta^2 \#\{\mathcal{M}D \ge \Theta\}\, d\Theta \le 12 \sum_\ell D_\ell \int_0^{2D_\ell} \Theta\, d\Theta = 24 \sum_\ell D_\ell^3. \tag{7.3}$$

□

Theorem 7.3 (Sublevel transfer)**.** *In this theorem (γ_k), N and D_ℓ are general: let $(\gamma_k)_{k\ge1}$ be any nondecreasing real sequence with finitely many terms in each bounded set, $N(t)=\#\{k:\gamma_k\le t\}$, and $a<b$ with $\ell_0=N(a)\ge1$, $\ell_1=N(b)$. Put $D_\ell=\gamma_{\ell+1}-\gamma_\ell$ for $\ell_0\le\ell\le\ell_1$. Let f be a measurable real function on $[a,b]$, and for k with $a<\gamma_k\le b$ put*

$$x_k^f=a+\int_a^b \mathbf{1}_{\{f(t)<k-\frac12\}}\,dt,\qquad q_f(t)=\bigl|\lfloor f(t)+\tfrac12\rfloor-N(t)\bigr|. \tag{7.4}$$

Here q_f is the integer discrepancy between the levels of f and of N, not the continuous error $f-N$; note that $q_f\le|f-N|+\frac12$. Then

$$\sum_{a<\gamma_k\le b}(x_k^f-\gamma_k)^2\le 2\cdot24^{1/3}\Bigl(\sum_{\ell=\ell_0}^{\ell_1}D_\ell^3\Bigr)^{1/2}\Bigl(\int_a^b q_f^4\,dt\Bigr)^{1/2}, \tag{7.5}$$

and $q_f^4\le(1+|f-N|)^4$. (The right side may be $+\infty$; its first factor is positive, since $\sum_{\ell=\ell_0}^{\ell_1}D_\ell=\gamma_{\ell_1+1}-\gamma_{\ell_0}>b-a$. We do not claim that the constant $2\cdot24^{1/3}$ is optimal.)

Proof. Fix k with $a<\gamma_k\le b$. Then $\ell_0+1\le k\le\ell_1$. (A term γ_k equal to a has index $\le\ell_0$ and is not counted. The last gap $D_{\ell_1}=\gamma_{\ell_1+1}-\gamma_{\ell_1}$, to the first term beyond b, is included in (7.5) because $\{t\in[a,b]:N(t)=\ell_1\}\subset[\gamma_{\ell_1},\gamma_{\ell_1+1})$ reaches b.) Since $N(t)<k-\frac12$ exactly when $t<\gamma_k$, $\gamma_k=a+\int_a^b\mathbf{1}_{\{N<k-1/2\}}$, and $x_k^f-\gamma_k=u_k-v_k$ with $u_k=\mathrm{meas}\{t\in[\gamma_k,b]:f(t)<k-\frac12\}$ and $v_k=\mathrm{meas}\{t\in[a,\gamma_k):f(t)\ge k-\frac12\}$.

Let $E_k\subset[a,b]$ be the set where $\mathbf{1}_{\{f<k-1/2\}}\ne\mathbf{1}_{\{N<k-1/2\}}$. By the bathtub principle [26, Thm. 1.14], a measurable set $U\subset[\gamma_k,b]$ of measure u satisfies $\int_U(t-\gamma_k)\,dt\ge\int_{\gamma_k}^{\gamma_k+u}(t-\gamma_k)\,dt=\frac{u^2}{2}$: on $U\setminus[\gamma_k,\gamma_k+u]$ the integrand is $\ge u$, on $[\gamma_k,\gamma_k+u]\setminus U$ it is $\le u$, and the two sets have the same measure. Likewise on the left. Hence

$$(x_k^f-\gamma_k)^2\le u_k^2+v_k^2\le 2\int_{E_k}|t-\gamma_k|\,dt. \tag{7.6}$$

Let $t\in[a,b]$ and $\ell=N(t)\in[\ell_0,\ell_1]$, so $\gamma_\ell\le t<\gamma_{\ell+1}$. If $t\in E_k$, the level $k-\frac12$ lies between $N(t)$ and $f(t)$: either $N(t)+1\le k\le\lfloor f(t)+\frac12\rfloor$ or $\lfloor f(t)+\frac12\rfloor+1\le k\le N(t)$. So at most $q_f(t)$ indices k have $t\in E_k$. If $k>\ell$, then $k\le\lfloor f(t)+\frac12\rfloor$ and $|t-\gamma_k|\le\gamma_k-\gamma_\ell=\sum_{r=\ell}^{k-1}D_r\le(k-\ell)(\mathcal{M}D)_\ell\le q_f(t)(\mathcal{M}D)_\ell$. If $k\le\ell$, then $k>\lfloor f(t)+\frac12\rfloor$ and $|t-\gamma_k|\le\gamma_{\ell+1}-\gamma_k=\sum_{r=k}^{\ell}D_r\le(\ell-k+1)(\mathcal{M}D)_\ell\le q_f(t)(\mathcal{M}D)_\ell$. Here $\mathcal{M}$ is taken with $A=\ell_0$, $B=\ell_1$: the index intervals lie in $[\ell_0,\ell_1]$ and contain ℓ. Therefore

$$\sum_{a<\gamma_k\le b}\mathbf{1}_{E_k}(t)|t-\gamma_k|\le q_f(t)^2(\mathcal{M}D)_{N(t)}. \tag{7.7}$$

Sum (7.6) over k, integrate, and apply Cauchy–Schwarz in t. Since $\{t\in[a,b]:N(t)=\ell\}\subset[\gamma_\ell,\gamma_{\ell+1})$, Hölder's inequality and Lemma 7.2 give

$$\int_a^b(\mathcal{M}D)_{N(t)}^2dt\le\sum_{\ell=\ell_0}^{\ell_1}D_\ell(\mathcal{M}D)_\ell^2\le\Bigl(\sum D_\ell^3\Bigr)^{1/3}\Bigl(\sum(\mathcal{M}D)_\ell^3\Bigr)^{2/3}\le24^{2/3}\sum_{\ell=\ell_0}^{\ell_1}D_\ell^3, \tag{7.8}$$

which proves (7.5). Finally $q_f\le|f-N|+\frac12$. □

Remark 7.4 (Quantile form). In this remark m_f and m_N are local. Let m_f be the image of Lebesgue measure on $[a,b]$ under f, a measure of mass $b-a$ on $\mathbb{R}$, and m_N the image under N, which is the sum of the point masses $\mathrm{meas}\{t\in[a,b]:N(t)=\ell\}\,\delta_\ell$, $\ell_0\le\ell\le\ell_1$. Then $x_k^f-a=\mathsf{m}_f\bigl((-\infty,k-\frac12)\bigr)$ and $\gamma_k-a=\mathsf{m}_N\bigl((-\infty,k-\frac12)\bigr)$: the left side of (7.5) compares the distribution

functions of m_f and m_N at the half-integer levels, the thresholds between consecutive integer counting levels (with multiplicities, N may skip levels). Equivalently, x_k^f is the point where the increasing rearrangement of f on $[a,b]$ ([15, §10.12], in increasing order), the generalized inverse of the distribution function, reaches the level $k-\frac12$, and γ_k is the same point for N. The left side of (7.5) is thus a discrete form of the squared L^2 distance between the two distribution functions. It is not the squared Wasserstein distance W_2^2, which compares their inverses, the increasing rearrangements of f and N. The right side has no counterpart known to the author: one factor is a moment of the gaps D_ℓ, which are, up to the truncation at a and b, the masses of the atoms of m_N (the atoms themselves sit at the integers ℓ), the other a moment of the pointwise difference $f-N$.

7.3 The arithmetic inputs

Lemma 7.5 (Third moment of the gaps)**.** $\sum_{T<\gamma_\ell\le 2T}(\gamma_{\ell+1}-\gamma_\ell)^3 \ll T/\log^2 T$ *for* $T\ge T_0$.

Proof. Write $D_\ell=\gamma_{\ell+1}-\gamma_\ell$ and $L_T=\log\frac{T}{2\pi}$. By Theorem 3.1, $D_\ell\to 0$, so there is T_0 with $D_\ell\le 1$ whenever $\gamma_\ell>T\ge T_0$. For $0<h\le 1$ let $\mathcal{A}_h=\{t\in[T,4T]: N(t+h)=N(t)\}$. For every $t\in\mathcal{A}_h$ such that $t,t+h\notin\{\gamma_k\}$ (a null set is excluded), (1.1) at t and $t+h$ gives $|S(t+h)-S(t)|=|N_0(t+h)-N_0(t)+\frac{\varpi(t+h)-\varpi(t)}{\pi}|\ge \frac{hL_T}{2\pi}-O\big(\frac1T\big)\ge\frac{hL_T}{4\pi}$ when $hL_T\ge 1$ and $T\ge T_0$. Theorem 3.8(b) with $\mathsf{k}=2$, $\alpha_{\mathrm F}=\frac12$ and $H=T$ (the endpoints of its ranges, admitted there; this is the case (α) of [12]), on $[T,2T]$ and on $[2T,4T]$, together with $\log(2+h\log 4T)\le 2\log(2+hL_T)$, gives, by Chebyshev's inequality,

$$\operatorname{meas}\mathcal{A}_h\le\Big(\frac{4\pi}{hL_T}\Big)^4\int_T^{4T}\big(S(t+h)-S(t)\big)^4dt\ll\frac{T\log^2(2+hL_T)}{(hL_T)^4}\qquad(1\le hL_T,\ 0<h\le 1).\tag{7.9}$$

If $T<\gamma_\ell\le 2T$ and $D_\ell\ge 2h$, the interval $(\gamma_\ell,\gamma_{\ell+1}-h)$ lies in $[T,4T]$ (as $\gamma_{\ell+1}\le\gamma_\ell+1\le 4T$) and in $\mathcal{A}_h$ (for t in it, $\gamma_\ell<t<t+h<\gamma_{\ell+1}$, so $N(t+h)=N(t)=\ell$), and has length $D_\ell-h\ge D_\ell/2$. These intervals are disjoint. Hence, with $v=hL_T$,

$$\sum_{\substack{T<\gamma_\ell\le 2T\\ D_\ell\ge 2v/L_T}}D_\ell\ll\frac{T\log^2(2+v)}{v^4}\qquad(1\le v\le L_T).\tag{7.10}$$

Split according to $D_\ell<2/L_T$ and $2^{j+1}/L_T\le D_\ell<2^{j+2}/L_T$ for $j\ge 0$ with $2^j\le L_T/2$. The largest such j is $J=\lfloor\log_2(L_T/2)\rfloor$, and $2^{J+1}>L_T/2$ gives $2^{J+2}/L_T>1$, so these ranges cover $[2/L_T,1]\ni D_\ell$; moreover for each of them $v=2^j$ satisfies $1\le v\le L_T$. The first part contributes at most $\frac{4}{L_T^2}\sum D_\ell\le\frac{4}{L_T^2}(T+1)$. The j-th part, with $v=2^j$, contributes at most $(2^{j+2}/L_T)^2\sum_{D_\ell\ge 2^{j+1}/L_T}D_\ell\ll\frac{16\cdot4^j}{L_T^2}\cdot\frac{T(j+2)^2}{16^j}=\frac{16\,T(j+2)^2}{4^jL_T^2}$, and the sum over j converges. □

Lemma 7.6 (Fourth moment of the counting error)**.** *For* $U=2^j$ *with* $j\ge j_0$, $\int_U^{2U}|\mathcal{F}(t)-N(t)|^4dt\ll U$.

Proof. On $[U,2U]$, $\mathcal{F}-N=(2-\frac{t}{U})(N_0+\Phi_{m_j}-N)+(\frac{t}{U}-1)(N_0+\Phi_{m_{j+1}}-N)$. By convexity of $|\cdot|^4$ it suffices to prove the bound for $N_0+\Phi_m-N=\Phi_m-S-\varpi/\pi$ with $m\in\{m_j,m_{j+1}\}$, so that $\sqrt{U}-1\le m\le\sqrt{2U}$. Put $X=p_m$, so that $p\le X$ means $p\in\{p_1,\dots,p_m\}$, and $Y=U^{1/200}$. The ranges $p<Y$ and $Y\le p\le X$ below partition the primes $p\le X$. For U large, $X\ge m\ge Y$ and $\log X\asymp\log U$ (Theorem 3.10: $X=p_m\asymp m\log m\asymp\sqrt{U}\log U$), hence $V:=\sum_{Y\le p\le X}\frac1p=\log\log X-\log\log Y+O(1)=O(1)$ by Mertens' theorem. Expanding (1.10),

$$\begin{aligned}\Phi_m(t)&=\frac1\pi\operatorname{Im}\sum_{p<Y}p^{-1/2-it}+\frac1\pi\operatorname{Im}\mathscr{D}_1(t)+\frac{1}{2\pi}\operatorname{Im}\mathscr{D}_2(t)+\Phi^{(3)}(t),\\ \mathscr{D}_1(t)&=\sum_{Y\le p\le X}p^{-1/2-it},\qquad\mathscr{D}_2(t)=\sum_{p\le X}p^{-1-2it},\end{aligned}\tag{7.11}$$

with $|\Phi^{(3)}(t)| \le \frac{1}{\pi}\sum_p \sum_{r\ge 3} \frac{1}{rp^{r/2}} < \infty$. By Theorem 3.8(a) with $\mathsf{k}=2$, $\alpha_{\mathrm{F}}=\frac{1}{5}$ and $H=T=U$ (its hypotheses hold: $U^{1/2+\alpha_{\mathrm{F}}}=U^{7/10}\le H=U$, and $U^{\alpha_{\mathrm{F}}/(20\mathsf{k})}=U^{1/200}=Y\le U^{1/2}=H^{1/\mathsf{k}}$), $\int_U^{2U}|S-\frac{1}{\pi}\operatorname{Im}\sum_{p<Y}p^{-1/2-it}|^4 \ll U$.

The square $\mathscr{D}_1^2=\sum_n \mathfrak{c}_n n^{-it}$ has $\mathfrak{c}_n = d_1(n)n^{-1/2}$, where $d_1(n)$ is the number of ordered pairs (p,p') of primes of $[Y,X]$ with $pp'=n$: $d_1(n)=2$ if $n=pp'$ with $p\ne p'$, 1 if $n=p^2$, 0 otherwise, so $d_1\in\{0,1,2\}$, whence $d_1(n)^2\le 2d_1(n)$, and $\sum_n d_1(n)=\#\{p\in[Y,X]\}^2\le m^2$.

Hence $\sum_n|\mathfrak{c}_n|^2=\sum_n d_1(n)^2/n \le 2\sum_n d_1(n)/n = 2V^2 = O(1)$ and $\sum_n n|\mathfrak{c}_n|^2 = \sum_n d_1(n)^2 \le 2m^2 \le 4U$, and Theorem 3.9 gives $\int_U^{2U}|\mathscr{D}_1|^4\ll U$. (The weight n of Theorem 3.9 is compensated by the factor n^{-1} in $|\mathfrak{c}_n|^2$: the relevant bound is $O(m^2)=O(U)$, not $O(X^2)=O(U\log^2 U)$.)

In the same way $\mathscr{D}_2^2=\sum_v \mathfrak{c}'_v v^{-2it}$ with $\mathfrak{c}'_v=d_2(v)/v$, where $d_2(v)\le 2$ counts the ordered pairs of primes of $[2,X]$ with product v, and $\sum_v d_2(v)=m^2$. Since $v^{-2it}=(v^2)^{-it}$, $\mathscr{D}_2^2$ is the Dirichlet polynomial in n^{-it} whose coefficient is $\mathfrak{c}'_v$ at $n=v^2$ and 0 at the other n, and Theorem 3.9 applies to it with $\sum_v|\mathfrak{c}'_v|^2=\sum_v d_2(v)^2/v^2\le 2\big(\sum_p p^{-2}\big)^2$ and $\sum_v v^2|\mathfrak{c}'_v|^2=\sum_v d_2(v)^2\le 2m^2\le 4U$, so $\int_U^{2U}|\mathscr{D}_2|^4\ll U$.

Finally $|\varpi|\le 0.53/t$ and $|\Phi^{(3)}|$ is bounded, so their $L^4[U,2U]$ norms are $\ll U^{1/4}$. Minkowski's inequality in $L^4[U,2U]$ combines the five pieces:

$$\big\|\Phi_m - S - \tfrac{\varpi}{\pi}\big\|_4 \le \Big\|S-\frac{1}{\pi}\operatorname{Im}\sum_{p<Y}p^{-1/2-it}\Big\|_4 + \frac{1}{\pi}\|\mathscr{D}_1\|_4 + \frac{1}{2\pi}\|\mathscr{D}_2\|_4 + \|\Phi^{(3)}\|_4 + \frac{1}{\pi}\|\varpi\|_4 \ll U^{1/4}. \tag{7.12}$$

Since $\mathsf{k}=2$ and $\alpha_{\mathrm{F}}=\frac{1}{5}$ are fixed, and $m\in\{m_j,m_{j+1}\}$ enters only through $m\le\sqrt{2U}$, all the implied constants are absolute. □

In the present argument, the number of primes is dictated by the fourth moment. In the proof above, the remainder in Theorem 3.9 for $\mathscr{D}_1^2$ is $\sum_n n|\mathfrak{c}_n|^2=\sum_n d_1(n)^2$, which lies between $\#\{p\in[Y,X]\}^2$ and $2m^2$. With this expansion and this direct application of Theorem 3.9, it is $O(U)$ exactly when $m=O(\sqrt{U})$, and the same holds for $\mathscr{D}_2^2$. This is why the primes are counted by number and not by size: the bound depends on m, not on $p_m\asymp\sqrt{U}\log U$. The restriction comes from the method. The present argument does not exclude the use of a larger number of primes.

7.4 Proof of Theorem 1.5

Finiteness, order and the prime budget are Proposition 7.1. Let $T\ge T_0$, $a=T/2$, $b=3T$, and $k\in\mathcal{K}_T$. By (1.1), Theorem 3.1 and Lemma 4.1, there is an absolute constant $C_0\ge 1$ with $|N(t)-N_0(t)|\le C_0\log t$ for $t\ge T/4$ and $T\ge T_0$. Since $k>N(T)$, for $0<t\le a$ it follows, by (7.2) and the monotonicity of N_0 on $[2\pi,\infty)$ (and $N_0\le\frac{7}{8}$ on $(0,2\pi]$),

$$\mathcal{F}(t)\le \max\big(\tfrac{7}{8},N_0(\tfrac{T}{2})\big)+\tfrac{1}{4}\sqrt{T} < N_0(T)-C_0\log T\le N(T)<k-\tfrac{1}{2}, \tag{7.13}$$

because $N_0(T)-N_0(T/2)\ge\frac{T}{2}\mathcal{L}(\frac{T}{2})\gg T\log T$. For $t\ge b$, similarly, $\mathcal{F}(t)\ge N_0(t)-\frac{1}{4}\sqrt{2t}\ge N_0(3T)-\frac{1}{4}\sqrt{6T} > N_0(2T)+C_0\log(2T)\ge N(2T)\ge k$ (the function $N_0(t)-\frac{1}{4}\sqrt{2t}$ increases for $t\ge 2\pi e$). Hence $\tau_k^{\mathrm{C}}=a+\int_a^b \mathbf{1}_{\{\mathcal{F}<k-1/2\}}dt$, and $\gamma_k\in(a,b]$.

Theorem 7.3 with $f=\mathcal{F}$ applies to the indices $k\in\mathcal{K}_T$, a subset of $\{k:a<\gamma_k\le b\}$. The gaps D_ℓ, $\ell_0\le\ell\le\ell_1$, have $\gamma_\ell\in(T/4,3T]$: $\gamma_\ell\le\gamma_{\ell_1}\le b=3T$, and $\gamma_{\ell_0}>T/4$ because, by Theorem 3.1, there is ℓ_* with $D_\ell<1$ for all $\ell\ge\ell_*$, and for $T\ge T_0$ one has $\ell_0=N(T/2)\ge\ell_*$, so $\gamma_{\ell_0}>\gamma_{\ell_0+1}-1>T/2-1\ge T/4$. Lemma 7.5 on the four dyadic intervals $(T',2T']$, $T'=T/4,T/2,T,2T$, which cover $(T/4,4T]$, gives $\sum_{\ell_0\le\ell\le\ell_1}D_\ell^3\ll T/\log^2 T$. The interval $[a,b]$ is covered by at most four blocks $[2^j,2^{j+1}]$ with $T/4\le 2^j\le 3T$, so Lemma 7.6 gives

$\int_a^b (1+|\mathcal{F}-N|)^4 \ll T$. By (7.5),

$$\sum_{k\in\mathcal{K}_T} (\tau_k^{\mathrm{C}} - \gamma_k)^2 \ll \Big(\frac{T}{\log^2 T}\Big)^{1/2} T^{1/2} = \frac{T}{\log T}. \tag{7.14}$$

For the mean-square bound, (1.1) with $S = O(\log T)$ (Theorem 3.1) and $\varpi = O(1)$ (Lemma 4.1) gives

$$\begin{aligned} \#\mathcal{K}_T = N(2T) - N(T) &= N_0(2T) - N_0(T) + O(\log T) \\ &= \frac{T}{2\pi}\big(\log T + 2\log 2 - \log(2\pi e)\big) + O(\log T) \ge \frac{T\log T}{8\pi} \qquad (T \ge T_0), \end{aligned} \tag{7.15}$$

because $N_0(2T) - N_0(T) = \frac{2T}{2\pi}\log\frac{2T}{2\pi e} - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ and $2\log 2 - \log(2\pi e) = -1.45\ldots$. Hence the mean of $(\tau_k^{\mathrm{C}} - \gamma_k)^2$ over $\mathcal{K}_T$ is at most $8\pi C/\log^2 T$, and $C' = 8\pi C$. □

Proof of Corollary 1.6. Chebyshev's inequality applied to (1.13). □

Remark 7.7. The uncorrected sequence τ_k is, for $k \ge 2$, the case $\Phi = 0$. Then q_f is governed by S, and (7.5), with Selberg's fourth moment (Theorem 3.11 with $\mathsf{k} = 2$, in dyadic form) in place of Lemma 7.6 and with the window $[T/2, 3T]$ localized exactly as in the proof of Theorem 1.5, only yields $\sum_{k\in\mathcal{K}_T}(\tau_k - \gamma_k)^2 \ll T\log\log T/\log T$. This is the right order: Theorem 1.7 gives the asymptotic equality, by a route that does not pass through the transfer inequality. The same value has a reading in terms of Gram points. If Δ_k^{G} is the integer with $g_{k+\Delta_k^{\mathrm{G}}-2} < \gamma_k \le g_{k+\Delta_k^{\mathrm{G}}-1}$ (g_m the Gram points, $\vartheta(g_m) = m\pi$), then $\vartheta(\gamma_k)/\pi \in (k + \Delta_k^{\mathrm{G}} - 2, k + \Delta_k^{\mathrm{G}} - 1]$ and $\vartheta/\pi = N_0 - 1 + \varpi/\pi$ give $|s_k + \Delta_k^{\mathrm{G}}| < 1$ for $\gamma_k \ge 14$. Selberg's moments of Δ_k^{G} over short ranges of k, in the form of Korolev [23, Thm. 4.2], are therefore consistent with Proposition 8.4: both give $\langle s_k^2\rangle_{\mathcal{K}_T} \sim \frac{1}{2\pi^2}\log\log T$, the former through the Gram intervals and the latter through the exact identity (8.5). The Euler phases of finitely many primes in Theorem 1.5 remove the factor $\log\log T$.

8 The mean square of $\tau_k - \gamma_k$

In this section F, u_k, Ψ_j, k_1 and k_2 are local. The symbols s_k, $G = N - N_0 = S + \varpi/\pi$ and $\mathcal{L} = N_0'$ are those of Section 6, and $\mathcal{K}_T$, $\langle\cdot\rangle_{\mathcal{K}_T}$ those of Theorem 1.5. Section 6 compared τ_k with γ_k one index at a time. Here the comparison is in mean square, and it rests on an identity that turns a sum over the imaginary parts into an integral of the counting function. Nothing about the zeros is assumed.

Theorem 8.1 (Exact defect identity)**.** *Let $1 \le k_1 < k_2$ be indices, let F be a real absolutely continuous function on a compact interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, and put $u_k = k - \frac12 - F(\gamma_k)$ for $k_1 \le k \le k_2$. For every integer $j \ge 1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \big(N(t) - F(t)\big)^{2j} F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(u_k) + \frac{(\frac12 + u_{k_1})^{2j+1} + (\frac12 - u_{k_2})^{2j+1}}{2j+1}, \tag{8.1}$$

where

$$\Psi_j(u) = \frac{(\frac12+u)^{2j+1} + (\frac12-u)^{2j+1}}{2j+1} = u^{2j} + \sum_{l=0}^{j-1} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1}\, u^{2l}. \tag{8.2}$$

In particular, since $\Psi_1(u) = u^2 + \frac{1}{12}$,

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \big(N(t) - F(t)\big)^2 F'(t)\,dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac12+u_{k_1})^3 + (\frac12 - u_{k_2})^3}{3}. \tag{8.3}$$

Proof. Existence of the integral. On $[\gamma_{k_1}, \gamma_{k_2}]$ the function N is a step function with finitely many jumps, since $N(\gamma_{k_2})$ is finite, F is bounded, and F' is integrable. Hence the integrand is integrable.

One gap. Fix k with $k_1 \le k < k_2$. Suppose first that $\gamma_k < \gamma_{k+1}$. Then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly k imaginary parts counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is absolutely continuous, as the composition of the C^1 function $x \mapsto -(k-x)^{2j+1}/(2j+1)$ with F, and its derivative is $(k - F(t))^{2j} F'(t)$ almost everywhere; no monotonicity of F is needed. By the fundamental theorem of calculus for absolutely continuous functions,

$$\begin{aligned}\int_{\gamma_k}^{\gamma_{k+1}} \big(k - F(t)\big)^{2j} F'(t)\, dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2} + u_k)^{2j+1} + (\frac{1}{2} - u_{k+1})^{2j+1}}{2j+1},\end{aligned} \tag{8.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. Now let $\gamma_k = \gamma_{k+1}$. The left side of (8.4) is 0. On the other hand $u_{k+1} = u_k + 1$, so the right side equals $\big((\frac{1}{2} + u_k)^{2j+1} + (-\frac{1}{2} - u_k)^{2j+1}\big)/(2j+1) = 0$. So (8.4) holds for every k, and the values of N at the finitely many imaginary parts do not affect the integrals.

Summation. Sum (8.4) over $k_1 \le k < k_2$, a finite sum. Each k with $k_1 < k < k_2$ contributes $(\frac{1}{2} + u_k)^{2j+1}$ from the gap starting at γ_k and $(\frac{1}{2} - u_k)^{2j+1}$ from the gap ending at γ_k, that is, $\Psi_j(u_k)$. The unmatched terms are $(\frac{1}{2} + u_{k_1})^{2j+1}$ and $(\frac{1}{2} - u_{k_2})^{2j+1}$. This is (8.1). In the binomial expansion of $(\frac{1}{2} + u)^{2j+1} + (\frac{1}{2} - u)^{2j+1}$ the odd powers of u cancel and u^{2l} has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (8.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. For $j = 1$, $\Psi_1(u) = u^2 + \frac{1}{12}$, and (8.3) follows. □

Remark 8.2. $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j}\, dx$ is the mean of x^{2j} over the unit interval centered at u, and $\frac{1}{12} = \int_{-1/2}^{1/2} x^2\, dx$. The constant is associated with each unit increment in the enumeration counting multiplicities; repeated imaginary parts are handled by the degenerate-gap cancellation in the proof. When F is not monotone, $N - F$ need not run through an interval of length one between consecutive zeros, and the identity remains exact with the integral counted with sign.

Corollary 8.3. *For $1 \le k_1 < k_2$,*

$$\sum_{k=k_1+1}^{k_2-1} s_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} G(t)^2 \mathcal{L}(t)\, dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + s_{k_1})^3 + (\frac{1}{2} - s_{k_2})^3}{3}. \tag{8.5}$$

Proof. Take $F = N_0$ in Theorem 8.1 with $j = 1$. The function N_0 is C^∞ on $(0, \infty)$ with $N_0' = \mathcal{L}$, $u_k = k - \frac{1}{2} - N_0(\gamma_k) = s_k$, and $N - N_0 = G$. □

Proposition 8.4. *As $T \to \infty$,*

$$\langle s_k^2 \rangle_{\mathcal{K}_T} = \frac{\log\log T}{2\pi^2}\big(1 + o(1)\big). \tag{8.6}$$

Proof. Let $T \ge T_0$, $k_1 = N(T)$ and $k_2 = N(2T) + 1$. Since $\gamma_k \le T$ exactly when $k \le N(T)$, and likewise at $2T$, the indices k with $k_1 < k < k_2$ are exactly those of $\mathcal{K}_T$, and $k_2 - k_1 - 1 = \#\mathcal{K}_T$. By Theorem 3.1 (Littlewood's zero-gap theorem, as in [39, Thm. 9.12]) the gaps $\gamma_{\ell+1} - \gamma_\ell$ tend to 0, so for $T \ge T_0$ one has $T - 1 \le \gamma_{k_1} \le T$ and $2T < \gamma_{k_2} \le 2T + 1$: indeed $\gamma_{k_1} \le T < \gamma_{k_1+1} \le \gamma_{k_1} + 1$, and $\gamma_{k_2-1} \le 2T < \gamma_{k_2} \le \gamma_{k_2-1} + 1$. Apply Corollary 8.3 and estimate the three terms on the right of (8.5).

The integral. By Theorem 3.1 and Lemma 4.1, $|G(t)| \le |S(t)| + |\varpi(t)|/\pi \ll \log t$ for $t \ge 2$, and $\mathcal{L}(t) \ll \log t$. The parts of the integral over $[\gamma_{k_1}, T]$ and over $[2T, \gamma_{k_2}]$, two intervals of

length at most 1, are therefore $O(\log^3 T)$. On $[T,2T]$ write $G^2 = S^2 + 2S\varpi/\pi + \varpi^2/\pi^2$. Since $|\varpi(t)| \le 0.53/t$,

$$\int_T^{2T} \Big|2S\frac{\varpi}{\pi} + \frac{\varpi^2}{\pi^2}\Big|\mathcal{L}\,dt \ll \frac{\log T}{T}\int_T^{2T}\big(|S(t)|+1\big)\,dt \ll \log^2 T. \tag{8.7}$$

Next, $0 \le \mathcal{L}(t) - \mathcal{L}(T) = \frac{1}{2\pi}\log\frac{t}{T} \le \frac{\log 2}{2\pi}$ on $[T,2T]$, so

$$\int_T^{2T} S^2\mathcal{L}\,dt = \mathcal{L}(T)\int_T^{2T} S^2\,dt + O\Big(\int_T^{2T} S^2\,dt\Big). \tag{8.8}$$

Theorem 3.11 with $\mathsf{k}=1$ gives $\int_0^T S^2\,dt = \frac{T}{2\pi^2}\log\log T\,(1+o(1))$. Write $\int_0^X S^2\,dt = \frac{X}{2\pi^2}\log\log X + R(X)$ with $R(X) = o(X\log\log X)$. Subtracting the integrals over $[0,2T]$ and $[0,T]$, using $R(2T)-R(T) = o(T\log\log T)$ and $\log\log 2T - \log\log T = O(1/\log T)$,

$$\int_T^{2T} S(t)^2\,dt = \frac{T}{2\pi^2}\log\log T\,\big(1+o(1)\big). \tag{8.9}$$

Altogether, as $\mathcal{L}(T)\to\infty$,

$$\begin{aligned}\int_{\gamma_{k_1}}^{\gamma_{k_2}} G^2\mathcal{L}\,dt &= \frac{T\mathcal{L}(T)}{2\pi^2}\log\log T\,\big(1+o(1)\big) + O(T\log\log T) + O(\log^3 T)\\ &= \frac{T\mathcal{L}(T)}{2\pi^2}\log\log T\,\big(1+o(1)\big).\end{aligned} \tag{8.10}$$

The other two terms. The second term is $\#\mathcal{K}_T/12$. For the third, Lemma 6.1 gives $|s_k| \le \max(|G_k^-|,|G_k^+|) \ll \log\gamma_k$, so the two cubes are $O(\log^3 T)$.

The size of $\mathcal{K}_T$. By (1.1), Theorem 3.1 and Lemma 4.1,

$$\begin{aligned}\#\mathcal{K}_T = N(2T)-N(T) &= N_0(2T)-N_0(T)+O(\log T)\\ &= T\mathcal{L}(T) + \frac{T}{2\pi}(2\log 2-1) + O(\log T) = T\mathcal{L}(T)\Big(1+O\Big(\frac{1}{\log T}\Big)\Big),\end{aligned} \tag{8.11}$$

because $N_0(2T)-N_0(T) = \frac{T}{2\pi}\big(\log\frac{T}{2\pi} - 1 + 2\log 2\big)$.

Conclusion. Dividing (8.5) by $\#\mathcal{K}_T$,

$$\langle s_k^2\rangle_{\mathcal{K}_T} = \frac{\log\log T}{2\pi^2}\big(1+o(1)\big) - \frac{1}{12} + O\Big(\frac{\log^3 T}{T\log T}\Big) = \frac{\log\log T}{2\pi^2}\big(1+o(1)\big), \tag{8.12}$$

since $\log\log T\to\infty$. □

Proof of Theorem 1.7. By the mean value theorem, as in the proof of Corollary 1.4, $s_k = N_0(\tau_k) - N_0(\gamma_k) = \mathcal{L}(\xi_k)(\tau_k-\gamma_k)$ for some ξ_k between γ_k and τ_k. By the proof of Corollary 1.4(a), $|\tau_k-\gamma_k| \le C$ for all $k\ge 1$, with C absolute. For $k\in\mathcal{K}_T$ one has $\gamma_k\in(T,2T]$, hence $\xi_k\in[T-C,2T+C]$ and $\mathcal{L}(\xi_k) = \mathcal{L}(T)+O(1) = \mathcal{L}(T)\big(1+O(1/\log T)\big)$, uniformly in $k\in\mathcal{K}_T$. Therefore

$$(\tau_k-\gamma_k)^2 = \frac{s_k^2}{\mathcal{L}(\xi_k)^2} = \frac{s_k^2}{\mathcal{L}(T)^2}\Big(1+O\Big(\frac{1}{\log T}\Big)\Big) \qquad (k\in\mathcal{K}_T), \tag{8.13}$$

and, averaging over $\mathcal{K}_T$ and applying Proposition 8.4,

$$\begin{aligned}\big\langle(\tau_k-\gamma_k)^2\big\rangle_{\mathcal{K}_T} &= \frac{\langle s_k^2\rangle_{\mathcal{K}_T}}{\mathcal{L}(T)^2}\Big(1+O\Big(\frac{1}{\log T}\Big)\Big)\\ &= \frac{4\pi^2}{\log^2(T/2\pi)}\cdot\frac{\log\log T}{2\pi^2}\big(1+o(1)\big) = \frac{2\log\log T}{\log^2 T}\big(1+o(1)\big),\end{aligned} \tag{8.14}$$

because $\log(T/2\pi) = \log T\,(1+O(1/\log T))$. Finally $\sum_{k\in\mathcal{K}_T}(\tau_k-\gamma_k)^2 = \#\mathcal{K}_T\cdot\langle(\tau_k-\gamma_k)^2\rangle_{\mathcal{K}_T}$ and $\#\mathcal{K}_T \asymp T\log T$, which gives $\sum_{k\in\mathcal{K}_T}(\tau_k-\gamma_k)^2 \asymp T\log\log T/\log T$. □

Remark 8.5. Theorem 1.7 is an asymptotic equality, so the factor $\log\log T$ is a property of the sequence τ_k and not of the method. Theorem 1.5 gives $\langle(\tau_k^{\mathrm{C}}-\gamma_k)^2\rangle_{\mathcal{K}_T}\le C'/\log^2 T$ for the points corrected by the Euler phases of the primes: the primes remove this factor from the proved mean-square upper order. With $F=N_0$ and the higher moments of Theorem 3.11, the identity (8.1) gives in the same way all the even moments of s_k (Proposition 8.6). Only the case $j=1$ is used here.

Proposition 8.6 (Higher moments of s_k)**.** *For every fixed integer $j\ge 1$, $\langle s_k^{2j}\rangle_{\mathcal{K}_T}\sim \frac{(2j)!}{j!(2\pi)^{2j}}(\log\log T)^j$ as $T\to\infty$.*

Proof. The proof is by induction on j, the case $j=1$ being Proposition 8.4. Take $k_1=N(T)$ and $k_2=N(2T)+1$, so that the sum in (8.1) runs over $\mathcal{K}_T$, and $u_k=s_k$. The boundary terms are $O(\log^{2j+1}T)$, because $s_k\ll\log\gamma_k$ by Lemma 6.1. On the left side, the intervals $[\gamma_{k_1},T]$ and $[2T,\gamma_{k_2}]$ have length $o(1)$ by Theorem 3.1, and the integrand is $O(\log^{2j+1}T)$ there. On $[T,2T]$, $\mathcal{L}(t)=\mathcal{L}(T)(1+O(1/\log T))$ and $G=S+\varpi/\pi$ with $|\varpi|/\pi\le 0.53/(\pi t)$ (Lemma 4.1). By Hölder's inequality and Theorem 3.11, $\int_T^{2T}|S|^{2j-i}t^{-i}\,dt\ll T^{1-i}(\log\log T)^{j-i/2}$ for $1\le i\le 2j$, so $\int_T^{2T}G^{2j}\,dt=\int_T^{2T}S^{2j}\,dt+o(T)$. Theorem 3.11 at $2T$ and at T gives $\int_T^{2T}S^{2j}\,dt\sim\frac{(2j)!}{j!(2\pi)^{2j}}T(\log\log T)^j$. Hence the left side of (8.1) is $\mathcal{L}(T)\,T\,\frac{(2j)!}{j!(2\pi)^{2j}}(\log\log T)^j(1+o(1))$. On the right side, $\Psi_j(s_k)=s_k^{2j}+\sum_{l=0}^{j-1}\binom{2j+1}{2l}\frac{4^{l-j}}{2j+1}s_k^{2l}$ by (8.2), and the induction hypothesis gives $\sum_{k\in\mathcal{K}_T}s_k^{2l}\ll\#\mathcal{K}_T(\log\log T)^l=o(\#\mathcal{K}_T(\log\log T)^j)$ for $l<j$. Since $\#\mathcal{K}_T=T\mathcal{L}(T)(1+O(1/\log T))$ (proof of Proposition 8.4), dividing by $\#\mathcal{K}_T$ gives the claim. □

9 Further consequences of ordinal localization

The results of this section follow from the estimates already established in this article. The final theorem, a certificate for simple critical-line zeros, holds for any real test function evaluated at the imaginary parts. No conclusion about RH or a positive proportion of simple zeros is inferred from the bounds presently available.

Proposition 9.1 (Second moment and asymptotic correlation of the arithmetic correction)**.** *Write*

$$\delta_k=\gamma_k-\tau_k,\qquad b_k=\tau_k^{\mathrm{C}}-\tau_k,\qquad \varepsilon_k=\gamma_k-\tau_k^{\mathrm{C}},\qquad n_T=\#\mathcal{K}_T,\tag{9.1}$$

where $\mathcal{K}_T=\{k:T<\gamma_k\le 2T\}$, as above. Then, without RH,

$$\frac{1}{n_T}\sum_{k\in\mathcal{K}_T}b_k^2=\frac{2\log\log T}{\log^2 T}\,(1+o(1)),\tag{9.2}$$

and

$$\frac{\sum_{k\in\mathcal{K}_T}\delta_k b_k}{\left(\sum_{k\in\mathcal{K}_T}\delta_k^2\right)^{1/2}\left(\sum_{k\in\mathcal{K}_T}b_k^2\right)^{1/2}}=1-O\!\left(\frac{1}{\log\log T}\right).\tag{9.3}$$

Furthermore,

$$\frac{1}{n_T}\sum_{k\in\mathcal{K}_T}\left|e^{i(\gamma_k-\tau_k)}-e^{i(\tau_k^{\mathrm{C}}-\tau_k)}\right|^2\ll\frac{1}{\log^2 T}.\tag{9.4}$$

Proof. The exact relation $\delta_k=b_k+\varepsilon_k$ yields $\|\delta-b\|_2=\|\varepsilon\|_2$ on $\mathcal{K}_T$. Theorem 1.5 gives $\|\varepsilon\|_2^2\ll n_T/\log^2 T$, whereas Theorem 1.7 gives $\|\delta\|_2^2\sim 2n_T\log\log T/\log^2 T$. By the reverse triangle inequality,

$$\big|\|b\|_2-\|\delta\|_2\big|\le\|\varepsilon\|_2=O\!\left(\frac{\|\delta\|_2}{\sqrt{\log\log T}}\right).\tag{9.5}$$

Put $\mathsf{D}=\|\delta\|_2$, $\mathsf{B}=\|b\|_2$ and $\mathsf{E}=\|\varepsilon\|_2$. Since $\mathsf{E}/\mathsf{D}=O((\log\log T)^{-1/2})=o(1)$, the preceding inequality gives $|\mathsf{B}/\mathsf{D}-1|\le\mathsf{E}/\mathsf{D}=o(1)$. Hence $\mathsf{B}^2=\mathsf{D}^2(1+o(1))$, and Theorem 1.7 proves

(9.2). To prove the alignment, use the same D, B and $\mathsf{E} = \|\delta - b\|_2$. The polarization identity gives

$$1 - \frac{\langle \delta, b \rangle}{\mathsf{DB}} = \frac{\mathsf{E}^2 - (\mathsf{D} - \mathsf{B})^2}{2\mathsf{DB}} \le \frac{\mathsf{E}^2}{2\mathsf{DB}} = O\Big(\frac{1}{\log\log T}\Big), \tag{9.6}$$

using $\mathsf{B}/\mathsf{D} \to 1$. Finally, $|e^{iv} - e^{iw}| \le |v - w|$ for real v, w; apply it with $v = \delta_k$, $w = b_k$ and sum the squares. □

Lemma 9.2 (Control of the derivative and of the endpoint phases)**.** *For T sufficiently large, the continuous, piecewise C^1 function $\mathcal{F} = N_0 + \Phi$ satisfies*

$$\sup_{T/2 \le t \le 3T} |\Phi(t)| \ll T^{1/4}, \qquad \int_{T/2}^{3T} |\mathcal{F}'(t)|^2 \, dt \ll T \log^2 T. \tag{9.7}$$

Derivatives are interpreted almost everywhere at the finitely many dyadic junctions.

Proof. On a dyadic interval $[U, 2U]$, the two prime counts in the definition of Φ are at most $\sqrt{2U}$. For every prime p, $\sum_{r \ge 1} 1/(rp^{r/2}) \le C_0 p^{-1/2}$ with an absolute C_0. Since $p_\nu \ge \nu + 1$, the triangle inequality gives $|\Phi_m(t)| \ll \sum_{\nu \le m} \nu^{-1/2} \ll \sqrt{m} \ll U^{1/4}$. Convex interpolation preserves this bound.

Differentiate Φ_m termwise. For a fixed m the remaining series over r and its derivative converge uniformly, because $\sum_{r \ge 1} (\log p) p^{-r/2} < \infty$ for each prime p:

$$\Phi_m'(t) = -\frac{1}{\pi} \sum_{\nu \le m} \sum_{r \ge 1} \frac{\log p_\nu}{p_\nu^{r/2}} \cos(rt \log p_\nu). \tag{9.8}$$

For $r = 1$, the cosine sum is the real part of the complex Dirichlet polynomial with frequencies p and coefficients $(\log p) p^{-1/2}$. The Montgomery–Vaughan mean-value inequality, Theorem 3.9, and $|\operatorname{Re} z|^2 \le |z|^2$ show that its integral square on $[U, 2U]$ is bounded by

$$\ll U \sum_{p \le p_m} \frac{\log^2 p}{p} + \sum_{p \le p_m} \log^2 p \ll U \log^2 U. \tag{9.9}$$

Here $p_m \ll m \log(2m) \ll \sqrt{U} \log U$ follows from Theorem 3.10 as used in Lemma 7.6. For $r = 2$, the cosine sum is likewise the real part of a complex Dirichlet polynomial with frequencies p^2, so the same inequality applies:

$$\int_U^{2U} \Big| \sum_{p \le p_m} \frac{\log p}{p} \cos(2t \log p) \Big|^2 dt \ll U \sum_p \frac{\log^2 p}{p^2} + \sum_{p \le p_m} \log^2 p \ll U \log^2 U. \tag{9.10}$$

The sum over $r \ge 3$ is uniformly bounded by absolute convergence of $\sum_p (\log p) p^{-3/2}$. On $[U, 2U]$, differentiating the interpolation weights adds $U^{-1}(\Phi_{m_{j+1}} - \Phi_{m_j}) = O(U^{-3/4})$, so the same estimate holds for Φ'. All these bounds hold uniformly for $m \in \{m_j, m_{j+1}\}$, since only $m \le \sqrt{2U}$ is used. Finally, $N_0'(t) = (2\pi)^{-1} \log(t/2\pi) = O(\log T)$ on $[T/2, 3T]$. The interval intersects a uniformly bounded number of dyadic blocks; summing their bounds proves (9.7). □

Theorem 9.3 (The corrected ordinal defect at the actual zeros)**.** *For $k \in \mathcal{K}_T$ define*

$$d_k^{\mathrm{C}} = k - \frac{1}{2} - \mathcal{F}(\gamma_k) = \Big(k - \frac{1}{2} - N_0(\gamma_k)\Big) - \Phi(\gamma_k). \tag{9.11}$$

Then, without RH,

$$\frac{1}{\#\mathcal{K}_T} \sum_{k \in \mathcal{K}_T} |d_k^{\mathrm{C}}|^2 \ll 1. \tag{9.12}$$

Proof. Theorem 8.1 applies to absolutely continuous F. The function $\mathcal{F}$ is absolutely continuous on every compact interval: it is continuous and piecewise C^1 with finitely many dyadic junctions. Take $k_1 = N(T)$ and $k_2 = N(2T) + 1$ in (8.3), with $F = \mathcal{F}$. The interior indices are exactly $\mathcal{K}_T$ and, as in the proof of Proposition 8.4, the two endpoints lie in $[T - 1, 2T + 1]$ for large T. The terms $|k - \frac{1}{2} - N_0(\gamma_k)|$ at those endpoints are $O(\log T)$ by Lemma 6.1, and Lemma 9.2 gives $|\Phi(\gamma_k)| \ll T^{1/4}$. Using $(a + b)^3 \le 4(a^3 + b^3)$ for $a, b \ge 0$, the cubic boundary terms are $O(T^{3/4} + \log^3 T) = o(T \log T)$.

Cauchy–Schwarz, Lemmas 7.6 and 9.2, and a covering by finitely many dyadic intervals give

$$\left| \int_{\gamma_{k_1}}^{\gamma_{k_2}} (N(t) - \mathcal{F}(t))^2 \mathcal{F}'(t)\, dt \right| \le \left(\int_{T-1}^{2T+1} |N - \mathcal{F}|^4 dt \right)^{1/2} \left(\int_{T-1}^{2T+1} |\mathcal{F}'|^2 dt \right)^{1/2} \ll T \log T. \tag{9.13}$$

The integral need not be nonnegative because $\mathcal{F}'$ may change sign. Taking absolute values in the quadratic identity yields

$$\sum_{k \in \mathcal{K}_T} |d_k^{\mathrm{C}}|^2 \le \left| \int_{\gamma_{k_1}}^{\gamma_{k_2}} (N(t) - \mathcal{F}(t))^2 \mathcal{F}'(t)\, dt \right| + \frac{\#\mathcal{K}_T}{12} + O(T^{3/4} + \log^3 T) \ll T \log T. \tag{9.14}$$

Division by $\#\mathcal{K}_T \asymp T \log T$ proves (9.12). □

Corollary 9.4 (Second moment of the prime phases at the imaginary parts)**.** *With $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ as in Proposition 8.4, one has, without RH,*

$$\frac{1}{\#\mathcal{K}_T} \sum_{k \in \mathcal{K}_T} \Phi(\gamma_k)^2 \sim \frac{\log\log T}{2\pi^2}, \tag{9.15}$$

and

$$\frac{1}{\#\mathcal{K}_T} \sum_{k \in \mathcal{K}_T} s_k \Phi(\gamma_k) \sim \frac{\log\log T}{2\pi^2}. \tag{9.16}$$

Proof. The relation $s_k = \Phi(\gamma_k) + d_k^{\mathrm{C}}$ is exact. Theorem 9.3 gives $\langle |d_k^{\mathrm{C}}|^2 \rangle_{\mathcal{K}_T} = O(1)$, while Proposition 8.4 gives $\langle s_k^2 \rangle_{\mathcal{K}_T} \sim (2\pi^2)^{-1} \log\log T$. The reverse triangle inequality in the Euclidean norm therefore gives $\langle \Phi(\gamma_k)^2 \rangle_{\mathcal{K}_T} = \langle s_k^2 \rangle_{\mathcal{K}_T} + O(\sqrt{\log\log T} + 1)$, proving (9.15). Finally, $\langle s_k \Phi(\gamma_k) \rangle = \langle s_k^2 \rangle - \langle s_k d_k^{\mathrm{C}} \rangle$, and Cauchy–Schwarz makes the last term $O(\sqrt{\log\log T})$. □

Theorem 9.5 (A deterministic certificate for simple critical-line zeros)**.** *In this theorem v_k is a local symbol, unrelated to v_k in the proof of Theorem 7.3. Let $\mathsf{F}_{\mathrm{test}}$ be any real-valued function defined at the imaginary parts, let $j \ge 1$ be an integer, and set*

$$v_k = k - \frac{1}{2} - \mathsf{F}_{\mathrm{test}}(\gamma_k), \qquad \mathcal{K}_T = \{k : T < \gamma_k \le 2T\}. \tag{9.17}$$

Let $N_{\mathrm{simple},1/2}(T, 2T)$ count the simple zeros on the critical line in this block, with no repeated imaginary parts. Then

$$N_{\mathrm{simple},1/2}(T, 2T) \ge \#\mathcal{K}_T - \sum_{k \in \mathcal{K}_T} (2v_k)^{2j}. \tag{9.18}$$

The right side is positive only if the defects v_k are individually small; the bounds of this article do not achieve this (Remark 9.6).

Proof. At an imaginary part γ of total multiplicity $m \ge 2$, the consecutive indices can be written as $k, k + 1, \dots, k + m - 1$. Because $\mathsf{F}_{\mathrm{test}}$ has the same value at all these indices, $v_{k+i} = v_k + i$. Among all m-tuples of consecutive real numbers spaced by one,

$$\min_{a \in \mathbb{R}} \sum_{i=0}^{m-1} (a + i)^2 = \sum_{i=0}^{m-1} \left(i - \frac{m-1}{2} \right)^2 = \frac{m(m^2 - 1)}{12}. \tag{9.19}$$

Hence $\frac{1}{m}\sum_{i=0}^{m-1}(2v_{k+i})^2 \geq (m^2-1)/3 \geq 1$. The monotonicity of normalized ℓ^p norms, for $2j \geq 2$, gives

$$\left(\frac{1}{m}\sum_{i=0}^{m-1}|2v_{k+i}|^{2j}\right)^{1/(2j)} \geq \left(\frac{1}{m}\sum_{i=0}^{m-1}|2v_{k+i}|^{2}\right)^{1/2} \geq 1. \tag{9.20}$$

Hence $\sum_{i=0}^{m-1}(2v_{k+i})^{2j} \geq m$. The functional equation and conjugation symmetry give $\rho = \beta + i\gamma \mapsto 1-\overline{\rho} = 1-\beta+i\gamma$. If $\beta \neq \frac{1}{2}$ these are distinct zeros with the same imaginary part. Thus every imaginary part of total multiplicity one corresponds to a simple zero on the critical line. The indices of $\mathcal{K}_T$ form a disjoint union of complete multiplicity blocks. The number of indices belonging to blocks of multiplicity at least two is therefore at most $\sum_{k\in\mathcal{K}_T}(2v_k)^{2j}$. Subtracting this quantity from $\#\mathcal{K}_T$ proves the claim. □

Remark 9.6. It is a certificate, not an unconditional positive-proportion result: Theorem 9.3 bounds the second moment of d_k^{C} but does not provide a sufficiently small explicit constant for (9.18) to be positive.

10 Conclusion

The principal conclusion is the unconditional, fully ordinally indexed construction τ_k^{C} from finitely many Euler phases. The sublevel measures remain strictly increasing even though the underlying prime-corrected counting function need not be monotone. Theorem 1.5 proves a root-mean-square localization error $O(1/\log T)$ over the complete list of zeros, including multiplicities.

The exact defect identity provides a separate unconditional asymptotic: the uncorrected Riemann–von Mangoldt points τ_k have mean-square error $2\log\log T/\log^2 T\,(1+o(1))$. The arithmetic correction removes the divergent factor $\log\log T$ from the proved upper bound of the mean square, which is $O(1/\log^2 T)$ for the corrected points; no asymptotic of the corrected error is claimed. Proposition 9.1 shows that the correction has the same leading second moment and is asymptotically aligned with the actual displacement vector.

Theorem 9.3 and Corollary 9.4 demonstrate the related bounded ordinal defect and the leading second-moment asymptotic of the prime phases at the imaginary parts.

Independently, the argument estimates yield $\limsup_k |\tau_k - \gamma_k| \leq 0.60832$ and the explicit bound $|S(T)| \leq 0.09783\log T + 0.7214\log\log T + 10.6$ for $T \geq e$. The improved leading coefficient comes essentially from pointwise reflection by the functional equation on the left of the critical line. As explained in Section 1.2, it improves the complete estimate of Bellotti–Wong only beyond the stated crossing height.

Theorem 9.5 gives a deterministic criterion for certifying simple critical-line zeros from individual ordinal defects. The available bounds do not satisfy the numerical threshold required to deduce RH, simplicity, or any stated positive proportion from this certificate.

Limitations of this work. The constant in Theorem 1.5 is not explicit, so the result gives no numerical guarantee at any finite height; it controls the mean square, not every individual imaginary part, and the explicit bound of Theorem 1.3 improves on Bellotti–Wong only for $\log T > 2425.8385\ldots$, as it trades the constant term for the leading coefficient (for comparison, (1.8) is smaller than the bound $|N(t)-N_0(t)| \leq 0.11\log t + 0.29\log\log t + 2.29 + 0.2/t_0$, $t \geq t_0 \geq e$, used by Yang [44, (4.29)] in the proof of his zero-free region, as soon as $\log t > 933.5$).

An explicit version of Theorem 1.5 would require explicit forms of Selberg's approximation and of Fujii's theorem.

A Computation of the constants

The constants $\Lambda(\mathfrak{e}_{\mathrm{B}})$, C_1, C_2, $\log(\zeta(c)/\zeta(2c))$, the constant term $C_3(100) = 10.5986\ldots$ at $T = 100$ (and the same sum with the exact differences, $10.5985\ldots$), $\Lambda(0)$, the crossing point $\log T = 2425.8385\ldots$ with [3] and the crossing point $\log t = 933.4678\ldots$ with [44, (4.29)] are computed in floating point by the following program (Python 3.13.17 with `mpmath` 1.4.1, working precision 40 decimal digits), and certified by the last program of this appendix.

It only evaluates integrals of piecewise linear functions, elementary constants and values of ζ at real points, and finds the two crossing points by bisection and by the secant method (`mpmath.findroot`).

The integrals are computed piece by piece, with breakpoints at the nodes, and also by the closed form (5.9) (function `closed_form`), for $\mathfrak{e}_{\mathrm{E}}^*$ at $(c, R) = (1.000002, 1.000005)$ and for $\mathfrak{e}_{\mathrm{B}}^*$ at $(c, R) = (1, 1)$; the program checks that the two evaluations agree.

The decimals printed by this program are floating-point values. Every constant and every numerical inequality used in the proofs is certified by the ball-arithmetic program at the end of this appendix. The output follows the program.

```
# Constants of the article (Appendix A).
# Bourgain profile and explicit profile: the angular integrals by
# quadrature and by the closed form, the constants C_1, C_2, log(zeta(c)/zeta(2c)), log zeta(sigma#),
# C_3(100) as defined in the article (with the costs 2C_1/T and 2C_2/(T log T)) and the same sum with the
     exact
# differences, Lambda(0), and the crossing points with Bellotti-Wong and with Yang's (4.29).
# Python 3.13.17, mpmath 1.4.1, 40 decimal digits.
import sys, mpmath
from mpmath import mp, mpf, quad, sqrt, log, pi, acos, cos, sin, zeta, nstr
mp.dps = 40
print('Python', sys.version.split()[0], ' mpmath', mpmath.__version__, ' mp.dps =', mp.dps)

def closed_form(nodes, c, R):
    """int_0^pi e*(c+R cos th) dth by the hinge formulas: nodes (sigma_i, e_i), sigma_0 = 1/2 < ... <
     sigma_n = 1"""
    n = len(nodes)-1
    xs = [a for a,_ in nodes]; es = [b for _,b in nodes]
    s = [(es[i+1]-es[i])/(xs[i+1]-xs[i]) for i in range(n)]            # slopes
    hinges = [(xs[i], s[i]-s[i-1]) for i in range(1,n)] + [(mpf(1), -s[n-1])]
    th_h = acos((mpf(1)/2 - c)/R)                                      # sigma = 1/2
    e0, s0 = es[0], s[0]
    I = (e0 + s0*(c-mpf(1)/2))*th_h + s0*R*sin(th_h)                   # linear part, sigma >= 1/2
    I += ((1+s0)*(mpf(1)/2-c) + e0)*(pi-th_h) + (1+s0)*R*sin(th_h)     # linear part of 1/2-sigma+e(1-
     sigma), sigma <= 1/2
    for a, coef in hinges:
        th_a = acos((a-c)/R);   I += coef*((c-a)*th_a + R*sin(th_a))          # int_0^th_h max(0, sigma-a)
        th_b = acos((1-a-c)/R); I += coef*((1-c-a)*(pi-th_b) + R*sin(th_b))   # int_th_h^pi max(0, 1-sigma-
     a)
    return I

def pl(nodes):
    """piecewise linear function through the nodes (sigma, e), increasing sigma, 0 to the right of the last
     node"""
    def e(s):
        if s >= nodes[-1][0]: return mpf(0)
        for (a,ea),(b,eb) in zip(nodes, nodes[1:]):
            if a <= s <= b: return ea + (eb-ea)*(s-a)/(b-a)
        raise ValueError(s)
    return e

def reflect(e):
    """reflected extension"""
    def es(s):
        s = mpf(s)
        if s >= 1: return mpf(0)
        if s >= mpf(1)/2: return e(s)
        if s >= 0: return mpf(1)/2 - s + e(1-s)
        return mpf(1)/2 - s
    return es

# --- Corollary 1.2: Bourgain profile via the A-process ---
```

```
k, l = mpf(13)/84, mpf(55)/84
nodesB = []
for j in range(8):
    nodesB.append((l-k, k)); k, l = k/(2*k+2), mpf(1)/2 + l/(2*k+2)
nodesB.append((mpf(1), mpf(0)))
from fractions import Fraction as Fr
kf, lf = Fr(13,84), Fr(55,84); nf=[]
for j in range(8):
    nf.append((lf-kf, kf)); kf, lf = kf/(2*kf+2), Fr(1,2)+lf/(2*kf+2)
print('Bourgain nodes (1.6):', [(str(a),str(b)) for a,b in nf])
eB = reflect(pl(nodesB))
brk = sorted(set([mpf(0)]+[1-a for a,_ in nodesB]+[a for a,_ in nodesB]+[mpf(1)/2]))
brk = [b for b in brk if 0 <= b <= 1]
A = quad(lambda s: eB(s)/sqrt(s*(2-s)), brk) / (2*pi*log(2))
print('Cor 1.2: Lambda(e_B) =', nstr(A, 20), ' Cor 1.4(a): 2*pi*Lambda(e_B) =', nstr(2*pi*A, 20))
AB_closed = closed_form(nodesB, mpf(1), mpf(1))/(2*pi*log(2))
print('Cor 1.2 by the closed form (5.9) with c = R = 1: Lambda(e_B) =', nstr(AB_closed, 20))
assert abs(AB_closed - A) < mpf(10)**-20   # the sigma-quadrature has an integrable singularity at 0

# --- explicit profile and its constants ---
nodesE = [(mpf(1)/2, mpf(27)/164)] + [(1 - mpf(m)/(2**m-2), mpf(1)/(2**m-2)) for m in range(4,10)] + [(mpf
    (1), mpf(0))]
print('explicit nodes (4.5):', [(nstr(a,8), nstr(b,8)) for a,b in nodesE])
eE = reflect(pl(nodesE))
c, R = mpf('1.000002'), mpf('1.000005')
sig1 = c + (c-mpf(1)/2)**2/R; delta = 2*c - sig1 - mpf(1)/2
D = log(R/(c-mpf(1)/2))
print('sigma1 =', nstr(sig1,15), ' delta =', nstr(delta,15), ' c-R =', nstr(c-R,5), ' D =', nstr(D,20))
# integral I = int_0^pi eE(c+R cos th) dth, split at the angles of the nodes
pts = set([mpf(0), pi])
for a,_ in nodesE + [(1-a,0) for a,_ in nodesE] + [(mpf(0),0)]:
    x = (a - c)/R
    if -1 < x < 1: pts.add(acos(x))
pts = sorted(pts)
I = quad(lambda th: eE(c + R*cos(th)), pts)
C1 = I/(2*pi*D); C2 = 1/(2*D)
print('I =', nstr(I,25)); print('C1 = I/(2 pi D) =', nstr(C1,25)); print('C2 = 1/(2D) =', nstr(C2,25))
I_closed = closed_form(nodesE, c, R)
print('I by the closed form (5.9) =', nstr(I_closed,25), ' C1 =', nstr(I_closed/(2*pi*D),25))
assert abs(I_closed - I) < mpf(10)**-25
print('log zeta(sigma1) =', nstr(log(zeta(sig1)),15), ' zeta(c) <=', nstr(c/(c-1),10), ' zeta(c)/zeta(2c)='
    , nstr(zeta(c)/zeta(2*c),12))

# --- the constant term C_3(100) ---
t0 = mpf(98)
eps_chi = (mpf(1)/3 + mpf('2.0005')/6)/t0 + mpf('0.87')/t0**2   # |R1|+|R2| + |phi1|+|phi2|
chi_fac = mp.e**eps_chi
K1 = 2*(1+mpf('84.5')/t0**2)**(1+mpf(27)/164) * (1 + (mpf('84.5')/t0**2 + pi/2)/log(t0))
K = chi_fac*K1
print('factor chi <=', nstr(chi_fac,8), '  K1 <=', nstr(K1,8), '  K = chi*K1 <=', nstr(K,8))
Kr = mpf('2.74'); assert K <= Kr and K1 <= Kr
T = mpf(100)
lzr = log(zeta(c)/zeta(2*c))
Eterm = ((640*delta-112)/(1536*(3*T-1)) + mpf(2)**-10)/2     # E/(2 pi)
C3base = C2*(lzr + log(1+3/T) + log(Kr)) + (log(zeta(sig1)) + (sig1-mpf(1)/2)/T)/pi + Eterm + mpf(1)/4
C3T = C3base + 2*C1/T + 2*C2/(T*log(T))                                  # C_3(T)
C3exact = C3base + C1*(log(T+2)-log(T)) + C2*(log(log(T+2))-log(log(T)))  # same sum, exact differences
print('log(zeta(c)/zeta(2c)) =', nstr(lzr,12), ' E/(2pi) <=', nstr(Eterm,8))
print('C_3(100) =', nstr(C3T,12), ' (with the costs 2C_1/T + 2C_2/(T log T));  with exact differences:',
    nstr(C3exact,12))
print('Lemma 4.1: 1/16+1/8+1/3 =', nstr(mpf(1)/16+mpf(1)/8+mpf(1)/3,8))
print('Cor 1.4(a): 2*pi*0.0968160064 =', nstr(2*pi*mpf('0.0968160064'),10))

# --- Lindelof profile (e=0 on [1/2,1]) ---
eL = reflect(lambda s: mpf(0))
AL = quad(lambda s: eL(s)/sqrt(s*(2-s)), [0, mpf(1)/2, 1])/(2*pi*log(2))
print('Lindelof: a_L =', nstr(AL,12), ' 2pi a_L =', nstr(2*pi*AL,12))
# --- crossing with Bellotti-Wong for S(T): 0.10076 L + min(0.24460 LL + 7.20844, 1.68845 LL + 1.50956) ---
from mpmath import findroot
def diff(L):
    L = mpf(L); BW = mpf('0.10076')*L + min(mpf('0.24460')*log(L)+mpf('7.20844'), mpf('1.68845')*log(L)+mpf
    ('1.50956'))
    ours = mpf('0.09783')*L + mpf('0.7214')*log(L) + mpf('10.6')
```

```
    return ours - BW
lo, hi = mpf(10), mpf(10**6)
for _ in range(200):
    mid = (lo+hi)/2
    if diff(mid) > 0: lo = mid
    else: hi = mid
print('crossing: (1.7) < Bellotti-Wong for log T >', nstr(hi,8))
# --- crossing with Yang's (4.29): |N - N_0| <= 0.11 log t + 0.29 log log t + 2.29 + 0.2/t_0 (t >= t_0 >= e)
     ---
yang = lambda L: mpf('0.11')*L + mpf('0.29')*log(L) + mpf('2.29') - (mpf('0.09783')*L + mpf('0.7214')*log(L
    ) + mpf('10.7'))
print('crossing: (1.8) < Yang (4.29) (without the term 0.2/t_0) for log t >', nstr(findroot(yang, 933),8))

# --- Remark: Bellotti-Wong left profile (linear on [0,1/2]) with c,R -> 1 ---
eR = pl(nodesE)
def eBWl(s):
    s = mpf(s)
    if s >= 1: return mpf(0)
    if s >= mpf(1)/2: return eR(s)
    if s >= 0: return mpf(1)/2 - s + 2*s*mpf(27)/164
    return mpf(1)/2 - s
brkE = sorted(set([mpf(0), mpf(1)/2, mpf(1)] + [a for a,_ in nodesE] + [1-a for a,_ in nodesE]))
ABW = quad(lambda s: eBWl(s)/sqrt(s*(2-s)), [b for b in brkE if 0 <= b <= 1])/(2*pi*log(2))
AEr = quad(lambda s: eE(s)/sqrt(s*(2-s)), [b for b in brkE if 0 <= b <= 1])/(2*pi*log(2))
print('limit c,R->1: BW linear-left profile =', nstr(ABW,10), ' reflected explicit profile =', nstr(AEr,10)
    )
```

```
Python 3.13.17  mpmath 1.4.1  mp.dps = 40
Bourgain nodes (1.6): [('1/2', '13/84'), ('139/194', '13/194'), ('173/207', '13/414'), ('773/854',
    '13/854'), ('820/867', '13/1734'), ('3387/3494', '13/3494'), ('1149/1169', '13/7014'), ('13921/14054',
    '13/14054')]
Cor 1.2: Lambda(e_B) = 0.096816006323848965603  Cor 1.4(a): 2*pi*Lambda(e_B) = 0.60831290843381374985
Cor 1.2 by the closed form (5.9) with c = R = 1: Lambda(e_B) = 0.096816006323848965603
explicit nodes (4.5): [('0.5', '0.16463415'), ('0.71428571', '0.071428571'), ('0.83333333', '0.033333333'),
     ('0.90322581', '0.016129032'), ('0.94444444', '0.0079365079'), ('0.96850394', '0.0039370079'),
    ('0.98235294', '0.0019607843'), ('1.0', '0.0')]
sigma1 = 1.25000275000025  delta = 0.25000124999975  c-R = -3.0e-6  D = 0.69314818055544532975
I = 0.426063763803919133874 0744
C1 = I/(2 pi D) = 0.097829231878517946338767 44
C2 = 1/(2D) = 0.721346479766175634597 9781
I by the closed form (5.9) = 0.426063763803919133874 0744  C1 = 0.097829231878517946338767 44
log zeta(sigma1) = 1.5249835586892  zeta(c) <= 500001.0  zeta(c)/zeta(2c)= 303964.594821
factor chi <= 1.006918   K1 <= 2.7166063   K = chi*K1 <= 2.7353997
log(zeta(c)/zeta(2c)) = 12.6246665092  E/(2pi) <= 0.00054053965
C_3(100) = 10.5986023483  (with the costs 2C_1/T + 2C_2/(T log T));  with exact differences: 10.5985454733
Lemma 4.1: 1/16+1/8+1/3 = 0.52083333
Cor 1.4(a): 2*pi*0.0968160064 = 0.6083129089
Lindelof: a_L = 0.0786252790902  2pi a_L = 0.494017198352
crossing: (1.7) < Bellotti-Wong for log T > 2425.8386
crossing: (1.8) < Yang (4.29) (without the term 0.2/t_0) for log t > 933.4679
limit c,R->1: BW linear-left profile = 0.1007368703  reflected explicit profile = 0.09782899554
```

The constant 1.878 of Remark 4.5(ii) is certified by the following program (Python 3.13.17 with `python-flint` 0.9.0, Arb ball arithmetic [21] at 120 bits with outward rounding, so that every printed value is an upper bound).

For $4 \le m \le 9$ (the variables `k` and `K` of the program are m and 2^{m-1}, as in [44, §3]) it covers $\log 3 \le \log t \le \log T_m$ by intervals $[a, b]$ of length $0.002(1 + a/10)$.

On each interval it encloses $|\operatorname{Log}(Q_0 + s)|/\log|Q_0 + s| = \big(1 + \arg(Q_0 + s)^2/\log^2|Q_0 + s|\big)^{1/2}$, $s = \sigma_m + it$, over the whole interval, takes $t_0 = e^a$ in [44, (3.12)], and bounds $0.618^{m/(K-1)} A(e^a)\, t^{(m-3)/(6K-6)}$ at $t = e^b$, where the exponent is positive. The maximum over all intervals and all m is 1.87781.

```
# Interval-arithmetic certificate for the constant 1.878 of the article (Appendix A).
# Python 3.13.17, python-flint 0.9.0 (Arb ball arithmetic, 120 bits, outward rounding).
# Rigorous redo of Yang (JMAA 534 (2024) 128124), Sec. 3.1, eqs. (3.9)-(3.13), for 4<=k<=9,
# with |Log(Q0+s)| = rho(s)*log|Q0+s| in the Phragmen-Lindelof conclusion (Fiori Cor.3 / Trudgian Lemma 3),
# A(t0) as printed in (3.12), t0 on a partition of [3, T_k].  On a piece t in [e^a, e^b], t0=e^a:
#   |zeta(sigma_k+it)| <= 0.618^{k/(K-1)} * rho_max(piece) * A(e^a) * t^{(1-sigma_k)/3} log t
```

```
#                    <= C_piece * t^{1/(2K-2)} log t,  C_piece = 0.618^{k/(K-1)} rho_max A(e^a) e^{b(k-3)
    /(6K-6)}.
from flint import arb, ctx
ctx.prec = 120
Q0 = arb('1.31'); c231 = arb('2.31')**2
def A(u):            # (3.12) with t0 = e^u
    q = c231*(-2*u).exp()
    return (q+1)**(arb(23)/42) * (1 + (q+1).log()/(2*u))
def rho(u0, u1, sk): # max of |Log(Q0+s)|/log|Q0+s| for s = sk+it, t in [e^u0, e^u1]
    t = u0.exp().union(u1.exp()); x = Q0 + sk
    L = (x*x + t*t).log()/2; Ar = (t/x).atan()
    return (1 + (Ar/L)**2).sqrt()
out = []
for k in range(4, 10):
    K = 2**(k-1); sk = 1 - arb(k)/(2*K-2)
    uT = (arb('2.6134')*(K-1) + arb('2.8876')*k)/(k-3)
    base = arb('0.618')**(arb(k)/(K-1)); ex = arb(k-3)/(6*K-6)
    a = arb(3).log(); worst = arb(0); n = 0
    while (a < uT):
        b = a + arb('0.002')*(1 + a/10)
        if b > uT: b = uT
        C = base*rho(a, b, sk)*A(a)*(b*ex).exp()
        if C.upper() > worst.upper(): worst = C
        a = b; n += 1
    out.append((k, float(uT.upper()), n, float(worst.upper())))
    print('k=%d  T_k=exp(%.3f)  pieces=%d  constant on [3,T_k] <= %.5f' % out[-1])
m = max(o[3] for o in out)
print('max over 4<=k<=9: %.5f  (<= 2: %s); for t>=T_k Yang Sec.3.2 gives 1.546' % (m, m <= 2))
```

```
k=4  T_k=exp(29.844)  pieces=6392  constant on [3,T_k] <= 1.54798
k=5  T_k=exp(26.820)  pieces=5997  constant on [3,T_k] <= 1.68940
k=6  T_k=exp(32.780)  pieces=6747  constant on [3,T_k] <= 1.77986
k=7  T_k=exp(46.214)  pieces=8113  constant on [3,T_k] <= 1.83178
k=8  T_k=exp(71.001)  pieces=9940  constant on [3,T_k] <= 1.86126
k=9  T_k=exp(115.401)  pieces=12125  constant on [3,T_k] <= 1.87781
max over 4<=k<=9: 1.87781  (<= 2: True); for t>=T_k Yang Sec.3.2 gives 1.546
```

The construction of Section 7.1 is implemented, as an illustration, by the following program (`numpy`, `mpmath` 1.4.1): it evaluates $\mathcal{F} = N_0 + \Phi$ on a grid of step 10^{-3} on $(0, 120]$, computes $\tau_k^{\mathrm{C}} = \mathrm{meas}\{t > 0 : \mathcal{F}(t) < k - \frac{1}{2}\}$ for $k \le 20$ by counting grid cells, and prints τ_k^{C}, the imaginary part γ_k (`mpmath.zetazero`) and the point τ_k of (1.2).

Theorem 1.5 bounds the mean square of $\tau_k^{\mathrm{C}} - \gamma_k$ for T large; the table makes no claim beyond what it prints.

```
# Casana's method (Section on the construction in the article): the k-th imaginary part of a
# zero of zeta is approximated by the sublevel measure
#     tauC_k = |{t > 0 : F(t) < k - 1/2}|,   F = N_0 + Phi,
# where N_0(t) = (t/2pi) log(t/(2 pi e)) + 7/8 is the Riemann-von Mangoldt main term and Phi is the finite
    Euler
# phase of the article: on the dyadic block 2^j <= t <= 2^{j+1}, Phi interpolates linearly between Phi_{m_j}
     and
# Phi_{m_{j+1}}, m_j = floor(2^{j/2}), Phi_m(t) = -(1/pi) sum_{nu<=m} arg(1 - p_nu^{-1/2-it}) (principal
    argument).
# This program evaluates tauC_k on a grid (step h) for k <= K and compares it with the imaginary part
    gamma_k (mpmath) and
# with the bare point tau_k, N_0(tau_k) = k - 1/2.  Illustration only: the article proves a mean-square
    bound
# (Theorem on the sublevel construction), not a bound for each k.  Python 3.13.17, numpy, mpmath 1.4.1.
import numpy as np
from mpmath import mp, zetazero, lambertw, mpf, exp, pi as mpi

K, TMAX, H = 20, 120.0, 0.001
primes = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71]

def N0(t):
    return t / (2 * np.pi) * np.log(t / (2 * np.pi * np.e)) + 7 / 8

def phi_m(m, t):
    """Phi_m(t) = -(1/pi) sum_{nu<=m} arg(1 - p_nu^{-1/2-it})."""
    s = np.zeros_like(t)
    for p in primes[:m]:
```

```
        z = p ** (-0.5) * np.exp(-1j * t * np.log(p))
        s -= np.angle(1 - z)
    return s / np.pi

def F(t):
    j = np.floor(np.log2(t)).astype(int)              # block index: 2^j <= t < 2^{j+1}
    out = np.empty_like(t)
    for jj in np.unique(j):
        sel = j == jj
        tt = t[sel]
        mj, mj1 = int(np.floor(2 ** (jj / 2))), int(np.floor(2 ** ((jj + 1) / 2)))
        lam = tt / 2.0 ** jj - 1                       # 0 at 2^j, 1 at 2^{j+1}
        out[sel] = N0(tt) + (1 - lam) * phi_m(mj, tt) + lam * phi_m(mj1, tt)
    return out

def tau(k):
    """Bare point: N_0(tau_k) = k - 1/2, by Lambert's W."""
    a = (k - mpf(1) / 2 - mpf(7) / 8)
    return 2 * mpi * a / lambertw(a / exp(1))

t = np.arange(H / 2, TMAX, H)                          # midpoints of the grid cells (0, TMAX]
Ft = F(t)
assert Ft[-1] > K - 0.5 + 2, "TMAX too small"
print('Casana method: tauC_k = |{t > 0 : N_0(t) + Phi(t) < k - 1/2}|, grid step h = %g on (0, %g]' % (H,
    TMAX))
print('%3s %12s %12s %12s %12s %12s' % ('k', 'tauC_k', 'gamma_k', 'tauC_k-gamma_k', 'tau_k', 'tau_k-gamma_k
    '))
mp.dps = 15
for k in range(1, K + 1):
    xk = H * np.count_nonzero(Ft < k - 0.5)
    gk = float(zetazero(k).imag)
    tk = float(tau(k))
    print('%3d %12.4f %12.4f %12.4f %12.4f %12.4f' % (k, xk, gk, xk - gk, tk, tk - gk))
```

```
Casana method: tauC_k = |{t > 0 : N_0(t) + Phi(t) < k - 1/2}|, grid step h = 0.001 on (0, 120]
  k       tauC_k      gamma_k tauC_k-gamma_k        tau_k tau_k-gamma_k
  1      13.5500      14.1347      -0.5847      14.5213       0.3866
  2      21.0590      21.0220       0.0370      20.6557      -0.3663
  3      24.9140      25.0109      -0.0969      25.4927       0.4818
  4      30.3330      30.4249      -0.0919      29.7394      -0.6855
  5      32.9760      32.9351       0.0409      33.6245       0.6895
  6      37.6060      37.5862       0.0198      37.2574      -0.3288
  7      40.9130      40.9187      -0.0057      40.7006      -0.2182
  8      43.4220      43.3271       0.0949      43.9940       0.6669
  9      48.0050      48.0052      -0.0002      47.1651      -0.8400
 10      49.7900      49.7738       0.0162      50.2337       0.4598
 11      52.9500      52.9703      -0.0203      53.2144       0.2441
 12      56.4150      56.4462      -0.0312      56.1189      -0.3273
 13      59.3500      59.3470       0.0030      58.9563      -0.3908
 14      60.7590      60.8318      -0.0728      61.7338       0.9021
 15      65.1240      65.1125       0.0115      64.4577      -0.6548
 16      67.1430      67.0798       0.0632      67.1330       0.0532
 17      69.5590      69.5464       0.0126      69.7640       0.2176
 18      72.0010      72.0672      -0.0662      72.3544       0.2872
 19      75.7050      75.7047       0.0003      74.9073      -0.7973
 20      77.0400      77.1448      -0.1048      77.4257       0.2808
```

All constants and numerical inequalities of this article are certified by the following program. It uses the ball arithmetic of Arb [21], through `python-flint` 0.9.0 (FLINT 3.6.0 [10]), with working precision 256 bits. Every quantity is a ball $[m \pm r]$ that contains the exact value, all operations round outward, and every inequality is checked by an assertion on the whole ball.

The angular integrals are computed both by the closed form (5.9) and by rigorous integration (`acb.integral`) on the arcs where the integrand is analytic. Statements valid on a range of t are certified by monotonicity together with balls at the endpoints, or by covering the range with balls. The values are printed with 20 significant digits.

```
# Certification of the numerical statements of the paper by Arb ball arithmetic.
# Arb (midpoint-radius balls with outward rounding) is used through python-flint.
# Every computed quantity is a ball that contains the exact value; every inequality
# is checked on the whole ball with assert (an Arb comparison is True only if it holds
```

```
# for every point of the balls).  Decimal constants of the paper are exact rationals.
# Digits followed by an ellipsis mean truncation: x = d... means d <= x < d + 10^-k.
# The angular integrals over Backlund's circle are computed both by the closed form (5.9)
# and by rigorous Arb integration (acb.integral) on the arcs where the integrand is analytic.
# Statements on ranges of t are certified by monotonicity (argued in a comment) plus
# endpoint balls, or by covering the range with balls; never by sampling.
# Checks are labelled by the numbering of the paper (theorems, equations, sections).
import sys, flint
from flint import arb, acb, ctx
from fractions import Fraction as Fr
from math import comb

ctx.prec = 256
print('Python %s, python-flint %s, FLINT %s, precision %d bits'
      % (sys.version.split()[0], flint.__version__, flint.__FLINT_VERSION__, ctx.prec))

def A(q):                                   # exact rational (or decimal string) -> ball
    q = Fr(q); return arb(q.numerator) / q.denominator
def show(name, x): print('      %s = %s' % (name, x.str(20, radius=True)))
NCHK = NCALL = 0
def check(ref, text, *conds):
    global NCHK, NCALL
    for c in conds: assert c is True, ref + ': ' + text
    NCHK += len(conds); NCALL += 1; print('  ok [%s] %s' % (ref, text))
def lt(x, q): return bool(x < A(q))
def le(x, q): return bool(x <= A(q))
def gt(x, q): return bool(x > A(q))
def ge(x, q): return bool(x >= A(q))
def tr(x, s):                               # x = s... (truncated digits)
    d = Fr(s); u = Fr(1, 10**len(s.split('.')[1]))
    return (ge(x, d) and lt(x, d + u)) if d >= 0 else (le(x, d) and gt(x, d - u))
def tiny(x): return bool(x.abs_upper() < A(Fr(1, 10**60)))
def cover(pred, a, b, n):                   # pred(ball) for n balls covering [a, b]
    a, b = A(a), A(b)
    return all(pred((a + (b - a)*i/n).union(a + (b - a)*(i + 1)/n)) for i in range(n))
PI, LOG2, HALF, E = arb.pi(), arb(2).log(), A(Fr(1, 2)), arb(1).exp()

# ---- integrals of reflected piecewise linear profiles over sigma = c + R cos(theta), 0 <= theta <= pi ----
def closed_form(nodes, c, R):               # hinge closed form (5.9); nodes (varsigma_i, e_i) exact
    c, R = A(c), A(R); n = len(nodes) - 1
    xs = [a for a, _ in nodes]; es = [b for _, b in nodes]
    s = [(es[i+1] - es[i]) / (xs[i+1] - xs[i]) for i in range(n)]
    hinges = [(xs[i], s[i] - s[i-1]) for i in range(1, n)] + [(Fr(1), -s[n-1])]
    th = ((HALF - c)/R).acos(); e0, s0 = A(es[0]), A(s[0])
    I = (e0 + s0*(c - HALF))*th + s0*R*th.sin() + ((1 + s0)*(HALF - c) + e0)*(PI - th) + (1 + s0)*R*th.sin
     ()
    for a, k in hinges:
        a, k = A(a), A(k); ta = ((a - c)/R).acos(); tb = ((1 - a - c)/R).acos()
        I += k*((c - a)*ta + R*ta.sin()) + k*((1 - c - a)*(PI - tb) + R*tb.sin())
    return I
def pieces(nodes, left=None):               # affine pieces (lo, hi, a0, a1) of e*(sigma) = a0 + a1 sigma
    P = [(None, Fr(0), Fr(1, 2), Fr(-1)), (Fr(1), None, Fr(0), Fr(0))]
    for (x0, e0), (x1, e1) in zip(nodes, nodes[1:]):
        s = (e1 - e0)/(x1 - x0); P.append((x0, x1, e0 - s*x0, s))
        if left is None: P.append((1 - x1, 1 - x0, Fr(1, 2) + e0 + s*(1 - x0), -1 - s))   # reflection
     (1.4)
    if left is not None: P.append((Fr(0), Fr(1, 2)) + left)                               # affine on [0,
     1/2]
    return P
def arb_integral(P, c, R):                  # rigorous integration piece by piece
    c, R = Fr(c), Fr(R); tot = arb(0)
    for lo, hi, a0, a1 in P:
        u1 = Fr(1) if hi is None else min(max((hi - c)/R, Fr(-1)), Fr(1))
        u2 = Fr(-1) if lo is None else min(max((lo - c)/R, Fr(-1)), Fr(1))
        if u1 == u2: continue
        f = lambda t, analytic: acb(A(a0) + A(a1)*A(c)) + A(a1)*A(R)*t.cos()
        tot += acb.integral(f, acb(A(u1).acos()), acb(A(u2).acos())).real
    return tot

print('\nBourgain profile')
k, l = Fr(13, 84), Fr(55, 84); nodesB = []; kap = []
for j in range(8):
    nodesB.append((l - k, k)); kap.append(k); k, l = k/(2*k + 2), Fr(1, 2) + l/(2*k + 2)
```

```
nodesB.append((Fr(1), Fr(0))); xB = [a for a, _ in nodesB]
check('(1.6)', 'the nine nodes are (lambda_j-kappa_j, kappa_j), (kappa_j, lambda_j) = A^j(13/84, 55/84),
    and (1,0) (exact)',
     [str(a) + ',' + str(b) for a, b in nodesB] == ['1/2,13/84', '139/194,13/194', '173/207,13/414', '
    773/854,13/854',
     '820/867,13/1734', '3387/3494,13/3494', '1149/1169,13/7014', '13921/14054,13/14054', '1,0'])
check('Section 5.2', 'the nodes increase, 1/2 <= varsigma_j, varsigma_7 < 1, kappa_j decrease (exact)',
     all(x < y for x, y in zip(xB, xB[1:])) and xB[0] == Fr(1, 2) and xB[7] < 1 and all(a > b for a, b in
    zip(kap, kap[1:])))
IB = closed_form(nodesB, 1, 1); IB2 = arb_integral(pieces(nodesB), 1, 1)
LB = IB/(2*PI*LOG2); show('Lambda(e_B), closed form (5.9)', LB); show('Lambda(e_B), Arb integration', IB2
    /(2*PI*LOG2))
show('2 pi Lambda(e_B)', 2*PI*LB)
check('(5.9)', 'closed form and Arb integration of e_B* agree to 1e-60', tiny(IB - IB2))
check('Section 5.2', 'theta_h = arccos((1/2-c)/R) = 2 pi/3 at c = R = 1', tiny((-HALF).acos() - 2*PI/3))
check('Corollary 1.2', 'Lambda(e_B) = 0.0968160063...', tr(LB, '0.0968160063'))
check('Abstract', 'Lambda(e_B) = 0.0968160... and 2 pi Lambda(e_B) <= 0.60832', tr(LB, '0.0968160'), lt(2*
    PI*LB, '0.60832'))
check('Section 5.2', 'Lambda(e_B) = 0.0968160063238..., 0.096816006 <= Lambda(e_B) <= 0.096816007',
     tr(LB, '0.0968160063238'), ge(LB, '0.096816006'), le(LB, '0.096816007'))
check('Corollary 1.4(a)', '2 pi Lambda(e_B) = 0.6083129... < 0.60832', tr(2*PI*LB, '0.6083129'), lt(2*PI*LB
    , '0.60832'))

print('\nLindelof profile')
I0 = closed_form([(Fr(1, 2), Fr(0)), (Fr(1), Fr(0))], 1, 1); L0 = I0/(2*PI*LOG2)
show('Lambda(0)', L0); show('2 pi Lambda(0)', 2*PI*L0)
check('Remark 5.2', 'int_{2pi/3}^{pi} (-1/2 - cos t) dt = sqrt3/2 - pi/6 equals the circle integral of 0*',
     tiny(I0 - (arb(3).sqrt()/2 - PI/6)), tiny(arb_integral(pieces([(Fr(1, 2), Fr(0)), (Fr(1), Fr(0))]),
    1, 1) - I0))
check('(5.7)', 'Lambda(0) = (3 sqrt3 - pi)/(12 pi log 2) = 0.0786252...', tiny(L0 - (3*arb(3).sqrt() - PI)
    /(12*PI*LOG2)),
     tr(L0, '0.0786252'))
check('Remark 5.2', 'Lindelof gives a <= 0.0786253', le(L0, '0.0786253'))
check('Proposition 6.2', '2 pi Lambda(0) = 0.49401...', tr(2*PI*L0, '0.49401'))

print('\nExplicit profile and Theorem 1.3')
sm = {m: 1 - Fr(m, 2**m - 2) for m in range(4, 10)}
nodesE = [(Fr(1, 2), Fr(27, 164))] + [(sm[m], Fr(1, 2**m - 2)) for m in range(4, 10)] + [(Fr(1), Fr(0))]
vs = [a for a, _ in nodesE] + [Fr(3)]; es = [b for _, b in nodesE] + [Fr(0)]
check('(4.5)', 'sigma_m = 1 - m/(2^m-2) = 5/7, 5/6, 28/31, 17/18, 123/127, 167/170 with exponents 1/(2^m-2)
     (exact)',
     [sm[m] for m in range(4, 10)] == [Fr(5, 7), Fr(5, 6), Fr(28, 31), Fr(17, 18), Fr(123, 127), Fr(167,
    170)])
check('Lemma 4.4', 'varsigma_i increase, e_i = 1/(2^{i+3}-2) nonincreasing, max e_i = 27/164, Q = 10 > 1-
    varsigma_i',
     all(x < y for x, y in zip(vs, vs[1:])), all(es[i] == Fr(1, 2**(i+3) - 2) for i in range(1, 7)),
     all(x >= y for x, y in zip(es, es[1:])), max(es) == Fr(27, 164), all(10 > 1 - v for v in vs))
c, R = Fr('1.000002'), Fr('1.000005'); sigF = c + (c - Fr(1, 2))**2/R; delF = 2*c - sigF - Fr(1, 2)
sig, dl = A(sigF), A(delF); D = (A(R)/(A(c) - HALF)).log()
show('sigma#', sig); show('delta#', dl); show('d = log(R/(c-1/2))', D)
check('Section 5.3', 'c - R = -3e-6, sigma# = 1.25000275..., delta# = 0.2500012499...',
     c - R == Fr(-3, 10**6), tr(sig, '1.25000275'), tr(dl, '0.2500012499'))
check('(3.5)', 'c-R < 1/2 < 1 < c < sigma# < c+R, c < R, 0 < delta# < 1/2, delta# in [1/4,5/8], [c-R,c+R]
    in [-0.01,3]',
     c - R < Fr(1, 2) < 1 < c < sigF < c + R, c < R, 0 < delF < Fr(1, 2), Fr(1, 4) <= delF <= Fr(5, 8),
     c - R >= Fr('-0.01'), c + R <= 3)
IE = closed_form(nodesE, c, R); IE2 = arb_integral(pieces(nodesE), c, R)
show('I_{e_E*}, closed form (5.9)', IE); show('I_{e_E*}, Arb integration', IE2)
check('(5.9)', 'closed form and Arb integration of e_E* agree to 1e-60', tiny(IE - IE2))
args = set([(Fr(1, 2) - c)/R] + [(a - c)/R for a, _ in nodesE[1:]] + [(1 - a - c)/R for a, _ in nodesE
    [1:]])
check('Section 5.3', 'the closed form uses fifteen arccos values at distinct rational points of (-1,1)',
     len(args) == 15, all(-1 < u < 1 for u in args))
C1, C2 = IE/(2*PI*D), 1/(2*D)
lzr = (A(c).zeta()/(2*A(c)).zeta()).log(); lzs = sig.zeta().log()
show('C_1', C1); show('C_2', C2); show('log(zeta(c)/zeta(2c))', lzr); show('log zeta(sigma#)', lzs)
check('(5.10)', 'C_1 = 0.0978292318785..., C_2 = 1/(2d) = 0.7213464797..., log(zeta(c)/zeta(2c)) =
    12.6246665...',
     tr(C1, '0.0978292318785'), tr(C2, '0.7213464797'), tr(lzr, '12.6246665'))
check('Section 1.2', 'Backlund quotient of e_E: 0.0978292318...', tr(C1, '0.0978292318'))
check('(5.11)', 'log(zeta(c)/zeta(2c)) <= 12.62467, sigma# - 1/2 <= 0.7501', le(lzr, '12.62467'), le(sig -
    HALF, '0.7501'))
check('Section 5.3', 'log zeta(sigma#) = 1.52498... <= 1.52499', tr(lzs, '1.52498'), le(lzs, '1.52499'))
```

```
N = 10**4; zb = sum(((-sig)*arb(n).log()).exp() for n in range(1, N)) + ((1 - sig)*A(Fr(2*N - 1, 2)).log())
    .exp()/(sig - 1)
show('log of the bound with N = 10^4', zb.log())
check('Section 5.3', 'footnote: N = 10^4 gives log zeta(sigma#) <= 1.5249836 (and the bound is >= zeta(
    sigma#))',
     le(zb.log(), '1.5249836'), bool(sig.zeta() <= zb))
m640 = 640*dl - 112; show('640 delta# - 112', m640)
check('Section 5.3', '640 delta# - 112 = 48.0007... > 0: the majorant of E/(2 pi) decreases in T',
     tr(m640, '48.0007'), gt(m640, 0))
T = A(100); Emaj = (m640/(1536*299) + A(Fr(1, 1024)))/2; show('E(100, delta#)/(2 pi) majorant', Emaj)
check('Section 5.3', 'E(T,delta#)/(2pi) <= ((640 delta#-112)/(1536*299) + 2^-10)/2 < 0.00055 at T = 100',
     lt(Emaj, '0.00055'))

print('\nLemma 4.3, Lemma 4.4 and the constant K')
check('Lemma 4.3', 'omega_1 in [1/4,0.505], omega_2 in [-0.005,1/4], y >= 49 (affine in sigma, endpoints
    exact)',
     (1 - Fr('-0.01'))/2 == Fr('0.505'), Fr('-0.01')/2 == Fr('-0.005'), Fr(98, 2) == 49)
check('Lemma 4.3', '|phi| <= (0.505*0.256/2 + 0.043)/y^2 <= 0.108/y^2 <= 0.435/t^2 (log(1+x) <= x, arctan
    series)',
     Fr('0.505')**2 <= Fr('0.256'), Fr('0.505')**3/3 <= Fr('0.043'), Fr('0.505')*Fr('0.256')/2 + Fr('0.043
    ') <= Fr('0.108'),
     4*Fr('0.108') <= Fr('0.435'))
sec2 = 1/((PI/2 + (A('0.005')/49).atan())/2).cos()**2; show('sec^2(ph z_2 / 2)', sec2)
check('Lemma 4.3', 'sec^2(ph z_i / 2) <= 2.0005 (ph z_2 <= pi/2 + arctan(0.005/49), ph z_1 < pi/2)', le(
    sec2, '2.0005'))
eps = A('0.6669')/98 + A('0.87')/98**2; show('0.6669/98 + 0.87/98^2', eps); show('e^0.0069', A('0.0069').
    exp())
check('Lemma 4.3', '2*2.0005/6 <= 0.6669; 0.6669/t + 0.87/t^2 <= 0.0069 for t >= 98 (decreasing); e^0.0069
    < 1.0070',
     2*Fr('2.0005')/6 <= Fr('0.6669'), le(eps, '0.0069'), lt(A('0.0069').exp(), '1.0070'))
check('Lemma 4.4', 'e^{105/492} < 2/0.618 (1/6 - 27/164 = 1/492), 105 > 66.7/2, zeta(3) < 2 log 3',
     bool(A(Fr(105, 492)).exp() < 2/A('0.618')), 105 > Fr('66.7')/2, bool(arb(3).zeta() < 2*arb(3).log()))
check('Lemma 4.4', '|t| <= 3: sqrt 18 <= 4.243, sqrt 13 <= 3.606, 4.243 + 2*4.243*3.606 < 34.9 < 49 <
    2*10.5 log 10.5',
     le(arb(18).sqrt(), '4.243'), le(arb(13).sqrt(), '3.606'), Fr('4.243') + 2*Fr('4.243')*Fr('3.606') <
    Fr('34.9'),
     gt(21*A('10.5').log(), 49))
# each factor of K_1(t) = 2(1+84.5/t^2)^{1+27/164}(1+(84.5/t^2+pi/2)/log t) decreases in t, so t = 98 is the
     worst case
t0 = A(98); K1 = 2*(1 + A('84.5')/t0**2)**(1 + A(Fr(27, 164)))*(1 + (A('84.5')/t0**2 + PI/2)/t0.log())
K = eps.exp()*K1; show('K_1(98)', K1); show('K = exp(0.6669/98 + 0.87/98^2) K_1(98)', K)
check('(4.8)', 'K_1(98) <= 2.7167', le(K1, '2.7167'))
check('Lemma 4.4', '1.0070 * 2.7167 <= 2.74 and K <= 2.74', Fr('1.0070')*Fr('2.7167') <= Fr('2.74'), le(K,
    '2.74'))
check('Remark 4.5(i)', 'each input has room: e^{577/492} < 2/0.618 on 3 < |t| <= e^577; above e^577 any C
    <= 1154 = 2*577',
     bool(A(Fr(577, 492)).exp() < 2/A('0.618')), 2*577 == 1154)
check('Remark 4.5(ii)', '(1 - sigma_m)/3 = 1/(2K-2) + (m-3)/(6K-6) with K = 2^{m-1}, exponent (m-3)/(6K-6)
    > 0, 4 <= m <= 9',
     all(Fr(m, 2**m - 2)/3 == Fr(1, 2**m - 2) + Fr(m - 3, 3*2**m - 6) and m > 3 for m in range(4, 10)))

print('\nconstant term C_3(100)')
L274 = A('2.74').log()
def C3(c1, c2, lr, ls, st):                 # the sum C_3(100): costs 2C_1/T + 2C_2/(T log T), or exact
    differences
    base = c2*(lr + (1 + 3/T).log() + L274) + ls/PI + st/(PI*T) + Emaj + A(Fr(1, 4))
    return (base + 2*c1/T + 2*c2/(T*T.log()),
            base + c1*((T + 2).log() - T.log()) + c2*((T + 2).log().log() - T.log().log()))
C3T, C3x = C3(C1, C2, lzr, lzs, sig - HALF)
C3d = C3(C1, C2, A('12.62467'), lzs, A('0.7501'))[0]
C3u = C3(A('0.0978292318786'), A('0.7213464798'), A('12.62467'), A('1.52499'), A('0.7501'))[0]
show('C_3(100)', C3T); show('C_3(100) with exact differences', C3x)
show('C_3(100) with 12.62467 and 0.7501', C3d); show('C_3(100) with all displayed values rounded up', C3u)
check('Section 5.3', 'C_3(100) = 10.598602..., and 10.598545... with exact differences',
     tr(C3T, '10.598602'), tr(C3x, '10.598545'))
check('Appendix A', 'C_3(100) = 10.5986..., 10.5985...', tr(C3T, '10.5986'), tr(C3x, '10.5985'))
check('Section 5.3', 'with the rounded values displayed C_3(100) <= 10.5987', le(C3d, '10.5987'), le(C3u, '
    10.5987'))
check('Section 5.3', '0.09783 > C_1, 0.7214 > C_2, 10.6 > C_3(100) (all variants)', lt(C1, '0.09783'), lt(
    C2, '0.7214'),
     lt(C3T, '10.6'), lt(C3x, '10.6'), lt(C3d, '10.6'), lt(C3u, '10.6'))
# C_3(T) is nonincreasing on [100, oo): each term is constant or decreasing (the majorant since 640 delta# -
     112 > 0).
```

```
# For e <= T <= 100 the right side of (1.7) increases in T, so its value at T = e is the minimum.
check('Section 5.3', 'e <= T <= 100: 2.5167 < 0.09783 log T + 0.7214 loglog T + 10.6; |varpi(T)|/pi <=
    0.53/(pi e) < 0.07',
     gt(A('0.09783') + A('10.6'), '2.5167'), lt(A('0.53')/(PI*E), '0.07'))
check('Corollary 1.2, (5.10), (5.11)', 'the inequalities used for Lambda(e_B), 2 pi Lambda(e_B), C_1, C_2,
    C_3(100), log zeta(sigma#) hold with margin >= 3e-10',
     *[ge(m, Fr(3, 10**10)) for m in (LB - A('0.096816006'), A('0.096816007') - LB, A('0.60832') - 2*PI*LB
    ,
       A('0.09783') - C1, A('0.7214') - C2, A('10.6') - C3T, A('1.52499') - lzs)])

print('\nlimit c, R -> 1 of the Backlund quotients')
LE1 = arb_integral(pieces(nodesE), 1, 1)/(2*PI*LOG2)
LBW = arb_integral(pieces(nodesE, left=(Fr(1, 2), -1 + 2*Fr(27, 164))), 1, 1)/(2*PI*LOG2)
show('profile of Bellotti-Wong (chord on [0,1/2])', LBW); show('reflected profile e_E*', LE1)
check('Section 1.3', 'quotients 0.10073687... (Bellotti-Wong profile) and 0.09782899... (reflected profile)
    ',
     tr(LBW, '0.10073687'), tr(LE1, '0.09782899'), tiny(LE1*2*PI*LOG2 - closed_form(nodesE, 1, 1)))
check('Section 1.3', 'the reflection lies below the chord on [0,1/2] (both piecewise linear: checked at all
     kinks, exact)',
     all(a - Fr(1, 2) + b <= Fr(1, 2) + (-1 + 2*Fr(27, 164))*(1 - a) for a, b in nodesE))

print('\nmain term, Gram points and the construction')
def N0(t): return t/(2*PI)*(t/(2*PI*E)).log() + A(Fr(7, 8))
check('Section 1.1', 'N_0(2 pi) = -1/8', tiny(N0(2*PI) + A(Fr(1, 8))))
check('Theorem 1.5', 'N_0(2 pi e) = 7/8, so k >= 2 iff tau_k >= 2 pi e (1/2 < 7/8 <= 3/2)', tiny(N0(2*PI*E)
    - A(Fr(7, 8))))
def theta(t): return acb(A(Fr(1, 4)), t/2).lgamma().imag - t/2*PI.log()
def dtheta(t): return acb(A(Fr(1, 4)), t/2).digamma().real/2 - PI.log()/2
def dtheta_pos(lo, hi, n):                  # theta' > 0 on [lo, hi]: |theta''| = |Im psi'(1/4+it/2)|/4 <=
    M on 40 balls
    lo, hi = A(lo), A(hi); M = max((acb(A(Fr(1, 4)), ((lo + (hi - lo)*i/40).union(lo + (hi - lo)*(i + 1)
    /40)))/2)
                                    .polygamma(1).imag/4).abs_upper() for i in range(40))
    return all(bool(dtheta(lo + (hi - lo)*i/n) > M*(hi - lo)/(2*n)) for i in range(n + 1))   # grid spacing
     (hi-lo)/n
show('theta(9.66) + pi', theta(A('9.66')) + PI); show('theta(9.67) + pi', theta(A('9.67')) + PI)
check('Section 1.1', "g_{-1} = 9.66...: theta' > 0 on [6.5, 9.67]; theta(6.5), theta(9.66) < -pi < theta
    (9.67)",
     dtheta_pos(Fr('6.5'), Fr('9.67'), 400), bool(theta(A('6.5')) + PI < 0),
     bool(theta(A('9.66')) + PI < 0), bool(theta(A('9.67')) + PI > 0))
check('Section 1.1', '|varpi|/pi < 1/2 at the Gram points: 0.53/(9.66 pi) < 1/2', lt(A('0.53')/(A('9.66')*
    PI), Fr(1, 2)))
g1 = acb.zeta_zero(1).imag; show('gamma_1', g1)
check('proof of Corollary 1.4', 'gamma_1 > 14 > 2 pi > e (rigorous isolation of the first zero)',
     gt(g1, 14), bool(A(14) > 2*PI > E))
check('Remark 7.7', '|varpi(gamma_k)|/pi <= 0.53/(14 pi) < 1/2 for gamma_k >= 14', lt(A('0.53')/(14*PI), Fr
    (1, 2)))
def mj(j):                                  # m_j = floor(2^{j/2}); for odd j the ball decides the floor
    if j % 2 == 0: return int(Fr(2)**(j//2))
    x = (A(Fr(j, 2))*LOG2).exp(); f = int(x.mid().floor().unique_fmpz())
    assert bool(x > f) and bool(x < f + 1); return f
check('Section 7.1', '(1/pi) arcsin(2^{-1/2}) = 1/4; m_j = 0 for j < 0, m_j nondecreasing, m_j <= 2 for j
    <= 3',
     tiny((1/arb(2).sqrt()).asin()/PI - A(Fr(1, 4))), all(mj(j) == 0 for j in range(-6, 0)),
     all(mj(j) <= mj(j + 1) for j in range(-6, 12)), max(mj(j) for j in range(-6, 4)) == 2)
check('Proposition 7.1', '4 <= 2 pi <= 8, m_2 = m_3 = 2, F(2 pi) <= N_0(2 pi) + 1/2 = 3/8 < 1/2; 4 pi e > 2
     pi^2',
     bool(A(4) <= 2*PI <= A(8)), mj(2) == 2, mj(3) == 2, lt(N0(2*PI) + HALF, Fr(1, 2)), bool(4*PI*E > 2*PI
    **2))
# d/dt [N_0(t) - sqrt(2t)/4] = (1/2pi) log(t/2pi) - 1/(4 sqrt(2t)) increases in t
dd = 1/(2*PI) - 1/(4*(4*PI*E).sqrt()); show("d/dt [N_0(t) - sqrt(2t)/4] at t = 2 pi e", dd)
check('Section 7.4', 'N_0(t) - sqrt(2t)/4 increases for t >= 2 pi e; 2 log 2 - log(2 pi e) = -1.45...',
     bool(dd > 0), tr(2*LOG2 - (2*PI*E).log(), '-1.45'))
check('Theorem 3.8', 'Fujii side condition: T >= 9 gives H >= 3 and H - H^{1/8} >= 3 - 3^{1/8} >= 1 (
    increasing in H)',
     bool(3 - (arb(3).log()/8).exp() >= 1))
check('Lemma 7.5', 'log(2 + h log 4T) <= 2 log(2 + h L_T) for h L_T >= 1, h <= 1: 6 > log(8 pi)',
     bool(A(6) > (8*PI).log()))

print('\nproof of Theorem 1.1: c = 1 + eta, R = 1 + 2 eta, 0 < eta < 1/100')
def sg(e): return 1 + e + (Fr(1, 2) + e)**2/(1 + 2*e)
ends = (Fr(0), Fr(1, 100))   # all quantities affine in eta: >= 0 at both ends and > 0 at one end gives > 0
    inside
```

```
check('Section 5.1', 'sigma# = 5/4 + 3 eta/2, delta# = 1/4 + eta/2, R/(c-1/2) = 2 (affine identities at
    three points)',
     all(sg(e) == Fr(5, 4) + Fr(3, 2)*e and 2*(1 + e) - sg(e) - Fr(1, 2) == Fr(1, 4) + e/2 and (1 + 2*e)/(
    Fr(1, 2) + e) == 2
         for e in ends + (Fr(1, 7),)))
check('Section 5.1', '(3.5) holds, delta# in [1/4, 5/8], [c-R, c+R] = [-eta, 2+3 eta] in [-0.01, 3] for 0 <
     eta < 1/100',
     all(-e < Fr(1, 2) and e >= 0 and Fr(5, 4) + Fr(3, 2)*e > 1 + e and 2 + 3*e > Fr(5, 4) + Fr(3, 2)*e
         and Fr(1, 4) <= Fr(1, 4) + e/2 <= Fr(5, 8) and Fr(1, 4) + e/2 < Fr(1, 2) and -e >= Fr('-0.01')
    and 2 + 3*e <= 3
         for e in ends))

print('\ncrossing points')
def ours(L): return A('0.09783')*L + A('0.7214')*L.log() + A('10.6')                # (1.7), L = log T
def bw1(L): return ours(L) - A('0.10076')*L - A('0.24460')*L.log() - A('7.20844')   # minus Bellotti-Wong
    Cor. 1.5,
def bw2(L): return ours(L) - A('0.10076')*L - A('1.68845')*L.log() - A('1.50956')   # first and second
    branch
def yg(L):    # Yang (4.29) without 0.2/t_0 minus (1.8), L = log t
    return A('0.11')*L + A('0.29')*L.log() + A('2.29') - A('0.09783')*L - A('0.7214')*L.log() - A('10.7')
def root(f, lo, hi):                        # f(lo) > 0 > f(hi); bisection on balls, returns a ball
    containing the root
    lo, hi = A(lo), A(hi); assert f(lo) > 0 and f(hi) < 0
    for _ in range(240):
        m = (lo + hi)/2; v = f(m)
        if v > 0: lo = m
        elif v < 0: hi = m
        else: break
    return lo.union(hi)
xb = root(bw1, 2000, 3000); xy = root(lambda L: -yg(L), 900, 1000)
show('crossing with Bellotti-Wong, log T', xb); show('crossing with Yang (4.29), log t', xy)
# Sign structure for L = log T >= 1.  For L >= 52 the first branch is the minimum (log 52 > 3.95 >
    5.69888/1.44385).
# d/dL bw1 = -0.00293 + 0.4768/L < 0 for L > 163, so bw1 > 0 on [163, xb) and bw1 < 0 on (xb, oo).
# On [1, 163], (1.7) exceeds one branch, hence the minimum: covered by balls.
check('Section 1.2', '(1.7) is smaller than Bellotti-Wong exactly for log T > 2425.8385... (unique crossing
     on log T >= 1)',
     tr(xb, '2425.8385'), bool(A(52).log() > A('3.95')), Fr('5.69888')/Fr('1.44385') < Fr('3.95'),
     Fr('0.4768')/Fr('0.00293') < 163, cover(lambda x: bool(bw1(x) > 0) or bool(bw2(x) > 0), 1, 163, 2000)
    )
check('Abstract', 'the complete bound is smaller only for log T > 2425.8385...', tr(xb, '2425.8385'))
check('Section 1', 'the complete estimate improves upon Bellotti-Wong only for log T > 2425.8385...', tr(xb
    , '2425.8385'))
# yg(L) <= 0.01217 L - 8.41 < 0 for 1 <= L <= 35.5; d/dL yg = 0.01217 - 0.4314/L > 0 for L >= 35.5: one sign
     change.
# The term 0.2/t_0 of Yang's bound is positive and only enlarges his bound.
check('Section 10', '(1.8) is smaller than Yang (4.29) as soon as log t > 933.5 (unique crossing on log t
    >= 1)',
     Fr('0.01217')*Fr('35.5') < Fr('8.41'), Fr('0.4314')/Fr('0.01217') < Fr('35.5'), bool(yg(A('933.5')) >
     0),
     lt(xy, '933.5'))
check('Appendix A', 'crossing points log T = 2425.8385... and log t = 933.4678...', tr(xb, '2425.8385'), tr
    (xy, '933.4678'))

# ---- exact rational arithmetic and identities (Table 1, Sections 1.2, 1.3, Lemma 4.1, Lemma 7.6, Theorems
    8.1, 9.5) ----
for stated, src in [('0.4320', '0.43200'), ('1.9167', '1.91662'), ('13.0788', '13.07873'), ('0.2907', '
    0.29062'),
                    ('7.0120', Fr('6.13695') + Fr(7, 8)), ('5.225', Fr('4.35') + Fr(7, 8)), ('2.873', Fr('
    1.998') + Fr(7, 8)),
                    ('8.08344', Fr('7.20844') + Fr(7, 8)), ('11.55', Fr('10.6') + Fr(7, 8) + Fr('0.07'))]:
    assert Fr(stated) >= Fr(src)            # Table 1: upper bounds rounded up
assert Fr('10.7') - Fr(1, 2) == Fr('10.2') and Fr('10.6') + Fr('0.07') <= Fr('10.7')
assert Fr('7.20844') + Fr('0.07') <= Fr('7.28') and Fr('8.08344') - Fr('7.20844') == Fr(7, 8)
assert Fr('0.10076') - Fr('0.09783') == Fr('0.00293') and Fr('0.1038') - Fr('0.10076') == Fr('0.00304')
assert Fr(1, 16) + Fr(1, 8) + Fr(1, 3) == Fr(25, 48) < Fr('0.53') and Fr(13**2, 2) == Fr('84.5')
assert Fr('1.87781') <= Fr('1.878') <= 2 and Fr('1.546') <= 2
assert Fr(1, 2) + Fr(1, 5) == Fr(7, 10) and Fr(1, 5)/40 == Fr(1, 200) <= Fr(1, 2) and Fr(1, 2)/20 == Fr(1,
    40)
assert all(((Fr(1, 2) + u)**(2*j+1) + (Fr(1, 2) - u)**(2*j+1))/(2*j+1)    # Theorem 8.1: Psi_j, degree 2j+1,
     2j+2 points
         == u**(2*j) + sum(Fr(comb(2*j+1, 2*i))*Fr(4)**(i-j)/(2*j+1)*u**(2*i) for i in range(j))
         for j in range(1, 9) for u in [Fr(i, 3) for i in range(2*j + 2)])
```

```
assert all(sum((i - Fr(m - 1, 2))**2 for i in range(m)) == Fr(m*(m*m - 1), 12) for m in range(1, 7)) and Fr
    (2*2 - 1, 3) == 1

print('\n%d checks, %d asserted conditions' % (NCALL, NCHK))
print('all checks passed')
```

```
Python 3.13.18, python-flint 0.9.0, FLINT 3.6.0, precision 256 bits

Bourgain profile
  ok [(1.6)] the nine nodes are (lambda_j-kappa_j, kappa_j), (kappa_j, lambda_j) = A^j(13/84, 55/84), and
     (1,0) (exact)
  ok [Section 5.2] the nodes increase, 1/2 <= varsigma_j, varsigma_7 < 1, kappa_j decrease (exact)
      Lambda(e_B), closed form (5.9) = [0.096816006323848965603 +/- 1.85e-22]
      Lambda(e_B), Arb integration = [0.096816006323848965603 +/- 1.85e-22]
      2 pi Lambda(e_B) = [0.60831290843381374985 +/- 1.57e-21]
  ok [(5.9)] closed form and Arb integration of e_B* agree to 1e-60
  ok [Section 5.2] theta_h = arccos((1/2-c)/R) = 2 pi/3 at c = R = 1
  ok [Corollary 1.2] Lambda(e_B) = 0.0968160063...
  ok [Abstract] Lambda(e_B) = 0.0968160... and 2 pi Lambda(e_B) <= 0.60832
  ok [Section 5.2] Lambda(e_B) = 0.0968160063238..., 0.096816006 <= Lambda(e_B) <= 0.096816007
  ok [Corollary 1.4(a)] 2 pi Lambda(e_B) = 0.6083129... < 0.60832

Lindelof profile
      Lambda(0) = [0.078625279090205377185 +/- 4.37e-22]
      2 pi Lambda(0) = [0.49401719835247279041 +/- 2.48e-21]
  ok [Remark 5.2] int_{2pi/3}^{pi} (-1/2 - cos t) dt = sqrt3/2 - pi/6 equals the circle integral of 0*
  ok [(5.7)] Lambda(0) = (3 sqrt3 - pi)/(12 pi log 2) = 0.0786252...
  ok [Remark 5.2] Lindelof gives a <= 0.0786253
  ok [Proposition 6.2] 2 pi Lambda(0) = 0.49401...

Explicit profile and Theorem 1.3
  ok [(4.5)] sigma_m = 1 - m/(2^m-2) = 5/7, 5/6, 28/31, 17/18, 123/127, 167/170 with exponents 1/(2^m-2) (
     exact)
  ok [Lemma 4.4] varsigma_i increase, e_i = 1/(2^{i+3}-2) nonincreasing, max e_i = 27/164, Q = 10 > 1-
     varsigma_i
      sigma# = [1.2500027500002499988 +/- 5.00e-20]
      delta# = [0.25000124999975000125 +/- 6.5e-24]
      d = log(R/(c-1/2)) = [0.69314818055544532975 +/- 4.74e-22]
  ok [Section 5.3] c - R = -3e-6, sigma# = 1.25000275..., delta# = 0.2500012499...
  ok [(3.5)] c-R < 1/2 < 1 < c < sigma# < c+R, c < R, 0 < delta# < 1/2, delta# in [1/4,5/8], [c-R,c+R] in
     [-0.01,3]
      I_{e_E*}, closed form (5.9) = [0.42606376380391913387 +/- 4.08e-21]
      I_{e_E*}, Arb integration = [0.42606376380391913387 +/- 4.08e-21]
  ok [(5.9)] closed form and Arb integration of e_E* agree to 1e-60
  ok [Section 5.3] the closed form uses fifteen arccos values at distinct rational points of (-1,1)
      C_1 = [0.097829231878517946339 +/- 2.33e-22]
      C_2 = [0.72134647976617563460 +/- 2.03e-21]
      log(zeta(c)/zeta(2c)) = [12.624666509201434702 +/- 2.16e-19]
      log zeta(sigma#) = [1.5249835586891991530 +/- 3.15e-20]
  ok [(5.10)] C_1 = 0.0978292318785..., C_2 = 1/(2d) = 0.7213464797..., log(zeta(c)/zeta(2c)) =
     12.6246665...
  ok [Section 1.2] Backlund quotient of e_E: 0.0978292318...
  ok [(5.11)] log(zeta(c)/zeta(2c)) <= 12.62467, sigma# - 1/2 <= 0.7501
  ok [Section 5.3] log zeta(sigma#) = 1.52498... <= 1.52499
      log of the bound with N = 10^4 = [1.5249835587005347825 +/- 2.23e-20]
  ok [Section 5.3] footnote: N = 10^4 gives log zeta(sigma#) <= 1.5249836 (and the bound is >= zeta(sigma#)
     )
      640 delta# - 112 = [48.000799999840000800 +/- 4.02e-21]
  ok [Section 5.3] 640 delta# - 112 = 48.0007... > 0: the majorant of E/(2 pi) decreases in T
      E(100, delta#)/(2 pi) majorant = [0.00054053964604218924279 +/- 1.55e-24]
  ok [Section 5.3] E(T,delta#)/(2pi) <= ((640 delta#-112)/(1536*299) + 2^-10)/2 < 0.00055 at T = 100

Lemma 4.3, Lemma 4.4 and the constant K
  ok [Lemma 4.3] omega_1 in [1/4,0.505], omega_2 in [-0.005,1/4], y >= 49 (affine in sigma, endpoints exact
     )
  ok [Lemma 4.3] |phi| <= (0.505*0.256/2 + 0.043)/y^2 <= 0.108/y^2 <= 0.435/t^2 (log(1+x) <= x, arctan
     series)
      sec^2(ph z_2 / 2) = [2.0002041024583719369 +/- 3.98e-20]
  ok [Lemma 4.3] sec^2(ph z_i / 2) <= 2.0005 (ph z_2 <= pi/2 + arctan(0.005/49), ph z_1 < pi/2)
      0.6669/98 + 0.87/98^2 = [0.0068956892961266139109 +/- 2.96e-23]
      e^0.0069 = [1.0069238598460768235 +/- 2.01e-20]
  ok [Lemma 4.3] 2*2.0005/6 <= 0.6669; 0.6669/t + 0.87/t^2 <= 0.0069 for t >= 98 (decreasing); e^0.0069 <
     1.0070
```

```
  ok [Lemma 4.4] e^{105/492} < 2/0.618 (1/6 - 27/164 = 1/492), 105 > 66.7/2, zeta(3) < 2 log 3
  ok [Lemma 4.4] |t| <= 3: sqrt 18 <= 4.243, sqrt 13 <= 3.606, 4.243 + 2*4.243*3.606 < 34.9 < 49 < 2*10.5
     log 10.5
      K_1(98) = [2.7166062888678559009 +/- 4.21e-20]
      K = exp(0.6669/98 + 0.87/98^2) K_1(98) = [2.7354038985273522633 +/- 4.22e-20]
  ok [(4.8)] K_1(98) <= 2.7167
  ok [Lemma 4.4] 1.0070 * 2.7167 <= 2.74 and K <= 2.74
  ok [Remark 4.5(i)] each input has room: e^{577/492} < 2/0.618 on 3 < |t| <= e^577; above e^577 any C <=
     1154 = 2*577
  ok [Remark 4.5(ii)] (1 - sigma_m)/3 = 1/(2K-2) + (m-3)/(6K-6) with K = 2^{m-1}, exponent (m-3)/(6K-6) >
     0, 4 <= m <= 9

constant term C_3(100)
      C_3(100) = [10.598602348349442860 +/- 1.76e-19]
      C_3(100) with exact differences = [10.598545473327798800 +/- 4.84e-19]
      C_3(100) with 12.62467 and 0.7501 = [10.598605175981063029 +/- 4.86e-19]
      C_3(100) with all displayed values rounded up = [10.598607226776234255 +/- 1.43e-19]
  ok [Section 5.3] C_3(100) = 10.598602..., and 10.598545... with exact differences
  ok [Appendix A] C_3(100) = 10.5986..., 10.5985...
  ok [Section 5.3] with the rounded values displayed C_3(100) <= 10.5987
  ok [Section 5.3] 0.09783 > C_1, 0.7214 > C_2, 10.6 > C_3(100) (all variants)
  ok [Section 5.3] e <= T <= 100: 2.5167 < 0.09783 log T + 0.7214 loglog T + 10.6; |varpi(T)|/pi <= 0.53/(
     pi e) < 0.07
  ok [Corollary 1.2, (5.10), (5.11)] the inequalities used for Lambda(e_B), 2 pi Lambda(e_B), C_1, C_2, C_3
     (100), log zeta(sigma#) hold with margin >= 3e-10

limit c, R -> 1 of the Backlund quotients
      profile of Bellotti-Wong (chord on [0,1/2]) = [0.10073687033405844994 +/- 1.61e-21]
      reflected profile e_E* = [0.097828995542983327128 +/- 4.25e-22]
  ok [Section 1.3] quotients 0.10073687... (Bellotti-Wong profile) and 0.09782899... (reflected profile)
  ok [Section 1.3] the reflection lies below the chord on [0,1/2] (both piecewise linear: checked at all
     kinks, exact)

main term, Gram points and the construction
  ok [Section 1.1] N_0(2 pi) = -1/8
  ok [Theorem 1.5] N_0(2 pi e) = 7/8, so k >= 2 iff tau_k >= 2 pi e (1/2 < 7/8 <= 3/2)
      theta(9.66) + pi = [-0.0014853253998891945298 +/- 2.22e-24]
      theta(9.67) + pi = [0.0006656101479862206876 1 +/- 1.32e-24]
  ok [Section 1.1] g_{-1} = 9.66...: theta' > 0 on [6.5, 9.67]; theta(6.5), theta(9.66) < -pi < theta(9.67)
  ok [Section 1.1] |varpi|/pi < 1/2 at the Gram points: 0.53/(9.66 pi) < 1/2
      gamma_1 = [14.134725141734693790 +/- 4.58e-19]
  ok [proof of Corollary 1.4] gamma_1 > 14 > 2 pi > e (rigorous isolation of the first zero)
  ok [Remark 7.7] |varpi(gamma_k)|/pi <= 0.53/(14 pi) < 1/2 for gamma_k >= 14
  ok [Section 7.1] (1/pi) arcsin(2^{-1/2}) = 1/4; m_j = 0 for j < 0, m_j nondecreasing, m_j <= 2 for j <= 3
  ok [Proposition 7.1] 4 <= 2 pi <= 8, m_2 = m_3 = 2, F(2 pi) <= N_0(2 pi) + 1/2 = 3/8 < 1/2; 4 pi e > 2 pi
     ^2
      d/dt [N_0(t) - sqrt(2t)/4] at t = 2 pi e = [0.11638015805286826912 +/- 1.48e-21]
  ok [Section 7.4] N_0(t) - sqrt(2t)/4 increases for t >= 2 pi e; 2 log 2 - log(2 pi e) = -1.45...
  ok [Theorem 3.8] Fujii side condition: T >= 9 gives H >= 3 and H - H^{1/8} >= 3 - 3^{1/8} >= 1 (
     increasing in H)
  ok [Lemma 7.5] log(2 + h log 4T) <= 2 log(2 + h L_T) for h L_T >= 1, h <= 1: 6 > log(8 pi)

proof of Theorem 1.1: c = 1 + eta, R = 1 + 2 eta, 0 < eta < 1/100
  ok [Section 5.1] sigma# = 5/4 + 3 eta/2, delta# = 1/4 + eta/2, R/(c-1/2) = 2 (affine identities at three
     points)
  ok [Section 5.1] (3.5) holds, delta# in [1/4, 5/8], [c-R, c+R] = [-eta, 2+3 eta] in [-0.01, 3] for 0 <
     eta < 1/100

crossing points
      crossing with Bellotti-Wong, log T = [2425.8385801225889471 +/- 2.63e-17]
      crossing with Yang (4.29), log t = [933.46789604929362221 +/- 4.21e-18]
  ok [Section 1.2] (1.7) is smaller than Bellotti-Wong exactly for log T > 2425.8385... (unique crossing on
      log T >= 1)
  ok [Abstract] the complete bound is smaller only for log T > 2425.8385...
  ok [Section 1] the complete estimate improves upon Bellotti-Wong only for log T > 2425.8385...
  ok [Section 10] (1.8) is smaller than Yang (4.29) as soon as log t > 933.5 (unique crossing on log t >=
     1)
  ok [Appendix A] crossing points log T = 2425.8385... and log t = 933.4678...

61 checks, 136 asserted conditions
all checks passed
```

B Correspondence with the sources

The following table lists every statement taken from the literature (Section 3 and the proofs), where it is stated in the source, and the normalization or adaptation used here. Within the table, k, α_{F}, H and h are as in Theorem 3.8.

used as	source and place	form there; adaptation here
Theorem 3.1	[39], §§9.3, 9.4, 9.12 (formula for N and one-sided limits at imaginary parts; $S = O(\log t)$; gaps $\to$ 0; the convention $S(t) = S(t^+)$ at $t \in \{\gamma_k\}$ is adopted in this paper)	$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t)$ with ϑ through $\operatorname{Log}\Gamma(\frac{1}{4} + \frac{it}{2})$; Lemma 4.1 rewrites $\frac{1}{\pi}\vartheta + 1 = N_0 + \varpi/\pi$
Theorem 3.2	[6, §5.11(ii)]	$\lvert\mathcal{R}(z)\rvert \le \sec^2(\frac{1}{2}\operatorname{ph} z)/(12\lvert z\rvert)$, $\lvert\operatorname{ph} z\rvert < \pi$; used at $z = \frac{1}{4} + \frac{it}{2}$ (Lemma 4.1) and at $z = \frac{1-s}{2}, \frac{s}{2}$ (Lemma 4.3)
Theorem 3.3(a)	[19], Thm. 1.1; [30], Thm. 1.1; both as quoted in [3, §2, display before (2.3)]	$0.618\lvert t\rvert^{1/6}\log\lvert t\rvert$ and $66.7\lvert t\rvert^{27/164}$ for $\lvert t\rvert \ge 3$; used only through (4.6), with constant 2 on $\sigma = \frac{1}{2}$ (Remark 4.5(i))
Theorem 3.3(b)	[44], Thm. 1.1; quoted in [3, (2.4)]	stated there as $1.546\lvert t\rvert^{1/(2^m-2)}\log\lvert t\rvert$ on $\sigma = \sigma_m$, $\lvert t\rvert \ge 3$, $m \ge 4$; used only for $4 \le m \le 9$, with 1.546 for $\lvert t\rvert \ge T_m$ (§3.2) and 1.878 for $3 \le \lvert t\rvert \le T_m$ (§3.1 corrected in Remark 4.5(ii)), hence constant 2
Theorem 3.3(c)	[29], Thm. 1.1	$\lvert\zeta(1+it)\rvert \le \log\lvert t\rvert$, $\lvert t\rvert \ge 3$; as stated
comment after Theorem 3.3	[3], (2.1); [2], Thm. 1.1; [25], Lemma 5	$58.096\log^{2/3} t$ is Leong's, not a direct consequence of Bellotti's 70.6995; not used
Theorem 3.4	[9], Cor. 3	equal powers of the logarithm on both lines; here $Q = 10$, $\Xi(s) = (s-1)\zeta(s)$, $\alpha_2 = 1$
Theorem 3.5	[3], Lemmas 2.1, 2.3, 2.4, which take (a), (b) from [16], Props. 3.5, 3.7 (for ζ_K, here $K = \mathbb{Q}$); $\mathcal{E}(T,d) = E(0,d,T)$ of [4], Def. 3.3 and Fig. 2, and (c) its Lemma 3.4 at $a = 0$	$\mathfrak{f}_n$ as in (3.3), with the factor $\frac{1}{2}$; (c) is the majorant of $\mathcal{E}(T,d)/\pi$ for $d \in [\frac{1}{4}, \frac{5}{8}]$; the decomposition of $\pi S(T)$ in the proof of Proposition 5.1 is that of [3, §3]
Remark 4.5(i)	[34], Thm. 1.1	$0.611\lvert t\rvert^{1/6}\log\lvert t\rvert$ for $\lvert t\rvert \ge 3$; mentioned only
Remark 4.5(ii)	[44], §3.1, (3.9)–(3.13), §3.2	recomputed with $\lvert\operatorname{Log}(Q_0 + s)\rvert$, constant 1.878 for $4 \le m \le 9$ (interval-arithmetic program, Appendix A)
Remark 4.5(iii)	[18], §2	$\lvert\zeta(1+it)\rvert \le \log\lvert t\rvert - 0.45$ for $\lvert t\rvert \ge 3$; mentioned only
Table 1	[43], p. 1; [1], (39), pp. 354–355, 375 (also for Grossmann); [40], Thm. 1; the other rows as listed in [3], Table 1	normalization $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$: $\frac{7}{8}$ added to the constants stated for $N - N_0$ or for S
leading coefficients in Section 1.3	[41], Thm. 1; [32], Cor. 1; [17], Cor. 1.2 and Table 1; [3], Cor. 1.5 and Table 1	0.111 in the preprint of Trudgian, 0.112 in its published version (as listed in both tables); 0.110; 0.1038; 0.10076
Theorem 3.6	[3, (1.2)], from [31]	$\lvert S(T)\rvert \le 2.5167$ for $0 \le T \le 30\,610\,046\,000$; used on $[e, 100]$
Theorem 3.7(a),(b)	[20], Lemma 2.8 and (7.57), p. 190	the A-process; $\zeta(\sigma + it) \ll \lvert t\rvert^{(\kappa+\lambda-\sigma)/2}\log\lvert t\rvert$ for $\frac{1}{2} \le \sigma \le \lambda - \kappa$

used as	source and place	form there; adaptation here
Theorem 3.7(c)	[5], §5, Thm. 6	$(\frac{13}{84}+\varepsilon, \frac{55}{84}+\varepsilon)$ is an exponent pair; ε kept in both coordinates
Theorem 3.8(a)	[11], §3, pp. 232–233, with Lemma 5; corrections [12, §5]	modulus 1; used with $\mathsf{k}=2$, $\alpha_{\mathrm{F}}=\frac{1}{5}$, $H=T=U$, $Y=U^{1/200}$ (Lemma 7.6) and, in Section 1.3, with $\mathsf{k}=1$, $\alpha_{\mathrm{F}}=\frac{1}{2}$, $H=T$, $Y=T^{1/40}$
Theorem 3.8(b)	[11], Main Theorem, p. 225 ($0<a_i\le\frac{1}{2}$, $T^{1/2+a_2}\le H\le T$); [12], §1 (α) (the case $q=1$, $H=T$) and §5	modulus 1, $\alpha_{\mathrm{F}}=a_2$; used with $\mathsf{k}=2$, $\alpha_{\mathrm{F}}=\frac{1}{2}$, $H=T$, $0<h\le 1$ (Lemma 7.5); $\log(3+x)\le\frac{\log 3}{\log 2}\log(2+x)$
Theorem 3.9	[28], Cor. 3	finite sums only, shifted to $[U,2U]$; applied to $\mathscr{D}_1^2$ and, as a Dirichlet polynomial in v^2, to $\mathscr{D}_2^2$; also in Section 1.3 and Lemma 9.2
Theorem 3.10	[27]; [38, Ch. I.1]	as stated
$(s-1)\zeta(s)=s-s(s-1)\int_1^\infty\{u\}u^{-s-1}du$	[39, (2.1.4)]	for $\sigma>0$; used for $\lvert t\rvert\le 3$ in Lemma 4.4 and Section 5.2
$\zeta(\sigma+it)\ll\log t$ for $1\le\sigma\le 3$	[39, Thm. 3.5]	proof of Theorem 1.1
bathtub principle	[26, Thm. 1.14]	used in Theorem 7.3 in the elementary form stated there
rising sun lemma	[35]	discrete form, proved in Lemma 7.2
Theorem 3.11	[37]; [39, Notes for Chapter 9, (9.25.1), p. 245]; quoted with remainder in [11, p. 225]	asymptotic form only; $\mathsf{k}=1$ in dyadic form (8.9) (Proposition 8.4), $\mathsf{k}=2$ in Remark 7.7, every k in Proposition 8.6
Gram points, Selberg's moments of Δ_k^{G}	[23, Thm. 4.2]	Remark 7.7 only, as a consistency check of Proposition 8.4
Gram's law for γ_k, its failures	[23, Def. 1.4 and p. 58]	Section 1.1 only
increasing rearrangement	[15, §10.12, pp. 276–278] (first edition)	defined there in decreasing order; terminology in Sections 1.3 and 7.2

Data availability

The programs of Appendix A and their outputs are reproduced in that appendix. No other data were used.

AI Declaration

The ideas of this article, the constructions and the strategy of the proofs are the author's. Generative AI tools were used during the work and the preparation of the manuscript: Claude (Anthropic), models Claude Fable 5.1, Claude Opus 5.5 and Claude Sonnet 5.5; ChatGPT (OpenAI), model Astra 6; and Microsoft 365 Copilot, based on the GPT-5 reasoning model. Under the author's direction, Claude and ChatGPT were used to draft and revise the LaTeX text, to check proofs, constants and citations against the sources, and to search the literature. The Python programs of Appendix A were written with Claude, which was also used to formalize parts of the proofs in Lean 4; the Lean 4 formalization is not presented in this article. Microsoft 365 Copilot was used only to review drafts of the manuscript, not to write the Python programs or the Lean 4 code. The author has checked the whole article and takes full responsibility for its content. No AI tool is an author.

Mathematical software

The computations of Appendix A use Python 3.13 with `mpmath` 1.4.1, `numpy` and `python-flint` 0.9.0 (Arb ball arithmetic [21], FLINT 3.6.0 [10]).

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